



2024

PROSPECTUS



Postgraduate Admissions - 2024

MPhil, ME, MS, LLM, PhD, MS (Surgery) & MD (Medicine)

Advanced Studies & Research Board
UNIVERSITY OF KARACHI



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& MD (Medicine)**

**Advanced Studies & Research Board
University of Karachi**

Message



Prof. Dr. M. Khalid Iraqi
Vice Chancellor

Dear Prospective Postgraduate Research Students!

I extend a very warm welcome to all of you for your decision to join the University of Karachi for your postgraduate research journey. Your decision to pursue research studies from one of the leading public sector universities of Pakistan will prove to be a journey of transformation in the real sense.

The University of Karachi offers a vibrant research culture, where innovation, creativity, and critical thinking culminate. Being a multidisciplinary university that offers postgraduate degree programmes in a variety of disciplines, research scholars are encouraged to share their unique perspective and set of skills, contributing to the rich tapestry of research. We are committed to provide you with an environment where you can learn to question, accept challenges, find solutions to problems, and make impactful contributions to your field of study.

With a focus on collaborative approach to research, we promote interdisciplinary projects that allow our research students to explore new horizons and address complex global challenges from a multitude of perspectives. In order to encourage our research scholars to collaborate with others, we organize workshops, seminars, and conferences to provide them the required exposure and multiple opportunities for collaboration.

Since embarking on a postgraduate research program is a journey of discovery, perseverance, and growth, it requires dedication, curiosity, and a passion for learning. As the Vice-Chancellor of the University of Karachi, I assure you that our institution is dedicated to providing the support and resources you need to succeed.

We look forward to welcoming you to our university with the hope that you will produce groundbreaking research in your specific field of inquiry. Wish you all the best!

Message



Prof. Dr. Anila Amber Malik
Convener
Postgraduate Admission Committee

I welcome all the potential research scholars to the postgraduate programmes offered in different disciplines at the University of Karachi. A university with a rich history of research tradition has produced eminent research scholars and renowned scientists in diverse fields and the legacy continues, as the university is committed to unlocking your research potential as prospective researchers at postgraduate level.

With a focus on contemporary trends in research, the postgraduate programmes offered at the University of Karachi provide rigorous exposure to research that is required for successful completion of the specific postgraduate degree programmes. I assure you that being part of the research community of the University of Karachi will prove to be an intellectually stimulating experience for you, as the interactions with the academicians and research scholars will enrich your experience and broaden your horizon as a researcher.

Being one of the largest public sector universities of the country, the postgraduate programmes offered at the University of Karachi are diverse and multidisciplinary in nature. The courses designed for postgraduate programmes across various disciplines aim to enhance your research skills. Besides this, our highly qualified faculty, with extensive research experience, is available to provide the supervision needed to produce quality research. If you want to excel in the research field of your interest, then the University of Karachi is the right place to join.

ADVANCED STUDIES & RESEARCH BOARD UNIVERSITY OF KARACHI

Members of the Advanced Studies and Research Board

Chairman

Prof. Dr. Khalid M. Iraqi

Vice-Chancellor

Prof. Dr. Anila Amber Malik

Convener Postgraduate Admission Committee

(Nominee of Syndicate)

MEMBERS:

Deans of the Faculties

Prof. Dr. Shaista Tabassum	Dean, Faculty of Arts & Social Sciences
Prof. Dr. Harris Shoaib	Dean, Faculty of Pharmacy & Pharmaceutical Sciences
Prof. Dr. Zahid Ali Zahidi	Dean, Faculty of Islamic Studies
Prof. Dr. Shagufta Shahzadi	Dean, Faculty of Education
Prof. Dr. Farida Islam	Dean, Faculty of Medicine
Prof. Dr. Masarrat Jahan Yousuf	Dean, Faculty of Science & Faculty of Engineering
Prof. Dr. Tahir Ali (Look After)	Dean, Faculty of Management and Administrative Sciences

Nominees of the Syndicate

Prof. Dr. Intikhab Ulfat	Department of Physics
Prof. Dr. Asma Manzoor	Director, Center of Excellence for Women's Studies

Nominees of the Academic Council

Prof. Dr. Shah Ali ul Qadar	Department of Biochemistry
Prof. Dr. Uzma Ashiq	Department of Chemistry
Prof. Dr. Taseer Ahmed Khan	Department of Physiology

Professors Emeritus

Prof. Dr. Atta-ur-Rahman, FRS	International Center for Chemical & Biological Sciences (ICCBS)
Prof. Dr. Manzoor Ahmed	Department of Philosophy

Prof. Dr. Abdul Waheed	Registrar
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INTRODUCTION

University of Karachi established in 1951, is one of the oldest institutions of higher education of Pakistan. It offers rigorous and vibrant graduate and post-graduate degree programs in the Faculties i.e. Arts & Social Sciences, Education, Engineering, Islamic Studies, Law, Management & Administrative Sciences, Medicine, Pharmacy & Pharmaceutical Sciences, and Science. The research and the teaching programs are supervised by about 400 qualified faculty members. These postgraduate programs are offered in 56 teaching departments and 20 research institutes and centers.

University of Karachi offers its graduate and post-graduate students well equipped laboratories, digital library and video-conferencing facilities in a multi-disciplinary environment. It is committed to uphold its tradition of producing top-quality graduates in cutting-edge areas of Business and Management Sciences, Bio-Medical & Engineering Sciences, Technologies, Social and Natural Sciences. The programs are designed to match the emerging needs of industries and research and development organizations, both in the public and private sectors.

During the last five years, University of Karachi has awarded more than 2500 MS(Surgery), MD(Medicine), M.Phil, ME and Ph.D degrees. Presently, around 50,000 students are enrolled in MS(Surgery), MD(Medicine), M.Phil LL.M. and Ph.D, programs. Faculty members at the University of Karachi are engaged in pursuing research in various disciplines and publish extensively in journals of international repute.

University of Karachi provides excellent opportunities to its postgraduate students and faculty members for scholarships, linkages, trainings and collaboration with the world's top universities in research projects with the universities/institutions/centers in the technologically advanced countries.

FACULTIES

- 1. Faculty of Arts & Social Sciences**
- 2. Faculty of Education**
- 3. Faculty of Engineering**
- 4. Faculty of Islamic Studies**
- 5. Faculty of Law**
- 6. Faculty of Management & Administrative Sciences**
- 7. Faculty of Pharmacy & Pharmaceutical Sciences**
- 8. Faculty of Science**
- 9. Faculty of Medicine**

Departments / Institutes / Centers

FACULTY OF ARTS & SOCIAL SCIENCES	
1.	Arabic
2.	Criminology
3.	Economics
4.	English
5.	History
6.	International Relations
7.	Islamic History
8.	Library & Information Science
9.	Mass Communication
10.	Persian
11.	Philosophy
12.	Political Science
13.	Psychology
14.	Sindhi
15.	Social Work
16.	Sociology
17.	Urdu

FACULTY OF EDUCATION	
1.	Education
2.	Special Education
3.	Teacher Education

FACULTY OF ENGINEERING	
1.	Chemical Engineering

FACULTY OF ISLAMIC STUDIES	
1.	Islamic Learning
2.	Quran & Sunnah
3.	Usooluddin

FACULTY OF LAW	
1.	School of Law

FACULTY OF MANAGEMENT & ADMINISTRATIVE SCIENCES	
1.	Public Administration
2.	Business Administration
3.	Commerce

FACULTY OF MEDICINE	
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FACULTY OF PHARMACY & PHARMACEUTICAL SCIENCES	
1.	Pharmacognosy
2.	Pharmacology
3.	Pharmaceutics
4.	Pharmaceutical Chemistry & Pharmacology (HEJ)
5.	Pharmacy Practice

FACULTY OF SCIENCE	
1.	Agriculture & Agribusiness Management
2.	Applied Chemistry & Chemical Technology
3.	Applied Physics
4.	Biochemistry
5.	Biotechnology
6.	Botany
7.	Chemistry
8.	Computer Science
9.	Food Science & Technology
10.	Genetics
11.	Geography
12.	Geology
13.	Health, Physical Education & Sports Sciences
14.	Physical Therapy
15.	Mathematics
16.	Microbiology
17.	Physics
18.	Physiology
19.	Statistics
20.	Zoology

INSTITUTES / CENTERS	
1.	Dr. A. Q. Khan Institute of Biotechnology & Genetic Engineering (Biotechnology) (KIBGE)
2.	Dr. Panjwani Center for Molecular Medicine & Drug Research (Molecular Medicine) (PMDC)
3.	HEJ Research Institute of Chemistry (Chemistry)
4.	Dr. M. Ajmal Khan Institute of Sustainable Halophyte Utilization (Botany)
5.	Marine Reference Collection and Resource Center
6.	National Nematological Research Center (Nematology)
7.	Dr. Zafar H. Zaidi Center for Proteomics (Biochemistry)
8.	Institute of Space Science & Technology
9.	Center of Excellence for Women's Studies
10.	Institute of Marine Science
11.	Institute of Environmental Studies
12.	Center for Plant Conservation
13.	Pakistan Study Center

DEPARTMENT OF ARABIC

INTRODUCTION

The Department of Arabic at the University of Karachi is renowned for its rich academic environment, strategically located in front of the magnificent Jamia Masjid Ibrahim. Department prides itself on its dedication to the study and promotion of the Arabic language and literature. The Department of Arabic, University of Karachi, was established in 1955. Since its inception, the Department is enjoying the patronage of renowned scholars of international repute. From the outset, the Department started functioning under the auspicious headship of the renowned scholar Professor Allama Abdul Aziz Memon. Highly qualified faculty members having vast experience enjoying good reputation are taking part in dispersal education.

The cautiously chosen syllabus considering the demands of the student and the existing period, offers excellent study of the ancient times and the contemporary era. It is indeed a confluence of ancient and modern Arabic Literature, Language and Linguistics. Besides academic activities, curricular and co-curricular activities are also stressed like extension lectures, debates, study tours, picnics and other functions to reveal, enhance and polish the abilities and talent of the students.

POSTGRADUATE PROGRAMS AVAILABLE:

1. M.Phil in Arabic
2. Ph.D in Arabic

PROGRAM OBJECTIVES

1. To provide in-depth knowledge & research opportunities in Arabic language & literature.
2. To develop critical and analytical skills in classical and contemporary Arabic texts.
3. To promote the understanding of Islamic literature & its relevance to modern contexts.
4. To prepare students for academic & professional careers in Arabic & related fields.

AIMS AND SCOPE OF THE PROGRAM

1. **M.Phil in Arabic:** Aimed at developing advanced research skills and comprehensive knowledge in various aspects of Arabic studies, including linguistics, literature, and Islamic texts.
2. **Ph.D in Arabic:** Focuses on original research and contribution to the field of Arabic studies, encouraging scholarly excellence and innovation.

SCOPE

The M.Phil and Ph.D programs in Arabic Language and Literature at the University of Karachi are designed to provide advanced academic and research training in the fields of Arabic linguistics, literature, and culture. Graduates from these programs will acquire in-depth knowledge and expertise that open doors to a wide array of professional opportunities.

KEY AREAS OF OPPORTUNITY

Teaching:

1. Graduates can pursue academic careers as professors, lecturers, and researchers in universities and colleges worldwide.
2. There is a growing demand for Arabic language educators in primary and secondary education, both within the country and internationally.

Translation:

1. Expertise in Arabic opens up opportunities in translation services across various sectors, including academic, diplomatic, and commercial fields.
2. Graduates can work as translators and interpreters for government agencies, multinational corporations, and non-governmental organizations.

Electronic Media:

1. With the rise of global media, there is an increasing need for professionals proficient in Arabic for roles in broadcasting, journalism, and content creation.
2. Graduates can contribute to Arabic-language news outlets, online platforms, and social media channels.

Diplomacy and International Relations:

1. Proficiency in Arabic is highly valued in diplomatic services, where graduates can serve as cultural advisors, language experts, and policy analysts.
2. Opportunities exist in international organizations, embassies, and consulates.

Research and Publishing:

1. Graduates can engage in scholarly research, contributing to the academic community through publications, conferences, and seminars.
2. There is potential for authoring books, articles, and research papers in Arabic language and literature.

Cultural and Heritage Organizations:

1. Employment opportunities are available in museums, cultural centers, and heritage sites that require expertise in Arabic language and literature.
2. Graduates can work as cultural consultants, curators, and archivists.

By pursuing an M.Phil or Ph.D in Arabic Language and Literature at the University of Karachi, students will be equipped with the skills and knowledge necessary to excel in these diverse and rewarding fields, thereby significantly enhancing their career prospects and contributions to the global community.

RESEARCH FACILITIES AVAILABLE

Department of Arabic boasts a state-of-the-art library equipped with a vast collection of books on Arabic language and literature, Islamic literature, classical and modern texts, Arabic poetry, Quran and Hadith studies. Additionally, we have a fully equipped computer lab with multimedia projectors, computers, laptops, internet access, printing and scanning facilities to support research and academic activities.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION/EXPERTISE

1. **Dr. Asif Saleem:** Ph.D in Arabic Language & Literature, specializing in Arabic Language & Literature.
2. **Dr. Muhammad Imran Latif:** Ph.D in Arabic Language & Literature, specializing in classical literature, Morphology, Syntax, Semantics and Rhetorics.
3. **Dr. Muhammad Abdul Hameed Nabeel:** Ph.D in Language & Literature, specializing in modern Arabic Literature.
4. **Dr. Muhammad Faezul Ebrar:** Ph.D in Quranic Arabic Literature, Arabic grammar specializing in Arabic and Western Literary Relations..
5. **Dr. Aziz-ur-Rehman Saif:** Ph.D in Arabic Language & Literature, specializing in Arabic Manuscript in Science of Hadith. Academic visitor as Research fellow at University of Edinburgh, UK, specializing in Arabic Manuscripts and Palaeography.

INFRASTRUCTURE

1. **Library:** A comprehensive collection of Arabic and Islamic literature.
2. **Computer Lab:** Equipped with modern technology and facilities.
3. **Classrooms:** Well-furnished with multimedia projectors.
4. **Faculty Offices:** Spacious and well-equipped to facilitate academic work.

AVENUES FOR RESEARCH

Dept. actively encourages research in various areas of Arabic, including but not limited to:

1. Arabic Linguistics
2. Classical and Modern Arabic Literature
3. Comparative Literature
4. Islamic Literature
5. Manuscript Studies
6. Morphology
7. Philology
8. Poetic Devices / Poetic Meter
9. Rhetorics / Rhyme
10. Semantics / Syntax / Syntactics
11. Translation Studies

RESEARCH PUBLICATIONS

1. The department publishes a bi-annual, research journal recognized by HEC in the Y category, titled "Journal of Arabic, University of Karachi."
2. The department also produces a quarterly research journal for students, titled "Al-Nakhla."

ADDITIONAL INFORMATION

The Department of Arabic provides a vibrant academic atmosphere that fosters intellectual growth and scholarly excellence. We regularly organize seminars, workshops, and conferences to enhance the academic experience of our students and faculty. Faculty members are engaged in cutting-edge research and have published extensively in reputable National and International Research Journals.

DEPARTMENT OF CRIMINOLOGY

INTRODUCTION

Department of Criminology was established in 2014. Presently Prof. Dr. Naima Saeed is the chairperson of the department. She is internationally recognized professor of criminology and department of criminology is one of the Significant Departments in the Faculty of Social Science University of Karachi. Criminology is directly linked to the protection of human life, and their security issues. This is an important branch of sociology.

POSTGRADUATE PROGRAMS AVAILABLE AT CRIMINOLOGY

1. M.Phil
2. MS
3. Ph.D

OBJECTIVES

The main objectives of M.Phil./ MS/ Ph.D. Criminology are aligned with the objectives of the university and the program mission statements which focuses, research methodology, interactive teaching, class presentation and article writing .

These are as follows:

1. To Acquired advanced knowledge in criminological theory, research methods, and criminal justice systems.
2. To conduct original research that contributes to the field of criminology.
3. To develop expertise in theoretical and methodological approaches to criminological research.
4. To prepare for a career in academia as a professor or researcher.
5. To prepare for careers in research institutions, think tanks, or policy organizations.
6. To prepare for leadership roles in criminal justice agencies or organizations.
7. To develop advanced research and writing skills.
8. To enhance critical thinking and analytical skills.
9. To develop expertise in specialized areas of criminology, such as crime prevention, corrections, or juvenile justice.
10. To develop the ability to conduct independent research.
11. To enhance critical thinking and problem-solving skills.
12. To build professional networks and relationships in the field of criminology.

AIMS AND SCOPE OF THE PROGRAM

Aims

1. Academic Leadership: To position the Department of Criminology at the University of Karachi as a leading institution in education and research in the field of criminology nationally and internationally as well .
2. Interdisciplinary Approaches: To foster interdisciplinary approaches to understanding crime, criminal behaviour and crime patterns and crime prevention strategies and techniques incorporating insights from sociology, law, and other relevant fields including forensic science.
3. Advancement of knowledge: To advance the body of knowledge in criminology through cutting-edge research and scholarly contributions by faculty and students.
4. Enhancement of Research Skills: To develop and enhance research skills and ability to design, conduct, and analyze research in criminology.
5. Specialization: To offer specialized knowledge in specific areas of criminology, such as crime prevention, corrections, or forensic criminology.
6. Policy Influence: To influence and shape public policy related to criminal justice and crime prevention through evidence-based research and active engagement with policymakers.
7. Holistic Development: To nurture well-rounded graduates who are not only knowledgeable in criminological theories and practices but also possess the ethical and moral grounding necessary to address the complexities of crime and justice.
8. Community Impact: To have a positive impact on the community by developing strategies and solutions to prevent and combat crime, thereby enhancing public safety and justice.

SCOPE

The scope of postgraduate programs in criminology, continues to evolve, incorporating new areas of study and interdisciplinary approaches to understand the complex issues surrounding crime criminal behaviour, crime prevention techniques and most importantly receiving end of crime Victim's related issues

Some key areas within the scope of criminology include:

1. Crime and Deviance: Understanding the nature, causes, and consequences of crime and deviant behavior.
2. Social and Environmental Factors: Examining the social, economic, political, and environmental factors that contribute to crime.
3. Globalization and Transnational Crime: Studying the global dimensions of crime and criminal justice.
4. Cybercrime and Digital Criminology: Examining the intersection of crime and technology.
5. Forensic Criminology and Investigation: Applying scientific principles and techniques to aid in criminal investigations.
6. Criminal Psychology and Psychiatry: Understanding the psychological and psychiatric aspects of criminal behavior.
7. Human Rights and Social Justice: Analyzing the relationship between crime, justice, and human rights.
8. Criminal Justice Systems: Studying the structure, function, and effectiveness of law enforcement, courts, corrections, and rehabilitation.
9. Criminal Law and Policy: Analyzing the development, implementation, and impact of criminal laws and policies.
10. Victimology: Studying the impact of crime on victims and the services available to support them.
11. Crime Prevention and Control: Exploring strategies and approaches to prevent and reduce crime.
12. Research Methods and Statistics: Developing and applying research methods and statistical techniques to study crime and criminal justice.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION / EXPERTISE

The faculty comprises two full professors, whose expertise and dedication are instrumental in guiding students through their academic journey. Additionally, the department benefits from the wisdom and experience of seasoned retired professors who continue to impart knowledge and mentor the next generation of criminologists. The department also provides teaching opportunities to its postgraduate students

Following are the Departmental Research Committee members with their area of expertise

1. Prof. Dr. Naima Saeed
Gender sexuality and crimes, Youth violence and crime, investigation, cybercrime, and police administrations
2. Prof. Dr. Ghulam Muhammad Burfat
Drug related crime, child abuse, policing, mental health issues, and theoretical criminology
3. Prof. Dr. Shaista Tabassum, Dean Faculty of Arts and Social Sciences.

RESEARCH FACILITIES AVAILABLE

The department has a well-stocked library; the department provides an enriching environment for scholarly pursuit. Each year, students engage in rigorous research projects as part of their M.Phil and Ph.D programs, contributing valuable insights to the field of criminology. The students can also get membership of LEJ digital library.

INFRASTRUCTURE

The department has adequate arrangement of class rooms for post graduate students who support conclusive learning environment.

AVENUES OF RESEARCH

Students graduating from the criminology department have ample opportunities of getting employment in various law enforcement agencies, banks, security institutions and educational sectors

DEPARTMENT OF ECONOMICS

INTRODUCTION OF THE DEPARTMENT

The Department of Economics, established in 1950-51, stands as one of the oldest and largest departments within the Faculty of Arts & Social Sciences, holding a prominent position in Pakistan's academic landscape. The department's pedagogical approach embodies a holistic integration of theory and practice, blending lectures, case studies, oral presentations, written reports, theses, and field visits. Faculty members are dedicated to bridging the gap between theoretical concepts and real-world applications, often inviting distinguished guest speakers from different institutions to enrich the learning environment. Continuous assessment methods, including quizzes, examinations, and presentations, ensure students thoroughly grasp and apply the knowledge imparted to them.

The alumni of the Department of Economics have embarked on flourishing career paths across various sectors including banking, civil services, research, teaching, and business, both domestically and internationally. This widespread success underscores the department's key role in encouraging adaptable and proficient professionals who excel in a range of endeavors.

POSTGRADUATE PROGRAMS AVAILABLE

1. M.Phil
2. Ph.D

PROGRAMS' OBJECTIVE

Programs' Objective: M.Phil. (Economics):

The objectives of the M.Phil. (Economics) include:

1. **Advanced Knowledge:** To provide students with a deep and advanced understanding of economic theories, concepts, and methodologies, enabling them to analyze complex economic issues critically.
2. **Research Skills:** To develop research skills and methodologies necessary for conducting original research in economics, including data analysis, econometrics, and quantitative research techniques.
3. **Critical Thinking:** To foster critical thinking and problem-solving abilities in the context of economics, allowing students to evaluate economic policies, trends, and real-world problems rigorously.
4. **Specialization:** To allow students to specialize in specific areas of economics, such as macroeconomics, microeconomics, development economics, international economics, or financial economics, according to their interests and career goals.

Programs' Objective: Ph.D. (Economics):

The objectives of the Ph.D. (Economics) include:

1. **Research Excellence:** Foster a culture of research excellence by providing students with the necessary tools, resources, and mentorship to conduct innovative and impactful research in various fields of economics.
2. **Interdisciplinary Collaboration:** Encourage interdisciplinary collaboration by promoting engagement with other academic disciplines and research centers, enabling students to tackle complex economic issues from diverse perspectives.
3. **Policy-Relevant Expertise:** Equip students with the analytical skills and theoretical knowledge to analyze and contribute to relevant economic policy debates, ensuring that graduates can make meaningful contributions to public and private sector decision-making.
4. **Professional Development:** Support students in their professional development by offering opportunities for teaching experience, conference participation, and networking, preparing them for successful careers in academia, government, industry, and international organizations.

AIMS AND SCOPE OF THE PROGRAM***Aims and Scope of the Program: M.Phil. (Economics)***

The M.Phil. (Economics) aims to cultivate a deep understanding of economic principles and enhance students' analytical and research capabilities. The program covers a broad scope, including advanced economic theories, quantitative methods, and applied economics. It prepares students for independent research, policy analysis, and further academic pursuits, providing a strong foundation for those considering a Ph.D. or careers in economic research and consultancy.

Aims and Scope of the Program: Ph.D. (Economics)

The Ph.D. (Economics) is designed to develop leading economists who can contribute original research to the field. The program aims to produce scholars with expertise in economic theory, econometrics, and applied economics, capable of addressing complex economic issues. The scope encompasses rigorous training in research methodologies, academic writing, and teaching, preparing graduates for influential roles in academia, policy-making, and international organizations.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION / EXPERTISE

The Department of Economics boasts a diverse and highly qualified faculty, each contributing extensive knowledge and expertise to their respective fields. The faculty members possess advanced degrees from prestigious institutions and have significant research accomplishments, including numerous publications in national and international journals, presentations at esteemed conferences, and contributions to academic literature through books and monographs. They specialize in a wide array of areas, including quantitative development analyses, international economics, development issues, macroeconomics, financial market analysis, environmental economics, and more. This depth of expertise ensures a rich learning environment for M.Phil. and Ph.D. students, fostering rigorous academic inquiry and robust research outputs. Below is a compilation of faculty members' names along with their qualifications, designations, and areas of specialization or expertise.

S. No.	Name of Faculty Member	Qualifications	Designation	Areas of Specialization / Expertise
1	Prof. Dr. Abdul Waheed	Postdoctoral Research, Nagoya University, Japan; Ph.D., Nagoya University, Japan	Professor	Development Economics, International Economics, Economic Modelling
2	Prof. Dr. Roohi Ahmed	Ph.D. (Economics), University of Karachi	Professor	International Economics, Development Economics, Econometrics
3	Dr. Nooreen Mujahid (FSL)	Ph.D. (Economics), University of Karachi	Associate Professor	Human Resource Management, Development Economics, Health Economics
4	Dr. Rumana Zaheer	Ph.D. (Economics), University of Karachi	Associate Professor	International Economics, Monetary Economics, Macroeconomics
5	Dr. Syed Tehseen Jawaid	Ph.D. (Economics), University of Karachi	Associate Professor	International Economics, Development Economics, Econometrics
6	Dr. M. Ahsanuddin	Ph.D. (Economics), University of Karachi	Assistant Professor	International Economics, Macroeconomics, Econometrics, Mathematical Economics
7	Dr. Safia Minhaj	Ph.D. (Economics), University of Karachi	Assistant Professor	Microeconomics, Macroeconomics, Transport Economics, Tourism Economics, Environmental Economics, International Economics,

8	Mr. Mirza Muhammad Shehzad	Enrolled in Ph.D. (Economics), University of Karachi; M.Phil. (Economics), Quaid-i-Azam University	Assistant Professor	International Economics, Stabilization, Macroeconomics, Monetary Economics, Poverty and Inequality
9	Mr. Muhammad Haseeb	Enrolled in Ph.D. (Applied Economics), AERC, University of Karachi	Assistant Professor	Macroeconomics, Mathematical Economics
10	Dr. Zeeshan Ateeq	Ph.D. (Economics), The University of Manchester, UK	Assistant Professor	Financial Market Analysis, Macroeconomics, International Economics
11	Dr. Muhammad Umair	Ph.D. (Economics), University of Karachi	Assistant Professor (T)	Environmental Economics, Labor Economics, International Economics, Development Economics, Data Analytics
12	Ms. Rabia Shakir	Enrolled in Ph.D. (Economics), University of Karachi; M.Phil. (Economics), University of Karachi	Lecturer	Money and Banking, Migration Economics, Statistics and Project Management

RESEARCH FACILITIES AVAILABLE (LIBRARY, LABORATORY, COMPUTER LAB)

The Department of Economics is an ideal place for study and research, offering a range of excellent facilities to support academic and research activities. The department has its own library, which boasts a comprehensive collection of around 2000 books and periodicals related to economics and related fields. For materials not found within the library, students and faculty can utilize inter-library loan facilities and access other major libraries, including the main library and the research library of AERC at the University of Karachi. The department's library is well-equipped to provide course materials as well as reference literature essential for students throughout their studies. Additionally, the department has recently inaugurated a dedicated research lab for M.Phil. and Ph.D. students, featuring 20 computers to facilitate advanced research and data analysis. These resources create a supportive environment that enhances learning and research capabilities for all students.

INFRASTRUCTURE

The Department of Economics provides a good infrastructure tailored to the needs of M.Phil. and Ph.D. students, ensuring a conducive environment for advanced learning and research. Renovated classrooms equipped with projectors facilitate interactive and dynamic teaching methods, enhancing the educational experience for students. Additionally, the department offers dedicated support staff to assist with various activities, including administrative tasks and academic support. This comprehensive infrastructure aims to foster a stimulating academic environment where students can engage deeply with their studies, collaborate with peers and faculty, and pursue innovative research endeavors.

AVENUES FOR RESEARCH

In the M.Phil. and Ph.D. programs, students are provided with a wide range of research opportunities guided by the distinguished faculty members. The faculty members, who have obtained qualifications from prestigious institutions worldwide such as Nagoya University, Japan, and The University of Manchester, UK, bring extensive expertise and experience to support students in their research endeavors. Specialized areas of focus include quantitative development analyses, international economics, financial market analysis, and environmental economics, among others. Through collaborative mentorship, students are encouraged to explore innovative research pathways, thereby contributing to the advancement of knowledge in economics and related fields. With access to state-of-the-art resources such as dedicated research labs and partnerships with industry leaders, the aim is to uncover fresh perspectives and solutions to complex global challenges. By embracing new research avenues, we endeavor not only to enrich academic dialogue but also to generate meaningful impacts on society and the economy.

DEPARTMENT OF ENGLISH

INTRODUCTION

The Department of English was established in 1955. It was located in Shivjee Muraji building in Nanakwana. Later in January 1960 when the university moved to its present location, the Department was housed on the first floor of the Old Block in the Faculty of Arts, now called the Faculty Arts and Social Sciences. Professor Dr. Ahsan Farooqui was the first head of the Department. The Department pursues excellence in English studies by creating and maintaining a culture of teaching and research based on values of creativity, academic honesty, and collaboration among its faculty and students. It offers a wide range of courses in Literary Studies and Applied Linguistics with a curriculum aligned with international standards. The courses have been carefully designed to facilitate the students in firmly grounding themselves in the discipline both theoretically and practically. From a rigorous exposure to the English literary canon to the pragmatic necessities of language planning and policy making, the Department seeks to equip its scholars with a simultaneous awareness of English language in its aesthetic, cultural, scientific and pragmatic dimensions.

POSTGRADUATE PROGRAMMES AVAILABLE

1. M. Phil. English (Literature)
2. M. Phil. English (ELT & Applied Linguistics)
3. Ph. D. English (Literature)

REGULAR FACULTY MEMBERS IN THE POSTGRADUATE PROGRAMME

1. **Dr. Iftikhar Shafi**, specializes in Literary Theory, Romantic Aesthetics, Comparative Literary Aesthetics
2. **Dr. Shumaila Shafket Ali**, specializes in Sociolinguistics, Cross-cultural Linguistics, Language Assessment
3. **Dr. Faisal Nazir**, specializes in Literary Theory, Postcolonial Literature and Theory, Pakistani Anglo-phone Literature
4. **Dr. Bushra Ahmed Khurram**, specializes in ELT and Qualitative Research
5. **Dr. Muhammad Moonis Azad**, specializes in Psychoanalytic, Feminist and Postmodern Literary Theory, Modern Drama
6. **Mr. Wajdan Raza**, specializes in Socio-phonology, World Englishes, Language Assessment
7. **Ms. Lubna Hasan**, specializes in Romantic Poetry and Literary Theory
8. **Mr. Babur Khan Suri**, specializes in Postmodern Fiction and Literary Theory

FACILITIES

1. Department Seminar Library
2. Dr. Kaleem-ur-Rahman Library
3. Conference Room with Audio-Visual facilities
4. Computer Room with Research Resources

DEPARTMENT OF HISTORY

INTRODUCTION TO THE DEPARTMENT

Department of History, established in 1953, is one of the oldest departments in Karachi University which is proud of having world renowned scholars such as Dr. Mahmood Hussain, Dr. Riaz ul Islam, Dr. I.H. Qureshi, Dr. M. H. Siddiqui and many more as its faculty members in the past.

POST GRADUATE PROGRAMS OFFERED:

The Department offers MS/ M.Phil (20 seats) and Ph.D (10 seats) degrees in its Post-Graduate Program.

Specific Research Areas include Historiography, Philosophy of History, Ancient and Medieval History, Modern/Contemporary History, Archeology, History of Religions, History of Sindh, Regional and Urban History (focused on Karachi), Socio-Economic and Cultural History, Historical evolution of civil society and social movements, History of Conflict and Conflict Management, Public Policy, Demographic Transitions etc. All courses offered at the Department emphasize upon the latest relevant theoretical and empirical developments.

OBJECTIVES OF THE PROGRAM

The Post-Graduate Program of the University of Karachi is aimed at providing the facilities to all sections of graduate students to pursue higher studies and quality research in various fields of knowledge. The Department of History staunchly follows this aim with its own specific objectives as follows:

1. To promote quality research in all fields of History and its auxiliary sciences such as Archaeology, numismatics, epigraphy, archival studies etc.
2. To foster learning in modern trends, discourse and techniques in ancient, medieval and contemporary history and historiography.
3. To develop research skills in regional, class and socio-cultural, political and economic history and policy studies.
4. To train students in ciphering primary and secondary sources and writing unbiased and intellectually sound texts.
5. To encourage research and intellectual interaction between various universities and institutes pursuing historical research across the world.

AIMS AND SCOPE OF THE PROGRAM

The Program aims at making the students as well as the policy makers and stake-holders in the country understand the significance of historical research and history itself not only in widening the knowledge of human past and in turn, the socio-political and economic issues of the present, but also to provide necessary data to formulate public policy for the future.

History provides multi-dimensional answer to the question: "Why we are what we are and how we can improve?" In the twentieth century, historical studies across the world have been an essential part of various research and policy projects including those exploring and investigating geo-political and geo-economic issues, ethnic conflicts, terrorism, democratic development, human rights violations and civil society dynamics only to name a few. In this context, the students with post graduate degrees in History are likely to develop strong positive attitudes and skills to get employed in teaching and research jobs in higher education institutions, think tanks, civil society organizations and media houses both at national and international levels.

AVAILABLE FACULTY

Departmental Research Committee:

1. Professor Dr. S.M. Taha – Specialization in Modern History, Public Policy, History of USA
2. Dr. Hina Khan – Specialization in Modern History of Europe, South and Central Asia, Post Cold-War World Politics and Civil Society, Democracy and human rights
3. Dr. Kiran S. Siddiqui – Specialization in Archaeology, Pre-Islamic and Islamic Art and Architecture, Ancient South Asia and Tourism
4. Dr. Moiz Khan – Specialization in Political History of Pakistan and its roots from Colonial Era; Demographic changes and Urbanization with special reference to Karachi

5. Dr. Farah Deeba Khan – Specialization in Socio-economic History of Handicraft Industry in Sindh; Modern Islamic trends in South Asia, Middle East and Central Asia
6. Dr. Humera Naz – Specialization in Medieval History of South Asia and Sindh; Gender History of South Asia and Historiography
7. Dr. Abdul Jabbar Khan – Specialization in Anthropology, Archaeology and History of Pakistan

Adjunct Professors

1. Professor Dr. Javed Hussain – Specialization in Archaeology and Anthropology
2. Professor Dr. Nasreen Afzal – Specialization in Historiography, Constitutional History, Colonial History and History of Sindh

RESEARCH FACILITIES AVAILABLE

1. **Dr. I.H. Qureshi Memorial Library** with about 10,000 books, journals, encyclopedia, the newspaper Dawn (regular) and a computer lab with internet facility and soft copies of a large number of valuable books.
2. **Dr. Riaz ul Islam Audiovisual Center** with a capacity of seventy people and internet and multimedia facilities for seminars, extension lectures and workshops etc.
3. **Teaching Museum** with a collection of genuine and replica artifacts to facilitate the students of Ancient and Medieval History and Archaeology

INFRASTRUCTURE

The Department has two lecture halls, a teaching museum and an audio-visual room for classes. Some of rooms are equipped with air-conditions and multi-media facilities.

AVENUES FOR RESEARCH

Completely dispelling the misconception that history is only a study of past kings and leaders, the Department offers twenty-first century research avenues focused on contemporary discourse on modernity and post-modernism; oriental and post-colonial interpretations of history; subaltern and class history; feminist and gender history; post-cold war concepts of the 'end of history' and clash of civilizations; history of public policy and civil society initiatives; religions and religious philosophies; constitutionalism and constitutional history, socio-economic and demographic history; environmental history; democracy and authoritarianism; state-society dynamics; conflict and conflict resolution; physical and cultural anthropology; regional and urban history; archaeological and historiographical studies focusing various time periods and region across the world.

DEPARTMENT OF INTERNATIONAL RELATIONS

INTRODUCTION OF THE DEPARTMENT

The Department of International Relations, University of Karachi is the oldest Department in the field of International Relations in Pakistan. It was established in 1958 and has the honor of offering Bachelor of Studies (BS), Master of Studies (MS), M.Phil and Ph.D. programs. Prior to its formation, the Department was under General History Department and Professor Dr. Mahmud Hussain was its founding Chairman. Prof. Dr. Muhammed Ahsen Chaudhry succeeded Prof. Hussain as the Chairman of the Department. He was followed by Prof. Dr. Ahmed Abdul Kadeer, Prof. Dr. Mujtaba Razvi, Prof. Shameem Akhtar, Prof. Dr. Talat A. Wizarat, Prof. Syed Sikander Mehdi, Prof. Dr. Khalida Ghous, Prof. Dr. Moonis Ahmar, Prof. Dr. Shaista Tabassum and Prof. Dr. Mutahir Ahmed. Presently, Dr. Naeem Ahmed is the Chairperson of the Department. The Department has a faculty of six full-time and seven cooperative teachers. Around 2000 students are studying at the undergraduate, graduate and post-graduate levels, both in the Morning and Evening programs.

Since its inception, the Department of International Relations has been active in organizing seminars, conferences and workshops. A number of books have also been published under the Department and several Research Projects funded by national and international think-tanks, policy institutes, and foundations have been completed. The syllabus of the Department has been revised in 2022 and new courses focusing on the conceptual and regional studies have been introduced at the undergraduate, graduate and post-graduate levels.

POST-GRADUATE PROGRAMS AVAILABLE

1. M.Phil
2. Ph.D.

PROGRAM'S OBJECTIVES

We are imparting the knowledge of I.R. with our core objectives:

1. To provide extensive theoretical knowledge and conducting in-depth quality research, become trend-setter and top-tier school of international relations in the region;
2. To establish a link between the world of academia, practitioners, and policy makers by providing an input on key local, regional, and global foreign policy, security, environment, sustainable development, conflict, and peace issues;
3. To provide training and nurture critical thinking among students in order to train them for future's challenging leadership.

REGULAR FACULTY MEMBERS WITH AREA OF SPECIALIZATION

1. Prof. Dr. Shaista Tabassum (Dean, Faculty of Arts & Social Sciences)
Area of Specialization: International Law, Nuclear Proliferation, Water Issues in South Asia, Human Rights
2. Dr. Naeem Ahmed, Chairman and Associate Professor
Area of Specialization: Domestic and regional security of South Asia with special reference to religious extremism, countering violent religious extremism and terrorism; Pakistan-US relations; Nuclear issue; Kashmir issue.
3. Dr. Nausheen Wasi, Assistant Professor
Area of Specialization: Theoretical aspects of international relations, issues of regional cooperation and integration with particular reference to South Asia, and Pakistan politics and foreign policy.
4. Mr. Faheem Uddin, Assistant Professor
Area of Specialization: Science and Technology; and Theories of International Relations
5. Mr. Muhammad Faisal Awan, Assistant Professor
Area of Specialization: Political Islam, Human Rights Discourse and Development Discourse.
6. Mr. Amjad Ali, Lecturer
Area of Specialization: Non-traditional security threats, climate change, and theories of IR.

RESEARCH FACILITIES

The Department has a Seminar Library which contains more than 5000 books on different topics related to international relations.

INFRASTRUCTURE

The Department has a Conference Room with sitting capacity of 75 persons. The CR is equipped with the latest multimedia system.

AVENUES FOR RESEARCH

There are several opportunities for the graduate students of International Relations. These include: think-tanks, research institutes, international organizations, government service, banking sector, NGOs, Media etc.

DEPARTMENT OF ISLAMIC HISTORY

INTRODUCTION

The Department of Islamic History is one of the oldest departments in Karachi University. Established in 1953, it has a distinguished tradition of fostering advanced studies and original research in the field of Islamic History. Professor Dr. Amir Hassan Siddiqui, the department's founder and first Head, laid the groundwork for this enduring tradition.

The Department of Islamic History offers a curriculum that explores the religious, political, cultural, and educational journey of Islam from its beginnings to the present day through rigorous academic and specialized review. Specially designed courses unravel Islamic History across distinct regions, including the Middle East, Central Asia, North Africa, Spain, Sindh, and Hind (the Indian Subcontinent), and other parts of the world. These courses span pivotal periods such as the Prophetic Era, the Era of the Rightly Guided Caliphs, the Umayyad, Abbasid, and the Ottoman Caliphates. In addition, the department offers specific courses on Muslim jurisprudence, Sufism, Muslim political thought, Art and architecture, as well as Modern trends and Muslim revivalist movements.

POSTGRADUATE PROGRAMS AVAILABLE AT ISLAMIC HISTORY

1. M.Phil
2. MS
3. Ph.D

PROGRAM OBJECTIVE

1. Our M.Phil/MS and Ph.D. program equips you with advanced training in historical research methods, analysis, and interpretation.
2. Through focused coursework and independent research, students will gain in-depth knowledge of a specific historical period or topic within Islamic History.
3. Students will also culminate their studies with a substantial thesis, making a unique contribution to the field of Islamic History.

AIMS AND SCOPE OF THE PROGRAM

1. To cultivate the next generation of distinguished researchers in Islamic History, equipped to not only illuminate the past but also illuminate its ongoing influence on the present world.
2. To equip students with advanced historical research skills, enabling them to critically analyze and interpret the past, and leverage those insights to address contemporary challenges and inform future decision-making.
3. To promote the development of original research that contributes fresh insights and advances the understanding of Islamic History, with the goal of enriching public discourse and promoting intercultural understanding.
4. Students of Islamic History have a wide range of opportunities in various social science fields. They have a very bright future in the profession of teaching, research and media.

RESEARCH FACILITIES AVAILABLE

The Seminar Library possesses a collection of about 3000 books. It provides reading and research facilities to the students of Islamic History. By utilizing top research journals and Islamic software alongside departmental computers, students can reap significant benefits in education and research.

FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION / EXPERTISE

Associate Professors

1. Dr. Muhammad Zubair (Chairman)
Middle East, Historiography
2. Dr. Muhammad Sohail Shafiq
Muslim Political Thought, Modern Trends in South Asia

Assistant Professors

1. Dr. Farzana Jabeen
Muslim Jurisprudence, Ottoman Caliphate, Muslim Revivalist Movements
2. Dr. Farah Naz Sohail
Umayyad Caliphate, Culture and Civilization
3. Ms. Suraiya Tariq
Middle East, Modern Muslim world
4. Dr. Uzma Perveen
Seerah Studies, Rightly Guided Caliphate, Indian subcontinent
5. Dr. Humaira Naz
Ottoman Caliphate, Muslim Art & architecture
6. Dr. Muhammad Faisal Hasan
Abbasid Caliphate, Muslim Political Thought

DEPARTMENT OF LIBRARY & INFORMATION SCIENCE

INTRODUCTION

The Department of Library and Information Science at the University of Karachi has played a leading role in establishing library education in Pakistan. It was the first ever to offer a professional degree in Pakistan. It was established in 1956.

POSTGRADUATE PROGRAMS AVAILABLE

1. MS
2. M.Phil
3. Ph.D

PROGRAMS OBJECTIVES

To create leaders and innovators who possess the courage, confidence, and conviction to make appropriate decisions in all kinds of circumstances using the knowledge base and skills for systematic analysis.

AIMS AND SCOPE OF THE PROGRAM

M.Phil

1. To provide research experiences for students, and enable them to become adept in developing and applying research techniques and in the interpretation of information derived from research and creative activities.

Ph.D

1. To develop substantive knowledge and research skills.
2. To be able to design and conduct original research in the area of specialization.
3. To contribute to academic discourse and prepare the researchers for further leadership roles.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION / EXPERTISE

1. Dr. Rafat Parveen Siddiqui
Associate Prof. (Chairperson)
School & Children librarianship, Publishing, Reference sources & services, international librarianship are her main areas of interest.
2. Dr. Naweed e Sahar
Assistant Prof.
Public Library, Bibliography, Indexing Management of Libraries
3. Dr. Muhammad Waseem Zia
Assistant Prof.
Expertise in digitization, automation of libraries and technological advancement in the field of library & Information Sciences.

Besides the above DRC members, two lectures with M.Phil degree are also part of the department's faculty.

RESEARCH FACILITIES AVAILABLE FOR RESEARCHERS (LIBRARY LABORATORY COMPUTER LAB)

The Department of Library and Information Science has a well-stocked seminar library, a practical lab for cataloging and classification, and a well-equipped computer lab with 34 PCs with the facilities of networking and the Internet. The department has a digital library with more than 2000 digital books for researchers. Multimedia, slide projectors, sound systems, and other equipment are available for research activities.

INFRASTRUCTURE

At present Department is located in the third block of the Faculty of Arts & Social Sciences. The department has two classrooms and two labs at ground level, along with a seminar library, Computer Lab, The Digital Library and three rooms for teachers are located on the first floor.

AVENUE FOR RESEARCH

The Department of LIS seeks to train its research postgraduates to develop sound research skills, to acquire the latest theoretical knowledge and become critical thinkers in the fields of study, and to conduct Research in their areas of specialization in a professional, expert and ethical manner.

The Master of Philosophy (M.Phil) and the Doctor of Philosophy (Ph.D) degrees involve the completion of coursework in addition to the preparation of research theses.

M.Phil graduates will have attained advanced knowledge in their chosen fields of study, as well as training in research techniques for dealing with complex issues. They will also be prepared to pursue research degrees at the doctoral level.

Ph.D degree holders will leave the University with up-to-date expertise in their specialist areas, and superior understanding of research techniques for resolving scholarly and practical problems. They will also be prepared to embark on academic or professional careers, and to teach university-level courses in their areas of specialization.

ANY OTHER RELEVANT INFORMATION

1. The department was the first ever to offer Professional degree in Pakistan
2. The department has established the first computer lab of cataloguing in Pakistan; this was also the first computer lab of the Faculty of Arts, University of Karachi
3. The former Chairperson of the department Professor Dr. Anis Khurshid was honored with the **President's Award for Pride of Performance**

DEPARTMENT OF MASS COMMUNICATION

ABOUT THE DEPARTMENT OF MASS COMMUNICATION

The Department of Mass Communication at the University of Karachi, established in 1955, has a rich history of excellence and a commitment to nurturing talented individuals with a passion for communication and media. With a diverse and accomplished faculty, we offer comprehensive programs that cover various aspects of Mass Communication, including Journalism, Development Support Communication, Public Relations, Advertising, Digital Media Marketing, and Film Production.

Our department prides itself on promoting critical thinking, creativity, and ethical practices in the media industry. We believe that effective communication is the cornerstone of societal progress, and we are dedicated to producing graduates who can effectively shape public discourse and positively impact the world.

POST GRADUATE PROGRAMS AVAILABLE

1. M.S. / M.Phil
2. Ph.D.

AIMS AND SCOPE

The Department of Mass Communication offers M.Phil. and Ph.D. programs in Mass Communication that explore social scientific and humanistic approaches to human communication research. The programs aim to create knowledge in both the broad area of human communication and specialty research areas.

The M.Phil. program emphasizes comprehensive training in research methodologies, ethical considerations, and practical skills for effective media practice, with courses such as Development Support Communication, Advertising and Marketing, Political Communication, Content Analysis Research, and Approaches to Mass Communication Theories and Effects. The Ph.D. program advances scholarly research and theoretical contributions, preparing students for leadership roles in academia and the media industry, with an in-depth exploration of topics like Information Communication Technology, International Communication and Globalization, Media Debates, and interdisciplinary integration. These programs collectively ensure a deep understanding of global and cultural perspectives, multidisciplinary integration, and ethical scholarship in mass communication.

The programs also offer a broad scope of market, social, and employment perspectives, including delivering quality education through coursework, research, and experiential learning for scholars aiming for careers in Media, Communications Management, Journalism, and Research. Upon completing a Degree, Scholars will find numerous opportunities in various fields, including TV channels, research and educational institutions, FM radio, media marketing companies, websites, film industries, news and magazine publishers, colleges, universities, digital media industries, telecommunication industries, NGOs, and public sector organizations.

OBJECTIVES

1. Understand and critique the major paradigms and theories that have shaped the communication field, considering their historical evolution and contemporary relevance.
2. Communicate ethically and responsibly across various social contexts, recognizing and respecting cultural differences and ensuring fairness and transparency.
3. Analyze and evaluate existing research, and design and conduct communication studies using quantitative, interpretive, critical, rhetorical, and performative methodologies.
4. Assess and critique the research methods used in published communication studies.
5. Design, facilitate, and assess applied communication activities in professional settings, grounded in conceptual frameworks and research methods.
6. Collaborate with research teams to develop and implement research projects.
7. Produce innovative research that contributes new insights to the communication field.

REGULAR FACULTY MEMBERS WITH THEIR SPECIALIZATION

S#	Faculty Member	Designation	Area of Specialization
1.	Prof. Dr. Asmat Ara	Chairperson & Professor	Research Methodology, Research Theories, Research Communication, Historical Research & Development Support Communication.
2.	Prof. Dr. Fouzia Naz	Professor	Advertising, Public Relations, Development Support Communication, Print and Electronic Media.
3.	Dr. Sadia Mahmood	Associate Professor	Research Communication, Research Methodology, Gender & Media, Media Discourse.
4.	Dr. Raana Afzal	Associate Professor	Political Communication, Public Relations, Advertising, Media Research, Media Theories, Journalism, Gender Studies, Digital Media Marketing.
5.	Dr. Naveed Iqbal	Assistant Professor	Research & Electronic Media.
6.	Dr. Syed Azfar Ali	Assistant Professor	Electronic Media, Advertising, Pakistan Affairs.

RESEARCH FACILITIES

The department has a computer lab furnished with hi-tech computers and multimedia. It provides students access to the most current technological resources essential for their research and academic endeavors. Our students are trained to use and have access to the upgraded labs for collecting various types of data, including continuous and asynchronous audience response, content analysis, experiments, and focused focus groups. The department publishes a bi-annual bilingual research journal recognized by HEC in the Y-category, titled 'Journal of Mass Communication,' which promotes research and theoretical implications within the field of Mass Communication.

INFRASTRUCTURE

The Department of Mass Communication has TV and Radio Studios, which are fully equipped and recently upgraded to adhere to the highest standards and technological devices. The department also features three well-maintained Audio-Visual, News and Advertising labs, and a seminar hall, all equipped with multimedia amenities to offer substantial technical support during classes. Additionally, a portable multimedia setup is available to facilitate interactive learning sessions and classes. The department's seminar library is comprehensively stocked with up-to-date books, research journals, magazines, newspapers, and periodicals.

RESEARCH AVENUES

1. Journalism
2. Advertising
3. Public Relations
4. Development Support Communication
5. Political Communication
6. Gender Studies
7. Health Communication
8. Communication History
9. Information Communication Systems
10. Visual Communication Studies
11. Philosophy, Theory, critique
12. Communication, Law, and Policy

DEPARTMENT OF PERSIAN

INTRODUCTION

The inception of the Department of Persian dates to 1955, driven by a vision to facilitate advanced studies and research in Persian language and literature, particularly focusing on works originating from the Indo-Pak subcontinent. Central to our mission is acquainting the younger generation with Iran's history, language, and literary contributions. Persian has historically served as the secondary language of the Islamic World, renowned in Europe for its unparalleled literary tradition, boasting some of the most esteemed epic, narrative, and lyrical poems in global literature. Moreover, it served as the lingua franca among the cultural and political elite of the early modern Indian Subcontinent.

Beyond its historical significance, Persian holds contemporary relevance as a vital modern language. Evolving since the mid-nineteenth century, Modern Persian stands as a dynamic, adaptable language, serving as the national language of Iran and spoken across neighboring nations such as Afghanistan (where it is known as Dari) and Tajikistan (where it is known as Tajiki), in addition to Bahrain, Uzbekistan, and Turkmenistan. Pakistan shares a profound connection with Persian, boasting a rich heritage deeply intertwined with the language. Thus, fostering an understanding of Persian language and literature is imperative to preserving our cultural and literary legacy.

POSTGRADUATE PROGRAMS AVAILABLE

1. M.Phil.
2. Ph.D.

MISSION OF THE DEPARTMENTS

Our department is dedicated to facilitating advanced studies and research in Persian Language and Literature worldwide, with a strong focus on works from Iran, Afghanistan, Tajikistan, and the Subcontinent (Pakistan and India).

MISSION STATEMENT OF THE PROGRAMS

M.Phil. Program

Our mission is to equip students with a robust understanding of Research Methodology, Literary Criticism, and the Art of Rhetoric and Prosody. By deepening their knowledge of Persian Language and Literature, we aim to foster the production of high-quality research, particularly focusing on the context of Sindh and Pakistan.

Ph.D. Program

Our mission is to acquaint students with the evolving trends and advancements in Persian Language and Literature. We strive to prepare them for independent research and scholarly pursuits, enabling them to produce exceptional research in the fields of Persian Language, Literature, and History, with a special emphasis on Iran, Pakistan, Afghanistan, and Tajikistan.

OBJECTIVES AND SCOPE

M.Phil. Program

Objectives:

1. Develop a thorough understanding of literary research methods, criticism, rhetoric, prosody, and communication skills to enhance collaborative research efforts.
2. Conduct critical studies of Persian poets and prose writers from Iran and the Indian subcontinent.
3. Encourage and motivate students to undertake research related to Persian Studies, with a particular focus on Pakistan, especially Sindh.
4. Produce high-quality research that highlights the role and contributions of Persian Language and Literature in the civilization, history, and literary development of Pakistan.

Scope:

1. Increase students' confidence and sharpen their memory.
2. Provide a broader perspective on the world, a valuable tool in today's interconnected society.
3. Prepare graduates to work as translators, interpreters, and instructors of Persian Language, allowing them to teach the language and literature in various educational institutions.
4. Equip graduates with skills to edit old Persian manuscripts, preserving cultural heritage.
5. Enable graduates to join government and non-government organizations, particularly in the Foreign Ministry, Archaeology, Museums, Tourism, Media, News, and Advertisement agencies.

Ph.D. Program**Objectives:**

1. Enable students to comprehend and analyze various styles of Persian poetry and prose.
2. Encourage students to present their research findings in public forums and publish them in academic journals.
3. Produce high-quality research in Persian Studies globally, contributing to the preservation of the subcontinent's cultural heritage.
4. Explore new dimensions and theories in language and literature to advance the field of Persian Studies.

Scope:

1. Enhance students' performance in other academic areas and increase their IQ levels, fostering innovation through continued immersion.
2. Prepare graduates to work as translators, interpreters, and instructors of Persian Language, enabling them to teach and conduct research in universities and research institutions.
3. Equip graduates with skills to edit old Persian manuscripts, preserving cultural heritage.
4. Enable graduates to join government and non-government organizations, particularly in the Foreign Ministry, Archaeology, Museums, Tourism, Media, News, and Advertisement agencies.
5. Empower graduates to protect and sustain cultural, ethical, and spiritual values, as well as contribute to the historiography of their country.

FACULTY MEMBERS

S. NO.	Name of Faculty Member	Designation	Area of Specialization and Expertise
1.	Dr. Ramazan Bamari	Professor	Classical Persian Prose, Cross Cultural Studies, History of Persian Literature
2.	Dr. Faizeh Zehra Mirza	Associate Professor	Classical and Modern Persian Poetry, Iqbal Studies, Cross Cultural Studies.
3.	Dr. Muhammad Nazir	Assistant Professor	Rhetoric, Prosody, Stylistics, Literary Criticism, Grammar, Classical and Modern Persian Poetry
4.	Dr. Faiza Khatoon	Lecturer	History of Classical Persian Poetry, Allama Iqbal, Molana Rumi

AVAILABLE RESEARCH FACILITIES AT THE DEPARTMENT**Departmental Library**

Our department boasts a comprehensive library housing over 8,000 books and journals dedicated to Persian language, literature, history, and culture. The library accommodates up to 100 students simultaneously, providing ample space for extensive research and study. In addition to the vast collection of books, the library also holds numerous research theses from M.Phil. and Ph.D. graduates, offering valuable resources for current students.

COUNCIL ROOM

The department features a well-equipped council room with modern furnishings and an efficient cooling system. This facility can comfortably host up to 20 individuals, making it ideal for research activities, seminars, and academic conferences. It serves as a collaborative space where students and faculty can engage in meaningful discussions and presentations.

INFRASTRUCTURE OVERVIEW

The Department of Persian is located on the first floor of the Arts and Social Sciences building, occupying a portion of the floor space. Here's a brief overview of our infrastructure:

1. **Classroom:** Equipped to accommodate up to 50 students, our classroom provides a conducive environment for lectures, discussions, and interactive sessions.
2. **Council Room:** A dedicated space for student-teacher interactions, fostering collaborative learning and mentorship opportunities for postgraduate students.
3. **Library:** Our departmental library caters to the academic needs of up to 50 students simultaneously, offering a wide range of resources in Persian language, literature, history, and culture.
4. **Departmental Office:** The central hub for administrative activities, including student inquiries, faculty coordination, and departmental management.
5. **Faculty Rooms:** Five rooms are allocated for faculty members, providing individual spaces for teaching preparation, research activities, and consultations with students.
6. **Washrooms:** Separate restroom facilities are available for male and female students.

Our infrastructure is designed to support a vibrant academic environment, facilitating teaching, learning, and research endeavors within the Department of Persian.

KEY RESEARCH AVENUES FOR POSTGRADUATE PROGRAMS

1. **Literary Research Methods and Criticism:**
 - Advanced methodologies in Persian literary research.
 - Critical analysis of Persian literary works from Iran and the Indian subcontinent.
 - Development of new literary criticism frameworks.
2. **Persian Poetry and Prose:**
 - In-depth study of various styles and forms.
 - Comparative analysis of prominent Persian literary figures.
 - Examination of thematic and stylistic evolutions.
3. **Historical and Cultural Studies:**
 - Research on the historical development of Persian language and literature.
 - Study of Persian literature's impact on the cultural history of Pakistan and the subcontinent.
 - Preservation and interpretation of historical Persian manuscripts.
4. **Translation and Interpretation:**
 - Techniques for translating Persian texts.
 - Professional training for translators and interpreters.
 - Contemporary and historical roles of interpreters.
5. **Manuscript Editing and Preservation:**
 - Methods for editing and preserving ancient Persian manuscripts.
 - Research on the significance of historical documents.
 - Development of digital preservation tools.
6. **Contemporary Applications:**
 - Applying Persian language skills in media, tourism, and international relations.
 - Roles in government and non-government organizations.
 - Cultural preservation efforts.
7. **Interdisciplinary Connections:**
 - Linking Persian studies with history, archaeology, and cultural studies.
 - Exploring Persian literature's influence on other traditions.
 - Collaborative research with other academic fields.

These research avenues aim to deepen understanding and foster innovation in Persian language and literature, preparing graduates for impactful careers.

DEPARTMENT OF PHILOSOPHY

INTRODUCTION

The Department of Philosophy was founded in 1952. It was the first department established in the Faculty of Arts, (now the Faculty of Arts and Social Sciences) at the University of Karachi. Dr. M.M. Ahmed was the founding Chairman of the department. From the very beginning, the department has been committed to fostering a wide variety of philosophical approaches and has gradually evolved towards developing links between philosophy and other disciplines. The members of its teaching staff include Professors and Lecturers. All teachers are strongly committed to teaching and research and have significantly contributed through their publications to the advancement of learning. Moreover, its senior faculty members have also served in important positions of responsibility in other areas. Dr. Zulfiqar Ali is the current Chairman of the department.

POSTGRADUATE PROGRAMS

The Department of Philosophy has been offering the following two postgraduate programs: Master of Studies (MS) with course work or /Master of Philosophy (M.Phil) (with course and research work) and Doctor of Philosophy (Ph.D.).

OBJECTIVES OF PROGRAMS

The M.Phil and MS programs aim to provide students with foundational research skills in philosophy, preparing them for advanced research and specialization in specific fields. The Ph.D. program further aims to develop students' ability to contribute to advanced research, complete comprehensive training, and perform independent research, thereby opening new career opportunities.

AIMS AND SCOPE OF PROGRAMS

The MS and M.Phil programs, lasting two to four years with eight to ten courses, aim to prepare students for further academic research in philosophy, providing a foundation for advanced studies like Ph.D. and postdoctoral research, with the MS accommodating those not pursuing research. The Ph.D. program, spanning three to seven years and 18 credit hours of coursework, seeks to advance philosophical study and train students for independent research.

REGULAR FACULTY

S.No	Name with Designation	Area of expertise
1.	Professor Dr. Zulfiqar Ali	Postmodernism, Political Theory, Postcolonialism
2.	Professor Dr. Abdul Wahab Suri	Political Theory, Rawls
3.	Professor Dr. Abdul Rahim Afaki	Islamic Philosophy, Hermeneutics
4.	Assistant Professor Dr. Muhammad Ateeq	Modernity, Analytical Philosophy, Karl Popper
5.	Assistant Professor Dr. Abdul Khaliq Aboya	Modernism, Educational Philosophy
6.	Assistant Professor Dr. Muhammad Irfan	Political Theory, Human Rights, Habermas
7.	Assistant Professor Dr. Mahboob ul Hassan Bukhari	Islamic Philosophy, Deleuze, Postmodernism
8.	Assistant Professor Dr. Amjad Ali	Political Theory, Marxism, Deleuze
9.	Lecturer Safia Bano	Philosophy Of Science, Greek Philosophy

RESEARCH FACILITIES

The Department of Philosophy has a separate library with over 2,000 books, equipped with Wi-Fi, providing access to the HEC central library and other global resources on philosophy.

DEPARTMENT OF POLITICAL SCIENCE

INTRODUCTION

The Department of Political Science at the University of Karachi is a leading institution for the study of politics, governance, and public policy in Pakistan. Established in 1952 the department has a rich legacy of academic excellence, research, and community engagement. and has been offering MS and Ph.D programs in Political Science and its allied fields

AIMS AND SCOPE

This mission of the department is to foster a deeper understanding of political phenomena, promote critical thinking, and prepare students for careers in public service, research, and leadership. The primary focus is on achieving excellence in research that is of both strategic and practical importance. To achieve these objectives, the department arranges tutorials, seminars and academic conferences along with regular classes.

CAREER OPPORTUNITIES

In addition to teaching and research positions in academic settings, Political Science degree holders are prepared for careers at all levels of government

1. Civil Service (CSS, PCS, etc.)
2. Diplomatic Service
3. International Organizations (UN, World Bank, etc.)
4. Non-Governmental Organizations (NGOs)
5. Research Institutes and Think Tanks
6. Public Policy and Administration
7. Journalism and Media
8. Law and Advocacy
9. Political Consulting and Campaign Management

By joining the Political Science department, students can gain a deeper understanding of the world and prepare for a meaningful career in the public sphere.

PROGRAM OFFERS

1. MS/M.Phil Program
2. Ph.D Program

FACULTY

The Department has recruited faculty of diverse academic backgrounds such as Politics, International Relations, defense and Strategic Studies, Government and Public Policy, and European Studies.

S.No.	Professors	Area of Research/ Files of Expertise
1.	Dr. Samina Saeed (<i>Chairperson</i>)	Western Political Thoughts , Comparative Politics
2.	Dr. Summer Sultana	Muslim Political Thought, International Relations
3.	Dr. Muhammad Ali	Comparative Constitutions , Diplomacy and Foreign Policy Analysis

Assistant Professors		
4.	Dr. Muhammad Kamran Khan	Government and Politics of Pakistan , International Organizations
5.	Dr. Syed Shameel A. Qadri	Public Administration and Management , Muslim Political Thought
6.	Dr. Shahla Tehseen	Pakistan Movement , Human Rights
	Dr. Suwaibah Qadri (T)	Organizational Theory and Behavior

FACILITIES

The department has a fully functional Seminar Library with over 2000 books. Most of the books are recent publications in the subject in both English and Urdu. The library also offers dailies, periodicals and research

journals for reading to its students. Moreover, there is a well-equipped specialized Library with latest books and periodicals in all the areas relevant to Public Policy.

COMPUTER LAB

The Computer Lab is equipped with high speed computers, printers and scanner facilities. These computers are networked with a server and a stand by server. Audio/Visual aids including multimedia option are used during presentations/ lectures. Internet access is also available to the students.

RESEARCH

Research work for M.Phil. / Ph.D. students is regularly conducted by the Faculty Members. Special interest and aptitude of the student as well as velocity of the topic is kept in mind. Topics being completed are political culture in Pakistan, Military and politics in Pakistan, Economic Human rights & their impact on contemporary constitutions, Islamization in Pakistan, problems of good governance, and importance of International Law.

DEPARTMENT OF PSYCHOLOGY

INTRODUCTION

Psychology is one of the leading departments of faculty of arts and social sciences. The Department of Psychology came into existence in 1954 under the Chairmanship of Professor Dr. Qazi Aslam. Eminent psychologists like Dr. Haseen Uddin Zuberi, Dr. S. M. Hafeez Zaid, Dr. Farrukh Z. Ahmed, Dr. Afzal Imam and Dr. M. Wasim led the department to its present academic heights. It has now become one of the leading departments of the Faculty of Arts and Social Sciences. The Department of psychology has a good strength of Qualified Faculty who contributing meaningfully for the Department's academics. The Department is proud of its graduate and post-graduate students who have completed their studies and research successfully and are now actively serving in prestigious public and private, teaching and research institutes, colleges, and Universities, corporate organizations and arm forces throughout Pakistan. Teaching is based on interactive class participation: group discussions, class presentations by students, assignments, research work etc. to maintain interest and make learning more effective. The use of audio visual aids brings out the creativity of teachers and students, and helps in improving their presentation skills. Faculty Members and Students both are engaged in continuing their Research Publications.

POST-GRADUATE PROGRAMS

The Department offers:

1. **M.Phil in Psychology**
2. **Ph.D in Psychology**

Our programs meet international standards and comply with the Higher Education Policy.

POST-GRADUATE PROGRAM OBJECTIVES

Psychology, as it teaches about human behavior, is one of the most important disciplines of our times. Speciation in this course will help our students understand the impact of individual, cultural and societal factors on our functioning.

These Programs facilitate Research based understanding for what makes people take extreme actions against themselves and society in general, with improvised measures and culturally relevant research for our people. This course aims to develop students aware of the latest scientific measures available in psychology. The major focus of M.Phil and Ph.D programs is to build students' knowledge about current advances in psychology and equip them with critical thinking and research techniques.

To develop sound research skills, to acquire the knowledge and become critical thinkers and to conduct enquiries in their areas of research in a professional, expert and ethical manner.

AIM AND SCOPE OF THE PROGRAM

The postgraduate program in psychology offers a broad and enriching scope for students. Graduates can work in Academics, Research, Armed forces, Security departments, Law enforcement, Clinical and therapeutic services, Rehabilitation, and Industrial and organizational settings. The program develops critical thinking, research skills, and practical expertise for addressing complex psychological concepts in the research.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATIONS

Professors:

1. Prof. Dr. Anila Amber Malik, (*Chairperson*) Ph.D in Psychology, (Mental Health, Psychometrics, Wellbeing, Interpersonal Relations, Extremism, Political Psychology)
2. Prof. Dr. Farah Iqbal, Ph.D in Psychology, (Psychology of Management, Education, Crime, Marketing and Social Development)
3. Prof. Dr. Qudsia Tariq, Ph.D in Psychology, Certified Client centered Therapist (Clinical Interventions, Terrorism and Violence, Community Challenges and Mental Health)
4. Prof. Dr. Amena Zehra Ali, Ph.D in Clinical Psychology, (Marital Issues, Child and Adolescent, Psychometrics, Emotional issues and Interventions)
5. Prof. Dr. Zi Asma Haneef Khan, Ph.D in Clinical Psychology, (Psychology of Religion, Clinical Assessment and treatment, Intervention based Research)

Associate Professors

1. Dr. Bushra Khan, Ph.D in Psychology, (Mental Health, Life Span Development, Psychometrics)

Assistant Professors

1. Dr. Anila Mukhtar , Ph.D in Psychology,(Parent Child/Adolescents Relationship, Conflict and Mental Health Issues, Gender Psychology, Market Research)
2. Dr. Saima Masoom, Ph.D in Psychology, Post doctorate in Childhood and Adolescents, Leicester University (Childhood and Adolescents, Applied Research, Test Construction, Teaching Methodologies)
3. Dr. Farhan Kamrani, Ph.D in Psychology, (Psychological assessment, Psychotherapy, Scale Development/ adaptation).
4. Dr. Rabia Riaz, Ph.D in Clinical Psychology, (Action Research with Marginalized Communities, Development of Culture, Specific Therapeutic Interventions).
5. Dr. Mahwish Mursaleen, Ph.D in Clinical Psychology, (Clinical Psychology, Mental Health).

Lecturers

1. Dr. Amreen Rao, Ph.D in Psychology, (Clinical Psychology, Child and Adolescent Psychology, Health Psychology).
2. Dr. Aasma Ashraf, Ph.D in Clinical Psychology, (Attachment Psychology, Adolescent Wellbeing, Childhood mental and developmental disorders).
3. Ms. Saba Abid, M.Phil in Psychology, (Counseling, Educational Psychology, Psychosocial Issues).
4. Ms. Aelia Zaidi M.Phil in Clinical Psychology, (Trauma Psychology, Substance Use disorders Developmental Psychology).

RESEARCH FACILITIES AVAILABLE

The research facilities in the Psychology Department are integral to fostering academic and scientific advancement. The seminar library is a cornerstone, offering extensive collections of latest books, journals, and e-database resources that cover a wide range of psychological titles. It provides access to the latest research articles, databases, and e-books, supporting in-depth literature reviews, research methodology and relevant scholarly work.

1. Department has laboratories which are equipped with cutting-edge tools for conducting experiments in various areas of psychology.
2. The computer lab is another critical resource. High performance computers with access to e-databases particularly HEC digital library that has enormous collections of online journals and books. These facilities create a conducive environment for comprehensive research in psychology.
3. Department has Psychological Testing laboratory that has variety of Updated Internationally Standardized Psychological tests and Batteries and Inventories. Students are facilitated for their training under control Conditions.
4. Department also have Research Labs which is well-equipped with Standardized tools and instruments through which students can facilitate their learning in control environment.

INFRASTRUCTURE

The infrastructure of a psychology department is designed to support advanced learning and research. It typically includes well-equipped lecture halls for interactive learning and discussions. Multimedia facilities are available for Classes. Department research labs facilitate psychological experiments. The department also have a comprehensive library with more than 3000 books, with an access to the latest journals and databases. Department has 03 communal spaces designed to foster collaboration and a sense of community among students and faculty. Additionally, these spaces often host departmental events, and social gatherings, enhancing the overall academic and social experience within the psychology department.

AVENUES FOR RESEARCH

Avenues for research in psychology are vast and diverse, reflecting the field's complexity and breadth. Key areas include: clinical psychology, social and cultural diversity, cognitive sciences, forensic aspect in psychology, educational and organizational psychology, mental health and wellbeing, terrorism and violence, community and health. Clinical Interventions, psychometrics, child and adolescent issues, and Political Psychology.

DEPARTMENT OF SINDHI

INTRODUCTION

The Department of Sindhi was established in 1972. The department was started with a master's program only, but the **M.Phil and Ph.D Programs** were started in 1974. Later on Honors program, certificate and diploma were also introduced. The Department also offers Sindhi as a compulsory subject in lieu of Urdu in B.A./B.Sc. and B.Com. (Honors) 1st and 2nd year. In 2007 like other Departments of University of Karachi, Sindhi Department also started **BS four years program (120 Cr Hrs) MS/ Ph.D.** While the Balochi Language certificate course is also administrated by Sindhi Department. The curriculum is regularly revised and updated by introducing the latest literature being produced by authors so that students can be kept well aware. Regular literary lectures are arranged to inspire the students and to make them aware of practical criticism, and develop skills of expressing their opinion. For the same purpose, scholars and critics are invited regularly for extension lectures. Interaction with literary figures is a regular feature of the department's activities.

Moreover, the department also organizes excursions for students to various historical places for their close first hand interaction with history thus giving students a chance to triangulate their knowledge from books, observations and their own interpretations.

PROGRAMS OFFERED BY THE DEPARTMENT

1. M.S./M.Phil./ Ph.D.

M.Phil and Ph.D Programs

Yes both the Programs are being offered by the Department of Sindhi. Department uses to conduct Seminars and workshops to guide the scholars.

PROGRAM OBJECTIVES

1. To promote the Language, Literature, Art and Culture.
2. To play active role for the betterment and mental positivity of the society.
3. To convey the message of Humanism, harmony and tolerance through research.
4. To represent the department as a center of communication, awareness and literature.
5. To make efforts for the character building of the students

AIMS AND SCOPE OF THE PROGRAM

1. To promote the Art, Culture, language and literature, which are the main sources of social awareness and stability to build the positive society.
2. To create enthusiastic young scholars, they may be very beneficial for the society.
3. To create and enhance the soft image of the society.
4. To conduct the research over the remaining aspects of the language and literature.
5. To enhance the research interest of the scholars

REGULAR FACULTY MEMBERS

Professors

1. Dr. Naheed Parveen (Chairperson) She is expert in classical Literature, Research Methodology and Prose.

Assistant Professors

1. Dr. Sher Muhammad: He is Expert in the study of Shah latif, Modern Poetry and Contemporary Literature.
2. Dr. Sajida Parveen: She is expert in Criticism, Syntax, Linguistic

FACILITIES

1. Well- stocked seminar library
2. Projectors for Presentations
3. Spacious and well equipped class rooms

INFRASTRUCTURE

The Department of Sindhi is consisting over Well- stocked seminar library, Two Spacious and well equipped class rooms; including one air conditioned room and separate spacious offices for each faculty member.

AVENUES FOR RESEARCH

The Department of Sindhi has wide range of research areas including,

1. Classical literature including Classical poetry and folk literature
2. History of literature
3. Philosophical back ground of the language and literature.
4. Modern literature
5. Contemporary literature
6. Linguistic and syntax
7. Comparative literature

OTHER RELEVANT INFORMATION***Career opportunities for the graduates and post graduates***

The graduates and post graduates of the department find various openings of their career including teaching at Secondary, Higher Secondary, colleges and Higher Educational Institutions i.e universities, with a boom in media industry, now they also join various electronic and print media outlets. They also compete in competitive examinations. After passing out no student remains unemployed.

DEPARTMENT OF SOCIAL WORK

INTRODUCTION

The Department of Social Work was established in 1961 and since then it has responded to the societal and corporate needs by developing trained professional social workers. The department is proud to have very successful alumni. The department strives to strike the right balance by creating a strong base of theoretical input combined with a rigorous Industrial /NGO exposure of nearly 12 credit hour, through the fieldwork training which is an important part of the curriculum. Through this training, Students are provided opportunities to integrate their theoretical knowledge with the practical world and get hands-on experience to deal with the variety of complex societal problem as challenges. The students are placed in government and non-government organizations for obtaining Knowledge in the fields of community development, school social work, and medical social work. The students of BS 1 Year are bound to work on research Projects (Theses) to be submitted at the end of the year.

FACILITIES

The Department of Social Work has a well-stocked Library with more than 1000 books, thesis, journals, reports and periodicals. The department is also equipped with Audio-Visual Lab named as "Abdul Sattar Edhi Research Lab."

SPECIFIC RESEARCH AREAS

The research activities are related to all the social problems at large. Community development, public health issues as well as institutional studies allied area

CAREER OPPORTUNITIES

Job placements for social work graduates are:

1. Ministry of Social Welfare and special Education
2. Women Development
3. Bat-al-Mal Department
4. Medical Social Service Project, CDGK, RCD project, Jails, etc.
5. Ministry of Population Welfare
6. Ministry of Labour Welfare
7. Educational Institutions
8. Research Organization
9. Child Protection. Drugs Rehabilitations
10. Health Care Organization and NGOs working in development Sectors
11. Administrative positions in National and International NGOs
12. Planning and Development organization
13. Urban and Rural Community Development Projects\
14. Family & Child Welfare
15. Criminology and Correctional Administration
16. School Social Work

Programmes Offered

1. M.Phil & Ph.D.

FACULTY MEMBERS

Professor

1. Dr. Muhammad Nadeemullah (Chairman)
2. Dr. Sakina Riaz
3. Dr. Syeda Farhana Sarfaraz

Assistant Professors

1. Dr. Samina Rauf
2. Dr. Sumaira Ishrat
3. Dr. Muhammad Arshad
4. Dr. Sabir Micheal
5. Dr. Muhammad Faisal Zia

DEPARTMENT OF SOCIOLOGY

INTRODUCTION OF THE DEPARTMENT OF SOCIOLOGY

The Department of Sociology was established in 1960 with the overarching goal of enhancing students' comprehension regarding social actions, empowering individuals, and shedding light on the roles of social institutions in shaping human development. It also offers critical insights and perspectives for addressing societal issues. Sociology's principal objective lies in conducting research to generate, accumulate, and disseminate scientific knowledge about society and various social phenomena. The field of sociology embraces a vast scope, stretching from the examination of fleeting encounters between individuals on the streets to the exploration of global societal processes. This discipline addresses a wide spectrum, encircling every facet of human social conditions, all forms of human relationships, and various manifestations of social behavior.

Sociology encompasses a wide array of areas of study, including but not limited to community development, social organization, human resource management, urban and rural development, demography, human rights, social psychology, Anthropology, industrial and health-related aspects, law, gender studies and economic development.

Graduates from the Department have pursued diverse career paths and have demonstrated excellence in their respective professions, many of them finding positions in both public and private institutions. The teaching faculty comprises highly experienced and qualified professors and teachers. Research is the distinguishing feature of the department. Every year five to ten students get their M.Phil and Ph.D degrees from the Department under supervision of our esteemed faculty members.

The Department of Sociology actively organizes a variety of events, including conferences, seminars, workshops, and fieldwork, as integral components of its academic activities.

POSTGRADUATE PROGRAMS AVAILABLE

1. M. Phil
2. Ph.D

PROGRAMS' OBJECTIVES

The Doctoral program in Sociology's main objective is to train excellent Sociologists for both academic and University careers, as well as careers as experts in major public and private organizations and NGOs.

AIMS AND SCOPE OF THE PROGRAM

The Ph.D. Sociology program aims to develop knowledge and provide research and learning opportunities in the field of Sociology. This program is specifically focus on providing a strong theoretical and methodological base to its participants. Students of this program will be able to enhance their critical thinking and analytical skills which will help them become independent researchers and knowledgeable observers of sociological phenomena to provide concrete findings to policy makers and to society in general.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION AND EXPERTISE

1. **Prof. Dr. Sobia Shahzad**
National and International social system, Formal Organization, Community development, Theory construction, Social research, Applied sociology, Cultural studies, Collective Behavior and Social Change, Sociological Theories, Criminology and Social problems of Pakistan
2. **Prof. Dr. Nabeel Ahmed Zubairi**
Research Methodology, Criminology, Statistics, Anthropology
3. **Dr. Naila Usman Siddiqi**
Community Development, Survey Research, Social System, Marriage and family, Sexual and Reproductive Health, Social Inequality, Youth issues.
4. **Dr. Kausar Parveen**
Social Psychology, Criminology, Juvenile Delinquency, Gender Studies, Community Development, Economic Development, Cultural Studies and Research Methodology, Social Problems of Pakistan

5. Dr. Munazza Madani

Social Inequality, Criminology, Education, Family and Relationships, Globalization and Development, Health and Medicine, Religion and Spirituality, Social Movements and Collective Behavior, Urban Sociology, Work and Organizations, Environmental Sociology, Digital Sociology

6. Dr. Kaneez Fatima Mamdani

7. Social Research, Theory Construction & model building, Human Resource Management & Development, Project Management, Statistics, Software's application in Social Research (SPSS, CsPro, Stata, R, Minitab), Poverty & Deprivation, Environmental Impact Assessment (EIA), Demography

8. Dr. Nosheen Raza

Demography, Medical Sociology, Industrial Sociology, Sociology of Education, Social Problems

RESEARCH FACILITIES AVAILABLE (LIBRARY, LABORATORY, COMPUTER LAB.)

1. The seminar library houses an extensive collection of over 1000 books, journals, magazines, and various national and international periodicals.
2. The Ethnological Museum plays a pivotal role in the study of art and culture, boasting a rich repository of over 500 precious artifacts. It functions as a cultural laboratory, facilitating in-depth exploration and research in this field.

INFRASTRUCTURE

Apart from seminal Library and ethnological museum the department is fully equipped with spacious airy class rooms, office that assists students and other facilities.

AVENUES FOR RESEARCH

Sociology is a broad field that encompasses the study of society, social institutions, social relationships, and human behavior within social contexts. There are numerous potential areas of research in sociology that scholars may explore. Some of these areas include:

1. Social Inequality
2. Family and Relationships
3. Health and Healthcare
4. Work and Organizations
5. Globalization and Transnationalism
6. Environment and Society
7. Technology and Society
8. Crime and Deviance
9. Education
10. Culture and Media

CAREER OPPORTUNITIES

Sociologists find employment in a wide range of fields, including Academia, Government sector, Nonprofit Organizations (national & International, both), Healthcare, Criminal Justice, Market Research, Social Services, Human Resources, Media and Journalism, Environmental Organizations, International Development, Market Analysis and Consulting, Social Research Firms, Public Relations, Technology and Data Analysis, Education Administration, Urban & rural Planning.

These are just a few examples, and the versatility of a sociology degree allows professionals to apply their knowledge and skills in various settings and industries.

Sociologist is employed in industries, trade, association labor unions, in research organization and administration. Newly emergent trends with Artificial Intelligence students may find careers in many other fields.

Furthermore Sociologist can work as hospital management administrator, Public Administration Assistant, Rural health outreach worker, Census research Assistant, Consumer Researcher, Data Analyst, Demographer Assistant, Market Researcher, Statistician, Survey Research Technician, Correctional Counselor, Criminal Investigator, Juvenile Court Worker, Parole Officer, Rehabilitation Counselor and Humanitarian Organization like UN, national and international NGOs.

DEPARTMENT OF URDU

INTRODUCTION

The Department of Urdu officially came into existence in 1955 when the University of Karachi decided to organize postgraduate classes in Urdu under its own management instead of through the colleges. At the time of its establishment, Moulvi Abdul Haq, 'Baba-e-Urdu', was made honorary Professor and the Chairman of the Department. Teaching responsibilities were carried out by the teachers of local colleges. Later, Dr. Ghulam Mustafa Khan, a prominent scholar and Professor at Urdu College Karachi, was appointed both as a cooperative teacher and as the Chairman of the department. In the same year, Dr. Abul Lais Siddiqui was appointed as a Reader and Chairman and Dr. Abdul Qayyum, Dr. Syed Shah Ali, and Qudratullah Fatimi joined as lecturers.

With the valuable efforts and abilities of these teachers as well as the scholars and teachers who joined the department later on, the department has kept making progress. The following famous academics have been associated with the Urdu department:

1. Dr. Syed Abul Khair Kashfi
2. Dr. Farman Fatehpuri
3. Dr. Aslam Farrukhi
4. Prof. Jameel Akhtar Khan
5. Dr. Hameed Fauq
6. Dr. Younus Hasni Shamim Ahmed
7. Prof. Sahar Ansari
8. Dr. Vaqar Ahmed Rizvi
9. Dr. Siddiqa Arman
10. Dr. Moinuddin Aqeel
11. Dr. Zafar Iqbal

The Department of Urdu has a prominent position in the University of Karachi for research. Over 100 scholars have completed their Ph.D. research program and more than forty projects are in progress. In addition to regular teaching and lectures, co-curricular, literary and academic activities and gatherings are also organized frequently.

AIMS & SCOPE

1. To gain advance knowledge & expertise
2. To produce original and substantial research
3. Professional development for and other related academia, publishing, writing skills
4. To promote literary traditions and culture

TEACHING AND ASSESSMENT

Teaching at the Department combines lectures, group Discussions and class presentations. Students are also given writing assignments.

DEGREES PROGRAMS

1. M.Phil.
2. Ph.D.

FACULTY MEMBERS

Professors

1. Dr. Uzma Farman Farooqui (Chairperson)
2. Dr. Tanzeem-ul-Firdous

Assistant Professors

1. Dr. Safia Aftab
2. Dr. Uzma Hassan
3. Dr. Shama Afroz
4. Dr. Muhammad Sajid Khan
5. Dr. Sadaf Fatima
6. Dr. Tahmina Abbas
7. Dr. Sadaf Tabassum
8. Dr. Ansar Ahmed
9. Dr. Zakia Rani
10. Dr. Khalid Ameen
11. Dr. Ayesha Naz

Lecturers

1. Dr. Muhammad Shakir
2. Mr. Muhammad Salman
3. Mr. Muhammad Rashid Ali Khan
4. Ms. Atiya Huma Siddiqui

OBJECTIVES

1. Conduct research to make a contribution the field of literary studies
2. Engage with wide range of texts & critical perspective
3. Explore how literature is connected with other academic fields
4. Enhance skills in scholarly writing and publish research in reputed journals
5. Provide professional development opportunities
6. Encourage development of interdepartmental thoughts & perspectives.

AVENUEUS

1. Academic careers
2. Research
3. Publishing Editing
4. Translation
5. Media and journalism
6. Cultural Institutions
7. NGOs
8. Literary Language Institutions
9. Online Communities
10. Archiving & preservation

AVAILABLE RESEARCH FACILITIES

Departmental Library

1. Library has capacity of hundred students may be avail at the same time. Library entertains more than 1000 books in addition to hundreds of research theses of M.Phil & Ph.Ds passed out students.
2. Council Room for Student & Teacher interaction is also available in the department for only post graduate students.

DEPARTMENT OF EDUCATION

INTRODUCTION OF THE DEPARTMENT

The Department of Education was established in 1984. The main goal of the Department is to prepare human capital in the field of Education as Teachers Educators Educational leaders and researchers. Innovative methods of teaching and learning emerging techniques of research and material development in the field of education have made this department paralled with the international standards of Education.

FACILITIES

The Department of Education has a seminar library as its primary resource containing contemporary books related to the discipline of Education Moreover the library also contains extensive resource material for research at all levels of higher education. The Department also offers hands-on training regarding equipment's and software to present research findings to a large segment of academic.

SPECIFIC RESEARCH AREAS

1. Research Methods in Education
2. Educational Psychology
3. Teaching Strategies
4. Behavioral Management
5. Curriculum Development
6. Trends and Issues in Education
7. Comparative Education
8. Classroom Assessment Test Development and Evaluation
9. Guidance and Counseling
10. Foundations of Education
11. School Management and Community Development
12. SPSS

CAREER OPPORTUNITIES

Education is an important discipline in Social Science and Humanities. Therefore graduate students of Education have ample opportunities in the competitive job market at both national and international levels. Research Industry Media Career Counseling, Schools, Teacher Training Institute, Human Resource Development Centers, Professional Institutes, UNICEF, UNESCO and NGO based organizations can be the potential job market for the graduate of education department.

POSTGRADUATE PROGRAMS AVAILABLE

1. Ph.D.
2. M.Phil.
3. M.S.

PROGRAMS OBJECTIVE

1. Define research; explain and apply research terms; describe the research process and the principle activities, skills and ethics associated with the research process.
2. Explain the relationship between theory and research.
3. Propose a research study and justify the theory as well as the methodological decisions. Including sampling and measurement.
4. Use appropriate online databases of research evidence and reference management software.
5. Know the innovative programmes, Monitoring and evaluation system.
6. To obtain the tools both technical and intellectual to perform statistical analyses and critiques of your own.
7. Control the operation of SPSS and manage your files and output.
8. Structure information effectively.
9. Perform role as curriculum planners and developers to meet the challenges and demands of the 21st century.
10. To gain competency in the critical evaluation of empirical analysis of the social world.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION/ EXPERTISE

- | | |
|------------------------|-----------|
| 1. Dr. Rizwana Muneer | Education |
| 2. Dr. Aziz un Nisa | Education |
| 3. Dr. Naila Siddiqua | Education |
| 4. Dr. Maroof Bin Rauf | Education |

Also the infrastructure supports to run a department successfully

DEPARTMENT OF SPECIAL EDUCATION

INTRODUCTION

The Department of Special Education at the University of Karachi was established in 1988 and provides graduate and postgraduate degrees for individuals interested in serving the special needs population. The Department strives to provide a vibrant, high-quality academic environment for future special education professionals. The faculty comprises experienced teachers with Ph.D. degrees who focus on developing autonomous and critical thinking skills among students.

POSTGRADUATE PROGRAMS AVAILABLE

The Department offers the following postgraduate degree programs:

1. MS/M.Phil in Special Education
2. Ph.D in Special Education

PROGRAM OBJECTIVES

Focusing on departmental mission, the objectives of the programs are to:

1. Train qualified professionals and activists to help persons with disabilities and their families.
2. Allow academics to seek the most recent knowledge via research in the field of special education with implications for individuals with disabilities.
3. Conduct research to help shape special education policies and practices at the university, provincial, and national levels.
4. To share current and relevant information through professional conferences, training programs, publications, and other media.
5. To increase awareness and advocate for children and adults with special needs from diverse backgrounds, while also encouraging collaborative partnership between families and professionals.

CAREER OPPORTUNITIES

Department of Special Education students continue pursuing various educational, vocational, and rehabilitative opportunities. Many graduates work as speech pathologists, audiologists, and instructors in special education facilities, while others find employment in schools, hospitals, institutions, and rehabilitation centers. To summarize, the passing graduates have a career in the following fields:

1. School teachers for Special Education
2. Itinerant Teachers for inclusive education
3. Administrators in Special Education Institution
4. Teaching jobs with attractive salaries abroad
5. Instructional supervisors
6. Subject specialists in education departments
7. Research Officers in different research organizations

FACILITIES

The Department of Special Education employs a lecture approach with an emphasis on classroom discussion and feedback, as well as audiovisual tools and professional assistance from resource individuals. The Department features a lecture, seminar, and audiovisual room, as well as a well-stocked seminar library with the most recent publications. Organize field trips for on-site observation of special persons. The Department offer adaptations during examinations; training in wheelchair and sign language; and accessible tools such as JAWS and DUXBury, white canes, Perkin's Brailers, Braille Frames, and Braille Stylus to those with visual impairments. In future, the department intends to grow as a research center for the teaching and learning requirements of students with disabilities. Students can ask the department for remote learning support.

FACULTY MEMBERS

Full Name	Qualification	Designation	Area of Specialization
Dr. Shagufta Shahzadi	Ph.D. (Special Education)	Professor	• Educational Psychology • Intellectual Disability Disorder • Research Method
Dr. Nasir Salman	Ph.D. (Special Education)	Professor	• Intellectual Disability Disorder • Quantitative Research • Inclusive Education
Dr. Humera Aziz	Ph.D. (Special Education)	Associate Professor	• Learning Disability • Intellectual Disability Disorder
Dr. Afshan Rahat	Ph.D. (Special Education)	Assistant Professor	• Hearing impairment
Dr. Saira Saleem	Ph.D. (Special Education)	Assistant Professor	• Visual impairment
Dr. Shaista Naz	Ph.D. (Special Education)	Assistant Professor (T)	• Communication Disorders • Qualitative Research
Ms. Saba Suleman	M.A (Special Education)	Lecturer	• Physical Handicap

DEPARTMENT OF TEACHER EDUCATION

INTRODUCTION

The Department of Teacher Education was established in 2012. The department aims at providing high quality teachers in line with National Professional Standards for Teachers in Pakistan for developing teachers for 21 century as reflective practitioners.

POSTGRADUATE PROGRAMS AVAILABLE

1. M.Phil.
2. Ph.D.

PROGRAMS' OBJECTIVES

1. To promote research culture
2. To promote research oriented activities
3. To enhance the research skills

AIMS & SCOPE OF THE PROGRAM

To build research society regarding faculty and learners its beneficial for educational industry

REGULAR FACULTY MEMBERS

Associate Professor & Chairperson

1. Dr. Safia Urooj (Ph.D. Education & Post Doctorate in Education)

Assistant Professors

- | | |
|---------------------------|--|
| 1. Dr. Intiaz Ahmed | (Ph.D. Education & Post Doctorate in Education) |
| 2. Dr. Muhammad Asim | (Ph.D. Education) |
| 3. Dr. Syeda Nazish Hasan | (Ph.D. Education) |
| 4. Dr. Intizar Hussain | (Ph.D. Education) |

RESEARCH FACILITIES AVAILABLE (LIBRARY, LABORATORY, COMPUTER LAB)

Available for research oriented activities

INFRASTRUCTURE

Department of Teacher Education has rich and comprehensive infrastructure for research activities for the researchers.

AVENUES FOR RESEARCH

1. Educational Planning and Supervision
2. Curriculum Planning and Development
3. Teaching Strategies
4. Leadership and Management
5. Educational Psychology
6. Educational Philosophy
7. Information and Communication Technology (ICT) in the field of Education
8. Teachers Training
9. Critical Thinking and Reflective Practices
10. Practicum and Micro-teaching
11. Child Development
12. General Method of Teaching
13. Classroom Management
14. Classroom Assessment
15. School Community and Teacher
16. Foundation of Education
17. Guidance and Counseling
18. Trends and Contemporary Issues in Education
19. Comparative Education

DEPARTMENT OF CHEMICAL ENGINEERING

INTRODUCTION OF THE DEPARTMENT

The Department of Chemical Engineering at the University of Karachi has a distinguished legacy of academic excellence and cutting-edge research since its establishment. It has been instrumental in producing skilled professionals who have significantly contributed to various industries, academia, and research institutions globally. Our department is committed to fostering a stimulating academic environment that encourages innovation, critical thinking, and practical application of chemical engineering principles to solve real-world problems.

POSTGRADUATE PROGRAMS AVAILABLE

The Department of Chemical Engineering offers a variety of postgraduate programs designed to advance knowledge and professional skills in specialized areas of chemical engineering. The available programs include:

1. Master of Engineering (M.E.) in Chemical Engineering
2. Doctor of Philosophy (Ph.D.) in Chemical Engineering

These programs are structured to provide students with comprehensive knowledge and research skills that prepare them for leadership roles in academia, industry, and research.

PROGRAM OBJECTIVES

The objectives of **M.E. (Chemical Engineering)** programme are:

1. **PEO 1:** Graduates will demonstrate the ability to effectively solve complex engineering problems and tasks using engineering, science, and statistical principles. They will be able to justify their recommendations through thorough analysis and evaluation of data.
2. **PEO 2:** Graduates will be able to assess the societal impact of their work, considering ethical, economic, global, and environmental factors. They will analyze the implications of their engineering solutions on various stakeholders and propose ethically sound and sustainable solutions.
3. **PEO 3:** Graduates will develop lifelong learning skills to stay updated with advancements in the field of chemical engineering. They will exhibit the capacity to analyze new technologies and systems critically, applying their engineering knowledge to adapt to evolving challenges and opportunities.
4. **PEO 4:** Graduates will emerge as effective leaders capable of working in diverse environments. They will demonstrate strong leadership qualities, interpersonal skills, and the ability to collaborate with multidisciplinary teams to achieve common goals.
5. **PEO 5:** The program will foster a culture of creative thinking, encourage a competitive environment, and promote active participation in professional activities. Graduates will uphold moral and ethical values, demonstrating integrity and professionalism in their professional endeavors.

The objectives of **Ph.D. (Chemical Engineering)** programme are:

1. **PEO 1:** Graduates will be able to conduct original, high-impact research in specialized areas of chemical engineering, fostering a deep understanding of theoretical frameworks, experimental methodologies, and computational techniques.
2. **PEO 2:** Graduates will be able to critically analyze complex engineering problems, develop innovative solutions, and apply advanced concepts across interdisciplinary domains, positioning graduates as leaders in addressing contemporary challenges in chemical engineering and related fields.
3. **PEO 3:** Graduates will be able to communicate effectively the research findings through scholarly publications, presentations, and collaborations with peers, industry partners, and the broader scientific community, enhancing the impact and visibility of their research contributions.

4. **PEO 4:** Graduates will be able to practice leadership roles in academia, industry, and government sectors by providing opportunities for mentorship, teaching experience, project management, and ethical decision-making, ensuring their readiness to drive positive change and contribute to the advancement of society through their expertise in chemical engineering.

AIMS AND SCOPE OF THE PROGRAM

The postgraduate programs in the Department of Chemical Engineering are designed to:

1. Provide a rigorous curriculum that integrates theoretical knowledge with practical applications.
2. Encourage research that contributes to advancements in chemical engineering, addressing global challenges such as energy sustainability, environmental protection, and industrial process optimization.
3. Develop technical expertise in areas such as process engineering, materials science, biochemical engineering, and energy systems.
4. Prepare graduates for careers in academia, research, and industry, with a focus on innovation and problem-solving.

REGULAR FACULTY MEMBERS WITH THEIR AREA OF SPECIALIZATION

The Department of Chemical Engineering boasts a diverse team of experienced and dedicated faculty members who are experts in various fields of chemical engineering. Some of our esteemed faculty members include:

1. Prof. Dr. Shagufta Ishteyaque
Specialization: Polymer Engineering, Environmental Engineering
2. Dr. Muhammad Yasir Khan
Specialization: Material Science, Environmental Engineering, Renewable Energy
3. Dr. Muhammad Furqan Ali
Specialization: Computational Fluid Dynamics, Process Optimization, CO₂ Capture, Global Warming
4. Dr. Adeel Ur Rehman
Specialization: Gas Hydrates, Wastewater Treatment, CO₂ Capture, Flow Assurance
5. Dr. Ahsan Abdul Ghani
Specialization: Environmental Engineering, Energy Systems, Waste Management, Material Science, Advanced Oxidation Processes
6. Dr. Obaid Ur Rehman
Specialization: Computation Fluid Dynamics, Crude Oil Processing, Numerical Modelling, Heat Transfer

RESEARCH FACILITIES AVAILABLE

The department is equipped with state-of-the-art research facilities that support a wide range of academic and industrial research activities:

1. Library: The department's library houses an extensive collection of books, journals, and digital resources in chemical engineering and related fields.
2. Laboratories: We have well-equipped laboratories for research in areas such as reaction engineering, materials science, environmental engineering, and fuel and energy engineering. These include pilot-scale reactors, analytical instruments, and specialized equipment for material characterization.
3. Computer Lab: The computer lab provides access to advanced simulation software and computational tools essential for process modeling, simulation, and data analysis.

INFRASTRUCTURE

The Department of Chemical Engineering is located in a modern facility designed to support both teaching and research. The infrastructure includes:

1. Lecture Halls: Equipped with multimedia facilities for effective teaching and presentations.
2. Research Labs: Dedicated spaces for conducting experimental research and project work.
3. Seminar Rooms: Used for workshops, conferences, and collaborative research activities.
4. Faculty Offices: Designed to facilitate interactions between students and faculty for academic guidance and research discussions.
5. Student Common Areas: Spaces for students to collaborate, study, and relax.

The Department has the infrastructure to support new trends in learning. All lecture rooms, labs, and classrooms provide support for e-learning, and faculty members frequently use these facilities. The infrastructure and facilities that support new trends in learning, along with their adequacy are given below:

1. Modern computer Labs with latest technologies are available in the department,
2. Latest Overhead Projectors,
3. Visual Aids,
4. Video CD's with latest software packages,
5. PA systems are available,
6. White Boards

RESEARCH FACILITIES

The University has a Video Conferencing facility that departmental faculty and students use frequently.

1. Libraries
2. Central Library: The University of Karachi has one Central Library with over two thousand engineering books. It has two sections, i.e., reference and circulation with textbooks, reference books, and research journals of all disciplines of Science, Engineering, Arts, Management, and Pharmacy.
3. Departmental Library
4. Seminar Library
5. Department of Chemical Engineering has a separate Seminar library accommodating 50 students. A qualified Seminar Librarian maintains it, and it contains around 3500 books.

Digital Library

Students have access to the following digital library:

1. HEC Digital Library (see Table 7.4 for available resources of HEC Digital Library)
2. HEC Ebrary (ProQuest Ebook Central)
3. SAGE Journals Online
4. IEEE Xplore

AVENUES FOR RESEARCH

Our department actively engages in various research initiatives and projects, offering numerous opportunities for postgraduate students to contribute to the field. Some key research avenues include:

1. Environmental Engineering: Projects addressing pollution control, waste management, and environmental impact assessment.
2. Advanced Materials: Studies on the development and application of novel materials, including polymers, nanomaterials, and composites.
3. Polymer Engineering: Studies on the development of biodegradable polymer films for food packing applications.
4. Interdisciplinary Research: Collaborations with other departments and industries to address complex, multifaceted engineering challenges.

These research activities are supported by national and international collaborations, providing students with the opportunity to work on impactful projects and gain valuable experience.

This prospectus information provides a comprehensive overview of the Department of Chemical Engineering at the University of Karachi, highlighting our commitment to excellence in education, research, and professional development. We invite prospective students to join our vibrant academic community and contribute to the future of chemical engineering.

DEPARTMENT OF ISLAMIC LEARNING

INTRODUCTION

The Department of Islamic Learning is one of the oldest departments of the University of Karachi and was founded in 1954. Its primary goal is to fulfill the academic needs of scholars in Islamic studies within Pakistan and abroad. The department promotes the study and research of Islamic thought at the MS/M.Phil. and Ph.D. levels and provides opportunities to study Islam at both graduate and postgraduate levels. The department is situated in the Islamic Studies building near Masjid-i-Ibrahim and aims to produce scholars for educational, religious, and social institutions both nationally and internationally.

POSTGRADUATE PROGRAMS AVAILABLE

1. MS/M.Phil. in Islamic Learning
2. Ph.D. in Islamic Learning

PROGRAMS OFFERED

1. M.Phil. & Ph.D.

PROGRAM(S) OBJECTIVES

The objectives of the MS/M. Phil. and Ph.D. programs in the Department of Islamic Learning are to:

1. Promote advanced study and research in Islamic thought.
2. Address modern and contemporary issues in Islamic studies.
3. Develop expertise in areas such as Islamic Economics, Comparative Religion, Islamic Sociology, and Muslim Political Thought.
4. Provide comprehensive knowledge in Islamic Sciences like Tafseer, Hadith, Seerah, Islamic History, Islamic Jurisprudence, Principles of Islamic Jurisprudence, Tasawwuf, Ethics, Philosophy, and Arabic.
5. Produce qualified scholars for academic, religious, and social institutions.

AIMS AND SCOPE OF THE PROGRAM

The aims of the program are to:

1. Develop a deep understanding of Islamic teachings and their application to contemporary issues.
2. Foster critical thinking and research skills in Islamic studies.
3. Prepare graduates for careers in academia, religious institutions, and other sectors requiring expertise in Islamic knowledge.
4. Contribute to the body of knowledge in Islamic studies through original research.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION / EXPERTISE

Professor

Dr. Zahid Ali (Dean, Faculty of Islamic Studies)

Associate Professor

Dr. M. Arif Khan Saqi (Chairman)

Assistant Professors

1. Dr. Taj Muhammad
2. Dr. Muhammad Atif Aslam
3. Dr. Jawad Haider
4. Dr. Umair Mehmood Siddiqi

Areas of specialization include Islamic thought, Tafseer, Hadith, Seerah, Islamic History, Islamic Jurisprudence, Principles of Islamic Jurisprudence, Tasawwuf, Ethics, Philosophy, Arabic, Islamic Economics, Comparative Religion, Islamic Sociology, etc.

RESEARCH FACILITIES AVAILABLE (LIBRARY)

The department's research facilities include the **Professor Iftikhar Ahmed Balkhi Memorial Library**, which offers a valuable collection of bibliographical and reference works, news sources, and current periodicals related to Islam and the Muslim world. The library also provides unique and primary reference books and Makhtootat essential for advanced research. Additionally, researchers have access to an extensive range of scholarly articles and journals through www.jstor.org, enhancing the research capabilities of students and faculty members.

INFRASTRUCTURE

The Department of Islamic Learning is in the Islamic Studies building near Masjid-i-Ibrahim. The infrastructure includes well-equipped classrooms, seminar rooms, and a specialized library that caters to the academic needs of students and faculty members.

AVENUES FOR RESEARCH

Faculty members are engaged in various research projects funded by the university and through individual initiatives. They have numerous publications and actively participate in debates, discussions, and talk shows on television, addressing national and international issues concerning Islam and the Muslim Ummah.

DEPARTMENT OF QURAN & SUNNAH

INTRODUCTION OF THE DEPARTMENT

The Department of Quran and Sunnah lies in the heart of the education hub of Karachi. Apart from offices, class rooms are available, it has a small seminar. The classrooms at Department of Quran and Sunnah are large and airy and all facilities at department are accessible to the students, faculty and course participants.

The Department was established in 2000. Master's Degree Program and now B.S./B.S. 3rd year, M.S. M.Phil. and Ph.D. program are available in this department. All the students who have passed / B.S. / B.S. 3rd Year/ M.A. or Shahdat-ul-A'liya / Shahdat-ul-A'almiya from Wafaq ul Madaris and other Madaris are eligible for admission. Those candidates who have done submit complete equivalence certificate issued by the University of Karachi.

POSTGRADUATE PROGRAMS AVAILABLE

1. M.S.
2. M.Phil.
3. Ph.D.

PROGRAMS' OBJECTIVES

It aims at producing students well aware with the knowledge of Islamic sciences with the specialized knowledge of Quran. The purpose behind this program is to combine religious education with modern sciences. To provide and develop high caliber leaders / scholars that can face the challenges of 21st century and will able to provide solution in the light of Quran and Sunnah.

AIMS AND SCOPE OF THE PROGRAM

After awarding M.Phil./ Ph.D. there is vast scope in Schools, Colleges, Universities, Madaras as teachers, as well as in other National and Multinational Institutions as Sarri'ah experts as Sahariah Advisor in Halal Food companies and in Islamic Banking.

REGULAR FACULTY MEMBERS THEIR AREAS OF SPECIALIZATION

1. Dr. Zeenat Haroon, Associate Professor (Seerat Un Nabi (S.A.W.)
2. Dr. Mufti Imran ul Haq, Assistant Professor (Tafseer ul Quran)
3. Dr. Syed Ghazanfar Aahmed, Assistant Professor (Hadith)

PUBLICATIONS

The Department had been publishing E-Research Magazine Bi-Annually in English/Arabic and Urdu since 2000. Now the Journal is recognize from HEC in "Y" category and indexed in international indexing and abstracting agencies.

RESEARCH ACTIVITIES

The research activities in the department are remarkable such as:

1. To write thesis on M.A. level is a part of research activities.
2. At least above 200 Ph.Ds. have been produced from the establishment of this department in 2000.
3. More than hundred students have been enrolled in M.S. / Ph.D.

DEPARTMENT OF USOOLUDDIN

INTRODUCTION

Department of Usool-ud-Din was established in 2000 to fulfill the need of scholars of Islamic Studies in various institute of Islamic Republic of Pakistan. Islamic Sciences like Tafseer, Hadith, Seerah, History of Islamic Jurisprudence, Comparative Study of Religions, Addawa & Its Method, Aqidah & Kalam, Al Ghaz wu Fikri, Islamic Psychology & Sociology, and Arabic are some of the courses that are being taught at the Department.

FACILITIES

The students of the department use seminar library facilities of the faculty of Islamic studies where a good collection of books and journals on Islamic Studies is available. The faculty of Islamic studies has also a computer lab for teaching and research activities.

RESEARCH ACTIVITIES

Thesis in M.A. & M.Phil/Ph.D. Programs is available for the research fellows.

PROGRAM AVAILABLE

1. MS / M.Phil.
2. Ph.D.

AIMS AND SCOPE (CAREER OPPORTUNITIES)

Students passing out from this department are providing their services in educational institutions, research institutes, religious institutions and armed forces. Master

FACULTY MEMBERS

Assistant Professors

1. Dr. Hafiz Muhammad Ismail (**Incharge**)
2. Teaching Experience in main subjects of Islamic Studies, specialization in Comparative study of Religion.
3. Dr. Muhammad Ishaq (Ph.D in Al- Fiqh Al- Islami.)
4. Dr. Muhammad Imran (Ph. D in Fiqh Al-Mu'amalat.)

SCHOOL OF LAW

INTRODUCTION

The University of Karachi, one of Pakistan's premier academic and research institutions, established the School of Law under its Faculty of Law in 2014. Building on the University of Karachi's rich heritage of research, the LL.M program at the School of Law equips students with essential drafting skills and robust research-based analytical and problem-solving abilities necessary for navigating the complexities of legal issues arising from global integration.

Benefitting from the wealth of research expertise and practical knowledge provided by our distinguished faculty, our LL.M program not only enhances the research acumen of lawyers but also opens remarkable career opportunities both nationally and internationally. Our graduates excel in litigation and transactional law practice, supported by the strong foundation they receive in our program.

Since its inception, the School of Law has produced exceptionally talented judges, litigators, transactional lawyers, and research scholars. Our students receive top-tier education and research supervision from highly qualified faculty members. The School of Law's curriculum is prescribed and approved by the Pakistan Bar Council and the Higher Education Commission of Pakistan (HEC). Currently, the Faculty of Law offers LL.B. (5 Years) and LL.M. (2 Years) programs.

POSTGRADUATE PROGRAM

1. **LL.M. 2 Years** (equivalent to M.Phil.)

OBJECTIVES OF LL.M PROGRAM

1. Deliver an advanced level of academic and professional training equivalent to an M.Phil. Enriching the legal education landscape.
2. Facilitate Career Growth and open doors to diverse and prestigious career opportunities in both national and international legal markets.
3. Promote Legal Innovation by encouraging pioneering research that drives legal advancements and policy development.
4. Support Professional Development which equip graduates with the expertise and skills needed to excel in their careers and contribute meaningfully to the legal profession and society.

AIMS AND SCOPE OF LL.M PROGRAM

The LL.M program at the School of Law, University of Karachi, aims to:

1. Cultivate Legal Mastery
2. Enhance Research Proficiency
3. Foster Professional Excellence
4. Uphold Ethical Standards

The scope of our LL.M program encompasses:

1. **Diverse Specializations:** Offering a broad array of courses covering traditional and emerging legal fields.
2. **Interdisciplinary Integration:** Encouraging cross-disciplinary studies to tackle multifaceted legal challenges.
3. **Global Orientation:** Providing a global perspective on legal practices and systems to prepare students for international careers.
4. **Practical Training:** Emphasizing experiential learning through moot courts, legal clinics, and internships.

REGULAR FACULTY MEMBERS

1. Dr. Abid Hussain, Assistant Professor (Senior Full-time Faculty)
2. Mr. Tauheed Ullah Siddiqui, Assistant Professor, School of Law, University of Karachi (Senior Full-time Faculty)

ACADEMIC BACKGROUND

Mr. Siddiqui is currently pursuing a Ph.D in Law at International Islamic University Malaysia, with an expected graduation in 2027. He holds an LLM from University of Karachi and LLB from International Islamic University Islamabad.

RESEARCH INTERESTS AND SPECIALIZATION

Mr. Siddiqui specializes in regulatory laws and statutes relating to Islamic Banking & Finance, Commercial laws, Islamic Jurisprudence and family laws. His Ph.D research focuses on statutory conflicts and consonance with Islamic banking principles and practices in Pakistan, aiming to identify the gaps in statutes which give legal cover to Islamic banking industry. His current research interests include:

1. Conventional financial lapses and alternative financial solutions.
2. Politico-judicial interplay in elimination of riba from the national economy over three decades.
3. Philosophical-legal foundations of the current monetary system

Mr. Siddiqui has published articles in reputable journals and participated at Global Forum on Islamic Economics and Finance under the patronage of the government of Malaysia.

PROFESSIONAL EXPERIENCE

In addition to his research, Mr. Siddiqui is an active faculty member at University of Karachi, where he teaches Family laws, Islamic commercial laws and Islamic Jurisprudence. He is known for his engaging teaching style and dedication to student success.

Mr. Siddiqui also brings foreign teaching experience to his academic role, having worked in Pakistan Islamia College, Abu Dhabi, United Arab Emirates, Al-wadi International, Jeddah, Saudi Arabia and International Online University, Doha, Qatar. He has also been visiting faculty at SZABIST, Hamdard University and Dadabouy Institute of Higher Education.

Dr. Sonia A. Syed

1. In-Charge, School of Law, BBA, LLB, M.Phil. /LLM, Ph.D Law
2. Academic Background: Dr. Sonia holds Ph.D in law from the School of Law, University of Karachi.
3. Research Interests and Specialization: Dr. Sonia specializes in corporate and business law. Her Ph.D research focuses on white-collar crime public corruption aiming to [brief description of research goals and significance]. Her current research interests include:
 - Financial investigation
 - Technology and white-collar crime
 - International economic law
4. Professional Experience: Dr. Sonia has more than 14 years of experience as a corporate lawyer. In addition to her research, Dr. Sonia is an active faculty member, where she teaches research methods, corporate law and contract Law.

Ms. Erum Ifrikhar

1. Advocate High Court/ Lecturer/ Student Adviser
2. LLB, M.Phil./LLM, MA (Political Science)
3. Academic Background: Masters in Law (L.L.M / M.Phil.) from University of Karachi, Masters in Political Science and Bachelors in Law (L.L.B) from University of Karachi.
4. Research Interests and Specialization: Specialization in Constitution, her research work based on judicial Activism; Protection of Fundamental Rights and Democracy in Pakistan. Her areas of interest/ expertise are Constitutional Law, Property Laws, Interpretation of Statutes and Legal System of Pakistan. Published number of articles in reputed International and National law journal.
5. Professional Experience: Eighteen years of teaching and research experience as a law faculty. Member of Sindh Bar Council and Karachi Bar Association. Certified Master Trainer of ADR–street law program for law colleges. Focal person law school incubator program for university of Karachi. Worked as a legal advisor of Brand Makers and Dee & Zee Web developers and designing company Also worked as an associate with Abrar Hassan & Co and Umer Qureshi.

Ms. Sana Khan

1. Lecturer, LLB, M.Phil./LLM, Ph.D Law (expected in 2024)
2. Academic Background: Ms. Sana has LLM in International and European Union Law from Vrije University Brussels, Belgium and did her BA Shariah and Law from international Islamic University Islamabad and also did her diploma in Global Development studies from Fontys University, Tilburg, The Netherlands.
3. Research Interests and Specialization: Sana Khan specializes in international law, Human rights law, Islamic law and EU law. Her Ph.D. research is about Pakistan International Human rights obligations under International covenant on civil and political rights (ICCPR). Her current research interests include:
 - Islamic law
 - Human rights law
 - International law
4. Sana Khan has published two articles in reputable law journals and presented his work at two international conferences.
5. Professional Experience: In addition to her research, Sana Khan is an active faculty member at University of Karachi. She also gives visiting lecture about human rights law nationally and international law as well.

Dr. Qamarabadi

1. (Full-time)

RESEARCH FACILITIES**Library**

The School of Law has a Seminar Library, containing extensive collection of law reports, textbooks, and law journals providing law students with research materials.

Computer Lab

Our Computer Lab is equipped with high-speed computers, printers, and scanners. Audio/visual aids, including multimedia options, are utilized during presentations and lectures to enhance learning.

Clinical Legal Education

Recognizing the limitations of traditional legal education, which focuses primarily on reading and classroom lectures with minimal practical exposure, the Faculty of Law has adopted an empirical approach. We place significant emphasis on clinical legal education to develop the methodical temperament of our students.

Traditionally Designed, Modernly Equipped Infrastructure

The School of Law at the University of Karachi is proud to offer its students a modern and inspiring learning environment in the newly constructed H.H. Dr. Syedna Mufaddil Building. The building is not only spacious and contemporary but also environmentally sustainable, nestled amidst lush greenery that offers a serene and conducive atmosphere for academic pursuits. Key Features of the H.H. Dr. Syedna Mufaddil Building are:

1. Spacious and Airy Classrooms
2. Seminar halls
3. Committee Rooms
4. Two Spacious Auditoriums
5. Male and female common rooms with dedicated prayers areas.

AVENUES FOR RESEARCH

Our research programs encourage students to engage in interdisciplinary studies, fostering a deep understanding of legal principles and their applications in real-world scenarios. The faculty's diverse research interests provide a rich environment for collaborative and individual research projects. The LLM degree at the School of Law is equivalent to an M.Phil., offering an advanced level of academic and professional training in various legal disciplines. This equivalency underscores the rigorous academic standards and comprehensive research opportunities available to our students.

EMPOWERING FUTURE LEADERS IN THE LEGAL PROFESSION

The School of Law at the University of Karachi is committed to fostering a supportive and intellectually stimulating environment. Our alumni network spans across the globe, with graduates holding esteemed positions in academia, judiciary, law firms, corporate sectors, and governmental organizations. The School's strong emphasis on ethics, professionalism, and continuous learning ensures that our graduates are well-prepared to tackle contemporary legal challenges and contribute meaningfully to the legal profession and society at large.

By choosing the School of Law at the University of Karachi, prospective students will benefit from state-of-the-art facilities designed to support their legal education journey and prepare them for successful careers in the legal profession

DEPARTMENT OF PUBLIC ADMINISTRATION

INTRODUCTION OF DEPARTMENT

The Department of Public Administration (DPA) prepares students to become exceptional and productive members of the society by serving in government, business and corporate sectors. With its scholarly and integrated programs emphasizing academic excellence and pragmatic approach, this department has gained a distinctive status in the University of Karachi since its inception in 1995.

POST GRADUATE PROGRAMS AVAILABLE

1. M. Phil with Course work / Thesis
2. Ph.D.

OBJECTIVES

The objective of DPA is to study, identify and resolve the issues related to the stewardship of common goods with an interdisciplinary approach. It works at the intersection of multiple constituents of social and management sciences including economics, business, law, political sciences and technology with local and global perspective along with contemporary practices.

AIMS & OBJECTIVE

M. Phil.

Objectives:

1. To develop advanced knowledge and understanding of public administration, drawing on multiple disciplines
2. To foster research skills and analytical thinking in public administration, incorporating insights from fields like public policy, local governance, entrepreneurship, medicine, business, research statistics, and more

Aims:

1. To provide advanced education in public administration, integrating multidisciplinary perspectives
2. To promote research and scholarship in public administration, informed by various fields
3. To develop professionals and researchers who can tackle complex public administration issues

Ph.D.

Objectives:

1. To produce original research contributions to public administration knowledge, drawing on multiple disciplines
2. To develop expertise in public administration research and scholarship, integrating insights from various fields
3. To prepare students for academic and professional leadership roles, equipped to address complex public administration challenges

Aims:

1. To advance knowledge and understanding of public administration through multidisciplinary research
2. To foster intellectual leadership and contributions to public administration scholarship
3. To develop professionals and researchers who can address complex public administration issues, informed by various disciplines

SCOPE

To advance knowledge and understanding of public administration through multidisciplinary research.

REGULAR/PERMANENT FACULTY MEMBERS WITH AREA OF EXPERTISE

1. **Prof. Dr. Khalid M. Iraqi (Vice Chancellor)**
 - Ph.D Thesis Title: Pakistan's Public Policy: Promise, Formulation & Performance.

- Areas of expertise: Politico-Administrative Leadership, Strategy, Public Policy
- 2. **Prof. Dr. Syed Shabib ul Hassan (On lien)**
 - Ph.D Thesis Title: Risk and Return Management: an Analysis of Financial Environment in Pakistan.
 - Areas of expertise: Financial Management
- 3. **Dr. Syed Shahid Zaheer Zaidi**
 - Ph.D Thesis Title: Local Governments Under Military Rule in Pakistan: Challenges and Prospects (A Case Study)
 - Areas of expertise: Local Government, Statistics, Research Methods
- 4. **Dr. Saima Akhtar (In charge)**
 - Ph.D Thesis Title: Strategies and Policies for High Performance Culture in Airlines: A Comparative Studies of PLA, Emirates and Singapore Airlines (1998 - 2006)
 - Areas of expertise: Communication Skills, Management, Marketing, Life Coaching
- 5. **Dr. Ghazal Khawaja Hummayun (Public Policy)**
 - Ph.D Thesis Title: Local Government And Devolution Plan 2000 In Pakistan: A Study Of Ideas And Realities
 - Areas of expertise: HRM, Public Policy, Local Government
- 6. **Dr. Mustafa Hyder (Good Governance)**
 - Ph.D Thesis Title: The Role of Judiciary in Good Governance of Pakistan
 - Areas of expertise: Organizational Behaviour, Leadership and Change Management, Civic Engagement, Sustainable Development Goals
- 7. **Dr. Tehmina Faisal (Entrepreneurship)**
 - Ph.D Thesis Title: Social Entrepreneurship, They Pathway to Social Change: A Case of Pakistan
 - Areas of expertise: HRM, Entrepreneurship, Civic Engagement, Sustainable Development Goals

RESEARCH FACILITIES AVAILABLE

The research philosophy of the faculty and research scholars at DPA is to bridge the gap between academia and practitioners. The faculty of DPA is actively engaged in national and international research projects and has won many research projects / grants. DPA also publishes HEC recognized bi-annual research journal Global Journal for Management and Administrative Sciences.

INFRASTRUCTURE FACILITIES

The state of the art building of DPA encompasses multiple 15 Class Rooms for Lectures, well-equipped Computer Lab, online Resource Room, Conference Hall, and an exclusive Research Lab for M. Phil. / Ph.D. students, APS Seminar Library housing relevant books, theses, journals, periodicals and other academic literature in soft and hard form, and flexible space for students' activities. Specific Research Areas

AVENUES OF RESEARCH

A comprehensive approach includes Research Projects and Collaborations with Industry and Academia, at national and international level. Our M.Phil and Ph.D students are rigorously engaged in teaching and research.

KARACHI UNIVERSITY BUSINESS SCHOOL (Business Administration)

INTRODUCTION

Karachi University Business School (KUBS) is one of the leading institutions in the country, disseminating the quality education and knowledge in the field of business and management very effectively. With the mission of catering the current needs of business and industry, the KUBS has been focusing on developing highly competent, professional and confident personnel, well equipped with latest technology and practices to manage and control all business functions successfully.

Considering the high contribution and involvement of research in business and industry, KUBS has special emphasis on business research activities, both for faculty and students. Almost all regular faculty members have doctorate degree and have been supervising M.Phil. and Ph.D. scholars for many years. Further, all Teaching Associates & Assistants are either Ph.D. or near to submit their doctorate thesis. Additionally, all students of BBA/BS and MBA are required to conduct an actual, practical research on current business issues, as part of their degree program (project/assignment) in one semester.

The KUBS has been serving as an important nursery for developing business scholars for academia and industry. Most of our Ph.D. students are faculty members of different business institutions of the city.

GOALS AND OBJECTIVES

The overall goal of KUBS is to produce qualified, professional and well competent business graduates and research scholars to highly competitive business and academic markets. Key specific objectives include:

1. To equip students with latest practices, procedures and technology exploited in business and industry
2. To enhance the personality traits of students to compete in the market effectively
3. To develop a research culture in all students in general, M.Phil. and Ph.D. scholars in particular.
4. To enable all students to face the challenges of business and industry successfully
5. To build the right image and reputation of KUBS and the University of Karachi among business, industry and markets worldwide.

PROGRAMS OFFERED

Currently, KUBS has been conducting following degree programs in the morning, evening and weekend shifts:

1. Ph.D. in Business Administration
2. M.Phil. in Business Administration
3. MBA – Finance & Investment, Marketing, HRM, Supply Chain Management, Project & Industrial Management and Islamic Banking & Finance
4. BBA / BS – Finance, Marketing, HRM
5. MBA Executive

REGULAR FACULTY MEMBERS

S.No.	Name	Designation	Qualification	Field of Expertise
1.	Dr. Tahir Ali	Professor	Ph.D.	Marketing & Management
2.	Dr. Danish Ahmed Siddiqui	Associate Professor	Ph.D.	Economics and Finance
3.	Dr. Muhammad Asim	Associate Professor	Ph.D.	Marketing & Quantitative methods
4.	Dr. Sohaib uz Zama	Assistant Professor	Ph.D.	Accounting and Finance

5.	Dr. Sarah Anjum	Assistant Professor	Ph.D.	Marketing & Management
6.	Dr. Syeda Nazneen Waseem	Assistant Professor	Ph.D.	HRM and Quantitative Methods
7.	Dr. Omer Ahmed Sheikh	Assistant Professor	Ph.D.	Economics and Finance
8.	Dr. Muhammad Muzammil	Assistant Professor	Ph.D.	Finance and Accounting
9.	Mr. Sheikh Muhammad Fakhr e Alam Siddiqui	Assistant Professor	Ph.D.in Process	Marketing and Management
10.	Mr. Muhammad Hammad	Lecturer	Ph.D. in Process	Accounting and Finance

INFRASTRUCTURE AND RESEARCH FACILITIES

KUBS is well equipped with adequate infrastructure. Suitable classrooms, with necessary fixture and furniture along with technological support are available. Computer Labs with internet and digital library facilities are present to cater the needs of all students in general, research scholars in particular.

DEPARTMENT OF COMMERCE

INTRODUCTION OF THE DEPARTMENT

The Department of Commerce University of Karachi has been imparting quality research oriented business education since 1974. In order to keep pace with the dynamic world of business and business education, the department of Commerce has brought a change in this curriculum which is in compliance with the courses which are demanded in the business world. The students are capable of acquiring good jobs both at national and international level.

CAREER OPPORTUNITIES

The mission of the department of Commerce is to equip students with analytical tools required for business management in the contemporary competitive world and to provide students with opportunities to explore new business vistas.

POSTGRADUATE PROGRAMS AVAILABLE

1. MS / M.Phil.
2. Ph.D.

PROGRAM'S OBJECTIVE

MS / M.Phil

1. To develop advanced research skills and expertise in a specific field.
2. To contribute original knowledge to the field through research.
3. To develop critical thinking, analytical, and problem-solving skills.
4. To enhance intellectual and scholarly abilities.
5. To prepare for careers in academia, research, or industry.

Ph.D:

1. To create original and significant contributions to knowledge in the field.
2. To develop expertise in research design, methodology, and implementation.
3. To demonstrate intellectual leadership and innovation.
4. To develop communication and presentation skills for disseminating research findings.
5. To prepare for careers in academia, research, industry, or leadership positions.

AIMS AND SCOPE OF THE PROGRAM

Aims of MS / M.Phil.

1. To provide advanced research training and education in a specific field.
2. To develop critical thinking, analytical, and problem-solving skills.
3. To foster intellectual curiosity and creativity.
4. To prepare students for careers in academia, research, or industry.

Aims of Ph.D.

1. To create original and significant contributions to knowledge in the field.
2. To develop expertise in research design, methodology, and implementation.
3. To demonstrate intellectual leadership and innovation.
4. To prepare students for careers in academia, research, or leadership positions.

Scope of MS / M.Phil.

1. Advanced research training.
2. Specialized coursework.
3. Original research project.
4. Contribution to the field.

Scope of Ph.D.

1. Original and significant research contributions.
2. Advanced research methodology and design.
3. Expertise in the field.
4. Leadership and innovation.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION/EXPERTISE:**Assistant Professors:**

- | | |
|--|-----------|
| 1. Dr. Zaeema Asrar Mohiuddin
In-Charge | Marketing |
| 2. Dr. Hasan Raza | Finance |
| 3. Dr. Sadaf Mustafa | Finance |
| 4. Mr. M.Arfeen Ahmed Siddiqui | Finance |

Lecturers

- | | |
|--------------------------|---------|
| 1. Ms. Ayesha Fareed | Finance |
| 2. Mr. M.Farrukh Aslam | Finance |
| 3. Ms. Rukhsinda Begum | Finance |
| 4. Ms. Syeda Kanza Nisar | Finance |

RESEARCH FACILITIES AVAILABLE

1. Library
2. Computer Lab
3. Auditorium

INFRASTRUCTURE

1. Seven (07) Classrooms

AVENUES FOR RESEARCH

When pursuing a Ph.D. in Commerce, students can choose from various specializations depending on their research interests, career goals and the focus their doctoral studies. Specializations in Ph.D. programs in Commerce may include but are not limited to:

1. Accounting: Focuses on advanced topics in financial accounting, managerial accounting, auditing, taxation, accounting systems, corporate law, accounting theory and applications.
2. Finance: Explores advanced concepts in corporate finance, investments, financial markets, banking, and financial management financial Institutions, international banking, working capital management.
3. Marketing: Involves in-depth research into consumer behavior, marketing strategy, advertising branding, and market research, marketing management, advertising and promotion, global marketing trends and brand management.
4. Management: Concentrates on organizational behavior, talent management, production management quantitative techniques in management, management of economics, employee relations, strategic HRM, and related topics.

FACULTY OF MEDICINE

MS (SURGERY) / MD (MEDICINE)

INTRODUCTION

The Faculty of Medicine, University of Karachi is a prestigious and well known in the country offering the major and well known postgraduate programs to the young medical fraternity to fulfill the current need of specialist and academicians in both basic and clinical sciences areas of medical and dentistry sections. Hence the Faculty of Medicine is contributing towards the national and international demands of the specialist medical professionals.

Looking at the statistics of the available medical professionals in the country, it is evident that it is 1.084 Physicians per 1,000 people in Pakistan (with almost 1:1 male female ratio) which is a bit lesser than the W.H.O recommendations. Adding to this, only 65% of the females come out for practicing the profession resulting in the scarcity of health care work force, further to add this the high cost of post-graduation program in private sector, the public sector universities remain as the only hope for the masses of medical professionals to materialize their dreams and to contribute towards the high national and international demands of the profession.

The faculty of medicine is primarily offering the postgraduate degree programs in the disciplines of Basic Medical Sciences, Clinical sciences and Dental Sciences and thus contributing in the national demands of specialists and academicians to serve the ailing community and teaching to medical and dental graduates. Besides this, faculty of medicine is also offering the different diploma courses of shorter duration in the same disciplines. In this context the major teaching institutions and tertiary care hospitals of repute have been affiliated to this faculty and hence to the University of Karachi.

POSTGRADUATE PROGRAMS

The MS (Surgery) / MD (Medicine) programs at the Faculty of Medicine, University of Karachi, are structured degree programs. They involve a semester-based examination system and include rigorous clinical training at hospital sites.

Faculty of Medicine, University of Karachi offers admission in MS / MD program in the following Specialties:

Doctor of Medicine (MD)

- *Anesthesiology*
- *Community Medicine*
- *General Medicine*
- *Nephrology*
- *Neurology*
- *Chest Medicine*
(Pulmonology)
- *Cardiology*
- *Psychiatry*
- *Gastroenterology*

Master of Surgery (MS)

- *General Surgery*
- *Gynecology & Obstetrics*
- *Ophthalmology*
- *Orthopedic Surgery*
- *Otorhinolaryngology*
- *Urology*
- *Oral & Maxillofacial Surgery*
- *Operative Dentistry*
- *Oral Pathology*

PROGRAMS' OBJECTIVES

Mission of the Postgraduate Program

To prepare the health care professionals in accordance with the international medical standards & valuable research thus to produce the quality academicians, researchers and clinicians in their specialties in the course of their specialized and patient centered training.

Objectives

The four years MD/MS training program yields the capability of the students transformed into being the expert in their relevant specialty so that they become good teachers, researchers and clinicians after completion of their successful training.

MD/MS training make possible a resident to equip with the *formal* as well as *advanced* techniques of learning. The formal training in their relevant specialties includes the history and Physical Examination, Case Presentations, Core Concepts of the subject, Test Interpretation, Diagnostic & Therapeutic Decision Making. Besides the formal training, the students are specially focused to be trained for their Communication Skills and Bioethics Learning of Patient Care, Self-directed Learning, Epitome of Preventive Medicine and Research & Scientific Knowledge pertaining to their discipline through comprehensive training.

AIM & SCOPE

Pursuing an MS or MD degree leads to significant career growth, offering numerous opportunities in healthcare and medical research. Graduates of MD programs are recognized as specialists in their respective fields. MD graduates have career prospects in various domains, including medicine, and pathology, with potential roles as cardiologists, gastroenterologist, psychiatrists, anaesthetists, hospital administrators, general surgeons, physicians, and nephrologists. The average annual salary for an MD graduate ranges between PKR 2,400,000 and PKR 3,500,000.

The Master of Surgery degree is in high demand and provides excellent job opportunities in both the private and government sectors. Opportunities are particularly abundant in the Middle East, where the annual salary ranges from PKR 4,800,000 to PKR 9,600,000. Graduates can be appointed as researchers, consultants, medical coders, cardiac surgeons, paediatric surgeons, general surgeons, ENT surgeons, orthopaedic surgeons, and trauma surgeons.

LIST OF FACULTY OF MEDICINE

S.NO	NAME	DESIGNATION	DEPARTMENT	QUALIFICATION
1.	Dr. Nadeem Baig	Professor	Anatomy	MBBS, M.PHIL, Ph.D
2.	Dr Shazia Fehmi	Assistant Professor	Anatomy	MBBS, M.PHIL
3.	Dr. Syeda Asiya Parveen	Associate Professor	Bio-Chemistry	MBBS, M.PHIL
4.	Dr. Nargis Anjum	Professor	Physiology	MBBS, M.PHIL, Ph.D
5.	Dr. Akbar Mughal	Associate Professor	Physiology	MBBS, M.PHIL Ph.D
6.	Dr. Gul Afshan	Professor	Physiology	MBBS, M.PHIL
7.	Dr. Uzma Tasneem	Associate Professor	Physiology	MBBS M.PHIL
8.	Dr. Roohi Ehsan	Associate Professor	Forensic Medicine	MBBS, DMJ
9.	Dr. Raffat Rasool	Assistant Professor	Forensic Medicine	MBBS, DMJ
10.	Dr. Wasiq Ahmed	Assistant Professor	Forensic Medicine	MBBS, DMJ
11.	Dr. Fareeda Islam	Professor	Pharmacology	MBBS, M.PHIL, P.HD
12.	Dr. Rana Mehmood	Associate Professor	Pharmacology	MBBS, M.PHIL P.HD
13.	Dr. Asia Rehman	Professor	Pharmacology	MBBS, M.PHIL
14.	Dr. Naila Zaheer	Professor	Pathology	MBBS, MCPS, FCPS
15.	Dr. Muhammad Sabir	Professor	Pathology	MBBS, M.PHIL
16.	Dr. Sheikh Sajjad Ahmed	Professor	Pathology	MBBS, MCPS, M.PHIL
17.	Dr. Rafia Anwar Us Salam	Associate Professor	Pathology	MBBS, M.PHIL
18.	Dr. Shahid Zafar	Associate Professor	Pathology	MBBS, M.PHIL
19.	Dr. Farhat Ali Jafri	Professor	Community Medicine/ Research Methodology	MBBS, MPH, C- Clinical Trials (UK)
20.	Dr. Khalid Shafi	Associate Professor	Community Medicine/ Research Methodology	MBBS, MPH, FCPS

21	Dr Gaiti Ara	Assistant Professor	Community Medicine/ Research Methodology	MBBS, FCPS
22	Dr Zohra Jabeen	Assistant Professor	Community Medicine/ Research Methodology	MBBS , MSPH
23	Dr.Sarwat Sultana	Assistant Professor	Community Medicine/ Research Methodology	MBBS,MCPS
24	Dr. Qaiser Jamal	Professor	Medicine	MBBS,FCPS
25	Dr.Aneela Altaf Kudwai	Professor	Medicine	MBBS, FCPS
26	Dr.Shaista Ahmed	Associate Professo	Medicine	MBBS, MCPS,FCPS
27	Dr.Mahnoor Khalil	Assistant Professor	Medicine	MBBS,FCPS
28	Dr.Firasat Waqar Khanzada	Assistant Professor	Medicine	MBBS,FCPS
29	Dr.Muhammad Adil Ramzan	Assistant Professor	Medicine	MBBS,FCPS
30	Dr.Jamal Ara	Assistant Professor	Medicine	MBBS,FCPS
31	Dr. Ayesha Javed	Assistant Professor	Medicine	MBBS,FCPS
32	Dr. Shabnum Shamim	Professor	Gynecology	MBBS, FCPS
33	Dr.Fouzia Ali	Assistant Professor	Gynecology	MBBS,FCPS
34	Dr.Sonia Rashid	Assistant Professor	Gynecology	MBBS,FCPS
35	Dr.Amna Begum	Assistant Professor	Gynecology	MBBS,FCPS
36	Dr.Shabnam Nadeem	Assistant Professor	Gynecology	MBBS,FCPS
37	Dr.Rahila Imtiaz	Assistant Professor	Gynecology	MBBS,FCPS
38	DR. Zubaida Masood	Associate Professor	Gynecology	MBBS,FCPS
39	Dr.Sarwat Sajjad Sethi	Assistant Professor	Gynecology	MBBS,FCPS
40	Dr.Shaista Rashid	Associate Professor	Gynecology	MBBS,FCPS
41	Dr. Asma Iqbal Ansari	Assistant Professor	Gynecology	MBBS, FCPS
42	Dr. Muhammad Jamaluddin	Professor	Surgery	MBBS, FCPS
43	Dr. Rahil Mehmood Ur Rehman	Professor	Surgery	MBBS, FCPS
44	Dr.Muhammad Aurangzeb	Associate Professor	Surgery	MBBS,FCPS
45	Dr.Rizwan Ahmed Khan	Associate Professor	Surgery	MBBS,FCPS
46	Dr.Bushra Tasneem	Assistant Professor	Surgery	MBBS,FCPS
47	Dr. Shakeel Ahmed	Assistant Professor	Surgery	MBBS, FCPS
48	Dr. Bushra Tasneem	Associate Professor	Surgery	MBBS, FCPS
49	Dr. Khalid Ashrafi	Professor	E.N.T	MBBS, FCPS
50	Dr. Ayub Musani	Professor	E.N.T	MBBS, FCPS
51	Dr. Faheem Ahmed Khan	Assistant Professor	E.N.T	MBBS
52	Prof. Dr M. Razzaq Dogar	Professor	ENT	MBBS, FCPS, MCPS, DLO
53	Dr. Amir Jalil	Professor	Orthopedics	MBBS, FCPS
54	Dr.Muhammad Naseem	Associate Professor	Orthopedics	MBBS,FRCS
55	Prof. Akhtar Baig	Professor	Orthopedics	MBBS, FCPS
56	Dr. Sultan Mustafa	Professor	Pediatrics	MBBS, FCPS, MCPS
57	Dr.Ibrahim Shakoor	Associate Professor	Pediatrics	MBBS,FCPS
58	Dr. Shaheen Masood	Assistant Professor	Pediatrics	MBBS, FCPS

59	Dr. Riffat Ghazala	Assistant Professor	Pediatrics	MBBS, FCPS
60	Dr. Atiya Rahman	Assistant Professor	Ophthalmology	MBBS, FCPS
61	Dr. Muhammad Raees	Assistant Professor	Ophthalmology	MBBS, FCPS
62	Dr. Erum Shahid Najeeb	Assistant Professor	Ophthalmology	MBBS, MCPS, FCPS
63	Prof. Dr. Muhammad Iqbal Afridi	Professor BPS	Behavioral Science /Psychiatry	MCPS, FCPS(Pak), FRCP(Ireland), FRCP(Edinburgh), FRCP (London).
64	Prof Imran Bashir Chaudry	Professor	Behavioral Science /Psychiatry	MBBS, MD
65	Dr. Nadeem Iqbal	Assistant Professor	Behavioral Science /Psychiatry	MBBS, FCPS
66	Prof Rozina Karmaliani	Professor	Medical ethics	BSN, Ph.D
67	Prof Tahira Ayuob	Professor	Medical ethics	BDS, FCPS
68	Dr. Nadia Yonus	Associate Professor	Medical ethics	MBBS, MBE, M. PHIL
69	Dr. Mubarak Ali	Professor	Radiology	MBBS, FCPS
70	Dr. Khurum Daniyal	Associate Professor	Nephrology	MBBS, FRCP
71	Dr. Nazish Butt	Associate Professor	Gastroenterology	MBBS, FCPS, MD
72	Dr Kamran Khan	Professor	Pulmonology	MBBS, MCPS
73	Prof. Dr. Navid Rashid Qureshi	Professor	Oral and Maxillofacial Surgery	BDS, FDSRCS(UK) / MSc. (Dental Radiology)
74	Prof. Dr. Syed Mehmood Haider	Professor	Oral & Maxillofacial Surgery	BDS, MSc. FFDRCSI, FDCRCS, FCPS, FDSRCS
75	Dr. Sufyan Ahmed	Associate Professor	Oral And Maxillofacial Surgery	BDS, FCPS
76	Dr. Mehwish Kashif	Associate Professor	Oral And Maxillofacial Surgery	BDS, FCPS
77	Dr Zahid Khan	Professor	Oral And Maxillofacial Surgery	BDS FCPS
78	Prof. Dr. Tahera Ayub	Professor	Oral And Maxillofacial Surgery	BDS, FCPS
79	Prof. Dr. Uzma Zareef	Professor	Oral And Maxillofacial Surgery	BDS, MCPS/MDS
80	Dr Sheeraz Hussain	Professor	Orthodontic	BDS, FCPS
81	Dr. Samia Siraj	Assistant Professor	Orthodontic	BDS, MCPS
82	Dr. Sana Adeeba Islam	Associate Professor	Community Dentistry	BDS, MCPS
83	Dr Muhammad Nasir	Associate Professor	Oral Biology	BDS, MCPS
84	Prof. Dr. Syed Yawar Ali Abidi	Professor	Operative Dentistry	BDS, FCPS
85	Dr Rehan Uddin Qureshi	Associate Professor	Operative Dentistry	BDS, MSC
86	Dr. Tazeen Zehra ,	Assistant Professor	Operative Dentistry/Periodontology	BDS FCPS
87	Dr. Asad Farooq	Associate Professor	Dental Materials	BDS, MDS
88	Dr. Affan Ahmed	Assistant Professor	Dental Material	BDS, MDS

RESEARCH FACILITIES

The students have been trained for principles and practices of research methods in special conjunction with Biostatistics & epidemiology through an academic board of experts under the name of DRC (Departmental Research Committee) whereby the students are not only evaluated but provided with guidance of conducting the research for preparing their synopsis and thesis of their chosen topic.

Institutional Review Board (IRB) is constituted by the Office of the Dean, Faculty of Medicine. The IRB's role is to facilitate student research is pivotal in fostering a culture of ethical research practices from the early stages of a researcher's career. By providing education, guidance, streamlined processes, and supportive feedback, IRB helps students navigate the complexities of ethical research, ensuring their projects are conducted responsibly and in compliance with applicable standards and regulations. This support not only protects research participants but also empowers students to become conscientious and skilled researchers.

INFRASTRUCTURE

The postgraduate degree program of MS/MD in various specialties and subspecialties at the Faculty of Medicine, University of Karachi, is a structured program that includes both Basic Medical Sciences education and Clinical training components. The clinical training is conducted at the university's affiliated institutions and clinical facilities.

In addition to the clinical and dental departments and faculties of Karachi Medical and Dental College (KMDC) / Abbasi Shaheed Hospital (ASH) and Liaquat College of Medicine and Dentistry (LCMD), these institutions also provide their resources to the Faculty of Medicine, University of Karachi. The affiliated institutions include:

1. Karachi Institute of Heart Diseases
2. Jinnah Postgraduate Medical College
3. Civil Hospital Karachi
4. Lyari Medical College & Hospital
5. Karachi University Hospital

These affiliations enhance the training and resources available to postgraduate students, ensuring a comprehensive education and practical experience in their chosen medical fields.

AVENUES FOR RESEARCH

The Faculty of Medicine at the University of Karachi offers a broad array of research avenues encompassing Basic Medical Sciences, Clinical Sciences, Public Health, Medical Education, Medical Humanities, Infectious Diseases, Non-Communicable Diseases, Maternal and Child Health, and Health Care. Each area focuses on advancing understanding and treatment of various medical conditions, improving public health, and enhancing the education and ethical considerations of healthcare professionals. Key disciplines include Anatomy, Physiology, Cardiology, Epidemiology, Molecular Medicine, Virology, Diabetes, Obstetrics, Medicine, highlighting the faculty's commitment to comprehensive medical research and innovation.

Renowned for its research excellence, the Faculty of Medicine at the University of Karachi consistently produces significant contributions to medical science and healthcare. By continually expanding its research focus, the faculty addresses critical health challenges and leverages emerging technologies to revolutionize healthcare delivery and improve patient outcomes.

ANY OTHER RELEVANT INFORMATION

The students have also been facilitated for their capacity building and increasing their understanding of the subject especially in context of Research Methodology, Epidemiology and Biostatistics, communication skills and medical writing through conducting the workshops by inviting the expert and external faculty of these areas.

DEPARTMENT OF PHARMACOGNOSY

INTRODUCTION

"Then eat from all the fruits and follow the ways of your Lord laid down [for you]." There emerges from their bellies a drink (honey), varying in colors, in which there is healing for people. Indeed in that is a sign for a people who give thought." (Sūrat l-nahl (The Bees): verse (69))

The Prophet Muhammad (PBUH) said, *"Make use of the two cures: honey and the Qur'an"*. [Ibn Majah & others]

Hazrat Abu Hurairah States – "I have heard from Rasool Allah (PBUH) that there is cure for every disease in black seeds except death and black seeds are Shooneez."

"Let food be thy medicine and medicine be thy food". Hippocrates, the father of modern medicine 2,500 years ago.

The Department of Pharmacognosy was established in 1973 as one of the four departments under the domain of Faculty of Pharmacy. It started off with meager resources and few faculty members. Today it has grown into a well-established department that is the backbone of pharmacy education in the Faculty of Pharmacy and Pharmaceutical Sciences.

Pharmacognosy is the oldest medical science since the days when pharmacognosists were physicians themselves. It is one of the five major areas of pharmaceutical education, an interdisciplinary subject which includes parts of botany, organic chemistry, biochemistry, pharmacology, zoology, anthropology, geography, ethnomedicine, nutraceuticals, cosmeceuticals, nutricosmetics, biotechnology etc. This discipline covers a wide range of topics such as sources of natural products, their chemical composition, medicinal value, distribution, ethno-pharmacological profile, biosynthetic pathways, isolation, evaluation, quality control, standardization etc. Pharmacognosy learning encompasses identification of plant drugs, phytochemistry, extraction methods, separation, purification, identification and structure elucidation of natural products, phytotherapy, biologics, allergens, nutraceuticals, cosmeceutical, nutricosmetics, herbal nanomedicines and toxic manifestation of natural products.

In Pakistan there has always been a growing interest in medicinal plants and their active chemical constituents are utilized in the modern system of medicine – allopathy and also in traditional system of medicine – Unani and homeopathic systems. Traditionally, most of the population depends on natural sources of medicine. A vast majority does not have access to modern medicine hence there is a huge scope to develop cheap drugs from locally available resources to meet the needs of the country.

The diversity of landscapes and climates in Pakistan provides for hundreds of plant and animal species other than huge marine resources that have largely remained untapped.

POST GRADUATES PROGRAMS AVAILABLE

1. M.Phil
2. Ph.D

AIMS AND SCOPE OF PROGRAMS

To fulfill the national requirement, Department of Pharmacognosy strives to provide professionals who could work in any Pharmaceutical industry or can be instrumental in modernizing herbal drug industry on scientific bases. Our graduates have an array of opportunities to work. They are developed to work in pharmaceutical, phytopharmaceutical, herbal nanomedicines, nutraceutical, nutricosmetics, cosmeceutical industries, hospitals, research and development, academia, regulatory bodies etc.

PROGRAM OBJECTIVE (M. PHIL)

The M. Phil. program is based on 08 course of 24 cr.hrs. including compulsory course i.e. Biostatistics and Computer Application in Pharmacy, Basics of Research Methodology the rest are elective (Advances in Spectroscopic Techniques, Structure Elucidation of Natural Products, Pharmacoeconomics, Analytical Pharmacognosy, Natural Toxicants, Advances in Pharmacognosy, Advance Phytopharmaceutical Analysis, Cell Biotechnology, Industrial Pharmacognosy, Biosynthesis of Natural Products, Phytochemistry and

Standardization of Phytomedicine). The outcome program is to enable the student to

- Have the knowledge of current advances in their field including spectroscopic techniques.
- Understand various biosynthetic pathway in plants leading to the production of medicinally important plant metabolites
- Understand how spectroscopic techniques are used to elucidate chemical structure of natural products
- To understand the pharmacoeconomic by identifying measuring and comparing the cost and consequences of drugs and services.
- Consults various resources to analyze standardize phytomedicines
- Have an understanding about the commercial and industrial application of natural products
- Recognize natural toxicants, their related toxicities and their medicinal value
- Produce and analyze data acquired during a course of study
- Utilize computer application to obtain current information about a topic, latest development in the field, utilize soft-wares and various computer programs to presents their work
- To educated about the Basics of Research Methodology.
- Work in both modern and herbal pharmaceutical organizations, nutraceutical, nutricosmetics and cosmeceuticals industries, ministry of health, drug regulatory bodies, small enterprises, biotechnology organizations, various drug regulatory bodies, research organizations and teaching institutions.

PROGRAM OBJECTIVE (PH.D)

The Ph. D. program is based on 06 courses of 18 cr.hrs. of which “Research Methodology” is mandatory and the rest are elective. The program is designed so the graduate is able to:

- Conceive, develop and execute a study plan that has relevance to current issues regarding health science
- Gather information from carious study tools and resources specially the information technology
- Carryout bioequivalence studies, utilize the internationally recognized guidelines, understand regulations and aspects of such studies.
- Have deep understanding of the agriculture of medicinal plants: how various practices affect their quality, sustainable methods of utilization etc.
- Understand and utilize various marine resources for drug development.
- Have in-depth knowledge of clinical uses of natural resources.
- To understand the concept of Integrative medicines.
- To explore the recent advancement in the pharmacoeconomics with reference to natural drugs.
- Understanding towards the current development of biopharmaceuticals and their application for the betterment of mankind.
- Present their work in the form of posters, research articles and seminars.
- Publish thesis and research work in national and international journals of repute.
- Work in both modern and herbal pharmaceutical organization, nutraceutical and cosmeceuticals industries, ministry of health, drug regulatory bodies, small enterprises, biotechnology organizations, various drug regulatory bodies, research organization and teaching institutions.

PHARMACOGNOSY FACULTY

Name	<i>Dr. Iqbal Azhar</i>
Designation	Professor and Chairman
Qualification	Ph.D., M. Pharm., B. Pharm. (University of Karachi)
Year of Association	1994
Subject of Interest	Natural Products
Area of Research Interest	Nutraceuticals, Cosmeceuticals, Phytopharmaceuticals
Email	iazhar@uok.edu.pk, iqbal_zhr@yahoo.com
Phone Number	(+92-21)99261300-7 Ext. 2202

Name	Dr. Muhammad Mohtasheemul Hasan
Designation	Associate Professor
Qualification	Ph.D., M.Phil., B.Pharm (University of Karachi)
Year of Association	2006
Subject of Interest	Pharmacognosy / Pharmaceutical Biology
Area of Research Interest	Phytopharmacology, Phytochemistry
Email	mohassan@uok.edu.pk
Phone Number	(+92-21) 99261300-7, Ext. 2202

Name	Ms. Farah Mazhar
Designation	Assistant Professor
Qualification	M. Phil, B. Pharm. (University of Karachi)
Year of Association	2000
Subject of Interest	Pharmacognosy, Natural Products
Area of Research Interest	Pharmacognosy, Terrestrial and Marine Natural Products, Evaluation and Standardization of Herbal Drugs, Bioassays
Email	famazhar@uok.edu.pk
Phone Number	(+92-21) 99261300-7 Ext. 2202

Name	Dr. Maryam Ahmed
Designation	Assistant Professor
Qualification	Ph.D., M. Pharm., B. Pharm. (University of Karachi)
Year of Association	2018
Subject of Interest	Pharmacognosy, Natural Products
Area of Research Interest	Phytomedicine, Phytochemical, Nutraceutical, Phytopharmaceutical, Phytopharmacology
Email	dr.maryamahmd@gmail.com
Phone Number	(+92-21) 99261300-7 Ext. 2202

Name	Dr. Safia Abidi
Designation	Assistant Professor
Qualification	Ph. D., M. Pharm., B. Pharm. (University of Karachi)
Year of Association	2013
Subject of Interest	Pharmacognosy
Area of Research Interest	Natural Products
Email	safiaabidi@yahoo.com
Phone Number	(+92-21) 99261300-7 Ext. 2202

Name	Dr. Salman Ahmed
Designation	Assistant Professor
Qualification	Ph. D. M.Phil., B.Pharm. (University of Karachi)
Year of Association	2013
Subject of Interest	Pharmacognosy
Area of Research Interest	Phytomedicine, Phytopharmacology, Medicinal Plants
Email	salmanahmed@uok.edu.pk
Phone Number	(+92-21) 99261300-7, Ext. 2202

BOARD OF STUDIES

1. **Dr. Iqbal Azhar**
Professor & Chairman
2. **Dr. Muhammad Mohtasheem ul Hasan**
Associate Professor
3. **Dr. Maryam Ahmed**
Assistant Professor
4. **Mr. Salman Ahmed**
Lecturer
5. **Dr. M. Izhar Hussain**
Director,
Institute of Health and Management Sciences
Dow University of Health Sciences
Karachi
6. **Dr. Zafar Alam Mahmood**
Country Manager, Pakistan, Gulf and Middle East Region
Colorcon Limited-UK,
Flagship House, Victory Way, Crossways
Dartford, Kent, DA2 6QD, UK.
A-178 Block-2 Gulistan e Johar Karachi
7. **Prof. Dr. Huma Sharif**
Institute of Pharmaceutical Sciences
Jinnah Sindh Medical University
Karachi.

RESEARCH FACILITIES

The following well equipped labs and facilities are available for the graduate students.

1. Phytochemical Analysis Lab.
2. Pharmacological / Biological Research Lab.
3. Quality Control and Standardization Lab.
4. Herbal Nanoformulations Lab.
5. Store
6. Extraction Rooms

INFRASTRUCTURE

1. Research Institute of Pharmaceutical Sciences
2. Undergraduate labs and offices in old block and new block of Faculty of Pharmacy and Pharmaceutical Sciences.
3. Conference Room New Block

AVENUES OF RESEARCH

Phytochemistry, phytotherapy, nutraceuticals, cosmeceutical, nutricosmetics, herbal nanomedicines, biopharmaceuticals, marine drugs.

DEPARTMENT OF PHARMACOLOGY

INTRODUCTION

The Department of Pharmacology is one of the important Departments of the Faculty of Pharmacy since 1973 not only sharing a major load at the under graduate level but also preparing students to fill the gap by suitably qualified manpower in the field of Pharmacy at Post graduate levels. Department of Pharmacology has so far produced the highest number of Ph.Ds in the faculty of Pharmacy and a large number of M. Phil and M. Pharm. The Department is composed of highly competent primary faculty, research faculty and technical support staff.

DEGREE PROGRAMS OFFERED

1. M.Phil (Morning Program)
2. Ph.D (Morning Program)

PROGRAMS' OBJECTIVES

The major academic objectives of the Department are to facilitate Basic and Applied Research, educate under graduate, graduate and professional students in various disciplines and provide academic excellence.

AIMS AND SCOPE OF THE PROGRAM

After completion of M.Phil/ Ph.D in the field of Pharmacology successful, candidates will be able to perform duties in following areas.

1. Research
2. Education
3. Hospital Pharmacist
4. Clinical Pharmacist
5. Pharmaceutical Industry
6. Drug testing laboratories

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION / EXPERTISE

Dr. Afshan Siddiq

B. Pharm., M. Phil, Ph.D (Kar)

Associate Professor & Chairperson

Year of association, 2006

Expertise: Biochemical Pharmacology, Systemic Pharmacology, Biochemistry, Toxicology

Dr. Syeda Afroz

B. Pharm., M. Phil., Ph.D (Kar)

Associate Professor

Year of association, 2006

Expertise: General Pharmacology, Neuropharmacology

Dr. Sadia Ghousia Baig

B. Pharm., M. Phil., Ph.D (Kar)

Assistant Professor

Year of association, 2006

Expertise: Biochemistry, Neuropharmacology, Biochemical Pharmacology

Dr. Nuzhat Sultana

B. Pharm., M. Phil., Ph.D (Kar)

Assistant Professor

Year of association, 2012

Expertise: Toxicology, Neurobehavioral Pharmacology, Biochemical, hematological and toxicological screening

Dr. Azra Riaz

B. Pharm., M. Phil., Ph.D (Kar)

Assistant Professor

Year of association, 2012

Expertise: Biochemical and hematological parameters coagulation, inflammation and Neurpharmacology

Dr. Saira Saeed Khan (On FSL)

B. Pharm., M. Phil. Ph.D (Kar)

Assistant Professor

Year of association, 2012

Expertise: Neurpharmacology, Biochemical and hematological studies.

Dr. Shadab Ahmed

B. Pharm., M. Phil. Ph.D (Kar)

Assistant Professor

Year of association, 2012

Expertise: Neuropharmacology, Physiology, Pathology, Biochemistry, Pharmacology

Dr. Sana Sarfaraz

B. Pharm., M. Phil. Ph.D (Kar)

Assistant Professor

Year of association, 2019

Expertise: Clinical pharmacology, toxicology

Dr. Adnan Iqbal

B. Pharm., M. Phil. Ph.D (Kar)

Lecturer.

Year of association, 2019

Expertise: Pharmacology and neurosciences

RESEARCH FACILITIES AVAILABLE (LIBRARY, LABORATORY, COMPUTER LAB)

1. Two Separate Research lab have been assigned in Research Institute of Pharmaceutical Sciences (RIPS) Faculty of Pharmacy & Pharmaceutical Sciences, University of Karachi for students which has basic advanced equipment and instruments required for research work.
2. A Seminar Library in Faculty of Pharmacy & Pharmaceutical Sciences, University of Karachi is well equipped. Latest pharmacology books and also computer facility.

INFRASTRUCTURE (OLD & NEW BUILDING)

Buildings		No. of Class Rooms		Seminar Library
-		2		-
No. of Office Room		No. of Staff Room		No. of Store Rooms
2		9		1
Laboratories				
No. of Computer Lab	No. of (RIPS) Research Lab.	No. Electric Lab.	No. of Microprocessor Lab.	Any other
Nil	2	Nil	Nil	Undergraduate Lab-03

AVENUE FOR RESEARCH

1. Neuropharmacology
2. Toxicology
3. Biochemical Pharmacology
4. Cardiac vascular pharmacology
5. Psychopharmacology

DEPARTMENT OF PHARMACEUTICS

INTRODUCTION OF THE DEPARTMENT OF PHARMACEUTICS

The Department of Pharmaceutics is one of the five constituent departments of the Faculty of Pharmacy and Pharmaceutical Sciences. It is the foremost discipline in the pharmaceutical sciences and covers about one-third of structured course work of the total Pharm D curricula, indicating the breadth of the field. Pharmaceutics is a multidisciplinary branch of pharmaceutical science that focuses on the development, design, formulation, and delivery of safe and effective pharmaceutical products. It encompasses a wide range of scientific and technological principles to create medications and dosage forms that optimize the delivery of active pharmaceutical ingredients (APIs) to patients.

The Department of Pharmaceutics at the University of Karachi is comprised of a diverse and dedicated team of individuals who contribute to the academic and operational excellence of the department. With a total of 14 esteemed faculty members and 13 non-teaching staff, the department embodies a collaborative and vibrant environment that fosters innovation, research, and education. The department offers 23 courses out of a total 71 courses being offered by the faculty for the PharmD program. The department also offers postgraduate course-work based MS and research intensive M.Phil and Ph.D programs. To date, the department has produced 125 M.Phil, 187 MPharm, and 67 Ph.D graduates in different disciplines of pharmaceutics.

POSTGRADUATE PROGRAMS AVAILABLE

1. M.Phil
2. Ph.D

PROGRAM'S OBJECTIVES

M.Phil Program Objectives

The M.Phil improves the career opportunities by inculcating the critical thinking and pharmaceutical analytical expertise in formulation design and development. The program prepares intellectual leaders for educational institutions of both public and private sector.

Ph.D Program Objectives

The Ph.D program in Pharmaceutics is the gateway to infinite research opportunities in drug discovery and pharmaceutical product development. The program offers comprehensive learning of nanomedicine, design, development and optimization of novel drug delivery system. Moreover, the research fellow will learn the research proposal writing, execution and finalization of data analysis and inference. The program offers different options to explore and collaborate and provides readiness to teach and conduct pharmaceutical research in every paradigm of pharmaceutical sciences

SCOPE OF THE PROGRAM

The scope of pharmacists is vast and dynamic, encompassing a wide array of roles that have a profound impact on healthcare and the pharmaceutical industry. As integral members of the healthcare team, pharmacists play a pivotal role in ensuring patient safety, accurate medication management, and optimal treatment outcomes. In community settings, pharmacists serve as frontline healthcare providers, offering expert guidance on medication usage, potential side effects, and drug interactions. In hospitals and clinics, they collaborate closely with medical professionals to devise personalized medication regimens, monitor patient responses, and mitigate potential risks. Beyond patient care, pharmacists occupy crucial positions within the pharmaceutical industry. They contribute significantly to drug and product research, development, and testing, striving to innovate and enhance therapeutic options. Pharmacists also engage in quality control, regulatory compliance, and pharmaceutical marketing, ensuring that medications meet the highest standards of safety and efficacy. Moreover, pharmacists excel in academia and research, driving scientific exploration, mentoring aspiring professionals, and advancing pharmaceutical knowledge. With their profound expertise, they also navigate the realms of informatics and technology, optimizing pharmacy information systems and healthcare technologies for improved patient care.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION/EXPERTISE

<u>Name</u>	<u>Designation</u>	<u>Areas of Specialization/Expertise</u>
Dr. Muhammad Harris Shoaib	Professor	Clinical Pharmacokinetics, Formulations development and optimization, Biopharmaceutics, Dosage form design. Simulation based in silico Pharmacokinetics modeling, Advance drug delivery system development.
Dr. S.M. Farid Hasan	Associate Professor & Chairman	Bioavailability, Bioequivalence, Pharmacokinetics, Formulation, Product development, Optimization with emphasis on solubility enhancement of poorly soluble drugs and Novel Drug Delivery System
Dr. Rabia Ismail Yousuf	Associate Professor	Pharmacokinetics, Formulations development and Optimization, Biopharmaceutics, Clinical Pharmacy. Simulation based in silico Pharmacokinetics modeling, Advance drug delivery system development.
Dr. Iyad Naeem Muhammad	Associate Professor	Pharmacokinetics, Formulation Development, Pharmaceutical Biotechnology, Pharmaceutical Microbiology
Dr. Farya Zafar	Associate Professor	Formulation Development, Pharmacokinetics, Biopharmaceutics and Novel Drug Delivery System.
Dr. Sabahat Jabeen	Assistant Professor	Formulation development, Physical pharmacy, Pharmacokinetics and drug stability.
Mrs. Rehana Saeed	Assistant Professor	Pharmacokinetics, Formulation Development, Hospital Pharmacy and Clinical Pharmacy
Dr. Faaiza Qazi	Assistant Professor	Formulation development and optimization, clinical pharmacy, pharmacokinetics, pharmacy practice, drug/ drug product stability. Designing and development of advanced drug delivery system.
Dr. Farrukh Rafiq Ahmed	Assistant Professor	Formulation designing, Nanomedicine, Population Pharmacokinetics, Advanced drug delivery system, simulation based in silico modeling for drug /drug product development.
Dr. Kamran Ahmed	Assistant Professor	Formulation and development of drug delivery system, Physical Pharmacy, Advanced drug delivery system.
Dr. Sadaf Farooqui	Lecturer	Novel Drug Delivery system, Drug Delivery Disposition and dynamics, Pharmaceutical Microbiology, Quality use of medicine.
Dr. Tazeen Hussain	Lecturer	Formulation Development, Novel Drug Delivery System, Pharmaceutical Microbiology, Pharmacy Practice.

RESEARCH FACILITIES AVAILABLE (LIBRARY, LABORATORY, COMPUTER LAB.) AND INFRASTRUCTURE

All the laboratories are well equipped with advanced and sophisticated industrial and analytical equipment. The classrooms are also furnished with modern facilities such HD enabled multimedia and audio systems. Following, are the library and laboratory facilities at the Department of Pharmaceutics:

1. Seminar Library
2. Natoli tableting and industrial laboratory.
3. Formulation development laboratory.
4. Pharmaceutical technology laboratory.
5. Industrial pharmacy laboratory.

6. Pharmaceutical analysis/bioanalysis laboratory.
7. Physical pharmacy and biopharmaceutics laboratory.
8. Pharmaceutical microbiology laboratory.
9. Computer laboratory.
10. Simulation laboratory (under development).

AVENUES FOR RESEARCH

The Department of Pharmaceutics is actively involved in the following research areas.

- Solid-state pharmaceutics
- Pharmaceutical microbiology
- Microneedles and transdermal drug delivery systems.
- Conventional and innovative modified-release oral formulations (osmotic-controlled, pellets and gastro-retentive tablets etc.) and immediate release drug delivery system (oral films, mouth dissolving tablets, orally disintegrating tablets, mouth dissolving pellets etc.)
- Intranasal drug delivery systems
- Biopharmaceutics and pharmacokinetics (bioavailability and bioequivalence studies)
- Population and clinical pharmacokinetics, and physiologic-based pharmacokinetics (PBPK)

DEPARTMENT OF PHARMACEUTICAL CHEMISTRY

INTRODUCTION

Department of Pharmaceutical Chemistry is one of the most important academic units of the Faculty of Pharmacy and Pharmaceutical Sciences. Department is dedicated to provide sound education in the field of Pharmaceutical Chemistry and deals with the physical and chemical aspects of Pharmaceuticals. Courses offer by the Department are based on organic chemistry, physical chemistry, analytical, instrumental and medicinal chemistry, with a focus on drug discovery and development. Currently, the department has six Professors, and three Assistant Professors. All are Ph.D. holders and actively engaged in teaching and research. The department aims to promote learning environment and training of students in the advanced specialized areas of the field.

FACILITIES

The department is comprised of two large buildings having spacious classrooms and laboratories. The curriculum focuses on practical experience supported by highly qualified and experienced professionals with well-equipped labs. Presently, four working labs are dealing with synthesis, analytical, instrumental and experimental work for postgraduate students. Three full-fledged research labs are operating in the Department and Research Institute of Pharmaceutical Sciences (RIPS) providing facilities regarding drug designing, drug development, drug stability, and analysis & method development.

SPECIFIC RESEARCH AREAS

The department has a rich history of excellence in research. Experienced and actively growing research groups are working in the disciplines of drug designing and development, product and process development of drug, analytical aspects of drug, nanotechnology, kinetic studies and drug stability. Researchers are working with different research institutes, industries and organizations which promote many areas of collaborative research studies. The department has the honor of producing the first M. Pharm, the first M. Phil., first Ph.D. of this Faculty and up till now 60 Ph. Ds and more than 170 M. Pharms and M. Phils have been produced by the Department.

CAREER OPPORTUNITIES

Graduate program of the department is dedicated to educate and train the students to produce active, skilled and responsible future pharmacist for academia, industry, hospitals and government setups. They can go in production, quality control, marketing and drugs regulatory department of different pharmaceutical industries. They can also work as part of a healthcare team comprising doctors, pharmacists and paramedical staff. They can play an effective role as drug experts, able to select best treatments for managing patient care. Apart from these, there is a number of opportunities available in administrative and general management sites in government health department. Pharmacists with M. Phil. and or Ph.D. degree can also join teaching and research organizations.

FACULTY MEMBERS

Professors

1. Dr. Faiyaz H. M.Vaid (Chairperson)
2. Dr. Mansoor Ahmed
3. Dr. Nousheen Mushtaq
4. Dr. Sohail Hassan
5. Dr. Asia Naz
6. Dr. Safila Sharif

Assistant Professors

1. Dr. Shazia Haider (T)
2. Dr. Rubina Siddiqui (T)
3. Dr. Urooj Nazim (T)

PROGRAMS OFFERED

1. M. Phil.
2. Ph.D.

DEPARTMENT OF PHARMACY PRACTICE

INTRODUCTION

The Department of Pharmacy Practice is a newly launched department of the Faculty of Pharmacy & Pharmaceutical Sciences, University of Karachi with a vision of strengthening the applied aspects of Pharmacy in order to meet the requirements of developing Pharmaceutical care plans to monitor patient's drug therapy, educate patients and health care professionals service of Pharmaceutical care, medication therapy management, community Pharmacy service, and certain specialized service in pediatrics, oncology, infectious diseases, cardiology, drug information service, etc. by providing trained health-care system pharmacists maintain drug distribution system total Parenteral Nutrition and to solve drug related issues, etc.

Under the guidance of esteemed and seasoned faculty, the Department of Pharmacy Practice is dedicated to producing skilled pharmacists and advancing the field through innovative research in various aspects of pharmacy practice. Our vision includes offering specialized programs that foster professional growth and excellence in the future.

Department of Pharmacy Practice executes and certifies 200 hours clinical clerkship program in different renowned tertiary care general and specialist hospitals to meet the Pharm-D. Curriculum and post graduate research requirements. In addition to this, one month training in a community pharmacy and Drug Information Center under the supervision of a registered pharmacist, is also part of the professional curricula.

DEGREE PROGRAM OFFERED

The Department of Pharmacy Practice is proud to offer a postgraduate degree program, M. Phil. in Pharmacy Practice, which has received approval from the Higher Education Commission (HEC) and meets all the required standards and criteria, ensuring a high-quality education and research experience for our students.

PROGRAMS' OBJECTIVES

The M.Phil. program in Pharmacy Practice is designed to pioneer innovative research in Clinical Pharmacy, Hospital Pharmacy, Community Pharmacy, Pharmaceutical Marketing, and Public Health. The department cores to develop and integrate educational and service models in pharmacy practice, bridging academia and real-world settings to cultivate leadership and expertise, providing visionary guidance to the profession and community.

AIMS AND SCOPE OF THE PROGRAM

M.Phil. in Pharmacy Practice at Faculty of Pharmacy and Pharmaceutical Sciences, University of Karachi is aimed to produce leaders for integrating education, research and practice by virtue of innovative and collaborative ventures within a diversified community and clinical health care systems across the country.

Pharmacy Practice is the demand of the current era in healthcare system. In hospitals, pharmacist take care of the patients by clinical interventions and rationalizing therapy, monitoring the outcomes and strategizing the clinical input to reduce adverse effects and report those adverse effects. Furthermore, via patient education and counseling, the pharmacist practitioner provides the social and behavioral contingency to the patients and general public.

The scope of M.Phil. Pharmacy practice is extending from Healthcare system, through Academia, and Regulatory Affairs of pharmacy practice to pharmaceutical research in Pharmacovigilance and Pharmacoepidemiology. These healthcare services as regarded as a pillar in healthcare system, hence, services to the society. In every sphere of healthcare system, these pharmacists are employed, and their services are procured.

Internationally, the widened scope of the pharmacy practice has made the post graduate studies in pharmacy practice a demand, M.Phil in pharmacy practice are procured in several countries and regions like in Middle East and USA etc.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION

1. **Prof. Dr. Harris Shoaib,**
Dean Faculty of Pharmacy and Pharmaceutical Sciences, being the In-charge of the Department.
mhshoaib@uok.edu.pk
Area of specialization are: Pharmaceutical care, Pharmacoeconomics, Clinical Pharmacokinetics, Pharmaceutical microbiology, Clinical Pharmacy
2. **Dr. Iyad Naeem Muhammad,**
Associate Professor and Coordinator
iyadnaeem@uok.edu.pk
Area of Specialization are: Pharmaceutical Care, Antimicrobial resistance, Nosocomial infections, Hospital Pharmacy Practices, Community Pharmacy
3. **Dr. Rabia Ismail Yousuf,**
Associate Professor
riyousuf@uok.edu.pk
Area of specialization are: Pharmaceutical care, Adverse Drug Reactions, Therapeutic Drug Monitoring, Clinical Pharmacy

RESEARCH FACILITIES

1. **Library:** The department boasts state-of-the-art facilities, complemented by a well-stocked faculty seminar library, featuring an extensive collection of up-to-date subject relevant books and journals. This resourceful library ensures that the academic and research requirements of the students and faculty are duly met, providing a conducive environment for learning and scholarly pursuits.
2. **Computer Facilities:** Sufficient computer facilities are available with a stable internet facility for the students.
3. **Alignment with Hospitals for post graduate research:** The department is aligned with renowned hospitals of the city for post graduate research projects. This is in collaboration with renowned health care providers, physicians and consultants onboard.

AVENUES FOR RESEARCH

For M.Phil Pharmacy practice following are some of the vast avenues to explore:

1. Medication safety and efficacy: Investigating optimal medication use, adverse drug reactions, and drug interactions.
2. Pharmacoeconomics and outcomes research: Evaluating the cost-effectiveness and health outcomes of medications and pharmacy services.
3. Pharmacy practice models and innovation: Exploring new models of care, such as community pharmacy-based services and innovative dispensing technologies.
4. Patient-centered care and counseling: Investigating effective communication strategies and patient education methods.
5. Disease state management: Developing and evaluating pharmacy-based interventions for chronic diseases, such as diabetes and hypertension.
6. Public health and health promotion: Examining pharmacists' roles in preventive care, health promotion, and disease prevention and its effects on Health related quality of life.
7. Pharmacy education and assessment: Investigating innovative teaching methods and assessment tools for pharmacy students.
8. Health services research: Analyzing healthcare systems, access, and equity, including pharmacists' contributions.
9. Digital health and technology: Exploring the impact of digital tools, such as telepharmacy and artificial intelligence, on pharmacy practice.
10. Policy and advocacy: Investigating the impact of policy changes and advocacy efforts on pharmacy practice and patient care.

DEPARTMENT OF AGRICULTURE & AGRIBUSINESS MANAGEMENT

INTRODUCTION

The Department of Agriculture and Agribusiness Management was established to offer comprehensive education and research opportunities in agriculture. It began with BS program in Agriculture in 2004, became a full-fledged department in 2006, and was renamed as Department of Agriculture and Agribusiness Management in 2008. The department offers specializations in Agribusiness Management, Plant Protection, and Plant Pathology. Agribusiness is a growing field in Pakistan, combining agricultural science with business management. This program helps students understand the economic principles of agribusiness, developing skills for managing agricultural enterprises. Specializations in plant protection and plant pathology teach students about managing pests and diseases in crops, promoting eco-friendly methods. The department is dedicated to preparing students for careers in agriculture, providing them with the knowledge and skills needed to enhance crop yields and support the agricultural economy.

DEGREE PROGRAM AVAILABLE

1. M.Phil.
2. Ph.D.

PROGRAM OBJECTIVES (MPhil in AGRICULTURE)

1. Enhance students' knowledge in **Agribusiness Management, Plant Protection and Plant Pathology**.
2. Develop research skills to solve agricultural issues.
3. Encourage innovative thinking for sustainable agriculture.
4. Prepare graduates for leadership roles in academia, industry, and public sector.

Program Aim

1. To provide advanced understanding and skills in agriculture, enabling students to contribute to research and development in sustainable agriculture and agribusiness.

Program Scope

1. Advanced courses in agribusiness management, plant protection, and plant pathology.
2. Research projects addressing real agricultural problems.
3. Training in modern research methods and techniques.
4. Opportunities for interdisciplinary collaboration.
5. Development of communication and leadership skills for career advancement.

PROGRAM OBJECTIVES (Ph.D. IN AGRICULTURE)

1. Provide in depth knowledge and research skills in **Agribusiness Management, Plant Protection and Plant Pathology**.
2. Encourage innovative research for agricultural advancement.
3. Develop independent researchers capable of high-quality studies.
4. Prepare graduates for leadership roles in academia, industry, and government.

Program Aim

1. To produce skilled researchers and experts who can advance agricultural science, develop sustainable solutions, and lead in agricultural technology and policy.

Program Scope

1. Advanced courses and seminars in agriculture and related fields.
2. Original research leading to a significant dissertation.
3. Training in advanced research methods and data analysis.
4. Opportunities to publish research and present at conferences/seminars *etc.*
5. Collaboration with faculty and industry experts.
6. Development of teaching and mentoring skills for academic careers.

FACULTY MEMBERS

S.No.	Name	Designation	Specialization
1.	Prof. Dr. Saleem Shahzad	Adjunct Professor	Plant Pathology
2.	Dr. Saboohi Raza	Associate Professor	Agriculture Biotechnology & Molecular Plant Pathology
3.	Dr. M. Faheem Akbar	Chairman & Associate Professor	Agriculture Entomology
4.	Dr. Rana Mazhar Abbas	Associate Professor	Horticulture
5.	Dr. Amjad Sultan	Assistant Professor	Pest Management & Biological Control
6.	Dr. Shagufta Sehar	Assistant Professor	Plant Protection & Agriculture Biotechnology
7.	Dr. Muhammad Shahid	Assistant Professor	Agriculture Entomology
8.	Dr. Abdul Qayoom Rajput	Assistant Professor	Plant Pathology

RESEARCH FACILITIES

The Department of Agriculture and Agribusiness Management offers a range of specialized laboratories:

- 1. Molecular Plant Pathology Lab**
 - Studies plant diseases at the molecular level.
- 2. Pest and Disease Research Lab**
 - Focuses on pests and crop diseases research.
- 3. Pesticides Toxicology Lab**
 - Examines the effects of pesticides.
- 4. Biological Control Lab**
 - Research on natural pest control methods.
- 5. Entomology Post Graduate Research Lab**
 - Advanced research on agricultural insects.
- 6. Seed Testing Lab**
 - Tests seed quality and viability.
- 7. Plant Tissue Culture Lab**
 - Cultivates plant tissues for genetic research.
- 8. Post-Harvest Management Lab**
 - Develops crop preservation techniques.

INFRASTRUCTURE

The Department of Agriculture and Agribusiness Management has excellent infrastructure to support education and research:

- 1. Classrooms:** state-of-the-art classrooms, each equipped with multimedia sound systems
- 2. Undergraduate lab:** for BS students
- 3. Seminar Library:** for text and reference books
- 4. Conference Room:** for meetings and trainings
- 5. Computer Laboratory with Digital Library:** access to online resources
- 6. Laboratories:** well-equipped specialized post-graduate labs for advanced research.
- 7. Agriculture Experimental Facilities:**
 - 18.18-acres agricultural field for research and trainings.
 - 0.5-acre apiculture field dedicated to beekeeping studies.
 - Green house for different field experiments.

These facilities provide a robust environment for both learning and research in agriculture.

AVENUES OF RESEARCH

The Department of Agriculture and Agribusiness Management offers diverse research opportunities. In Plant Protection, researchers study insects affecting crops and develop pest management strategies, while plant pathology investigates plant diseases, both aiming to create eco-friendly control methods. Agribusiness research explores the economic and business aspects of agriculture to enhance farm management and agribusiness practices. These research avenues address critical agricultural challenges and promote sustainable agriculture.

DEPARTMENT OF APPLIED CHEMISTRY & CHEMICAL TECHNOLOGY

INTRODUCTION

The department of Applied Chemistry and Chemical Technology was established in 1971. The courses were designed keeping in view the need of the chemical industries of Pakistan and with due consideration of courses of foreign and local Universities offering similar program. Now advancements in syllabus with special emphasis has been put on important chemical industries of Pakistan, for instance, fertilizers, refineries, cement, sugar, polymers, pharmaceuticals, textiles, etc. Due consideration has been given to important aspects of chemical engineering like unit operations, unit processes, quality control & instrumentation, thermodynamics reaction engineering, design engineering, materials engineering, maintenance and environmental engineering, etc. Hence it is hoped that the

Our program provides the knowledge, skills and resources for continued learning and professional development over a lifetime.

An ability to identify, formulate, and solve chemical related problems producing original and substantial research.

TEACHING AND RESEARCH ASSESMENT

Teaching and research is supported by lectures, colloquiums, and group discussion and seminar presentations. Online sources and webinars are also arranged to cultivate expertise and collaboration at international forum.

AIMS, SCOPE

Research graduates of this department will meet the needs of the chemical industries and institutes of Pakistan.

DEGREE PROGRAMS

1. M.Phil.
2. Ph.D.

DEPARTMENTAL RESEARCH COMMITTEE MEMBERS

1. Dr. Mohib Reza Kazimi
2. Dr. M. Ashraf Kamal
3. Dr. Syed Imran Ali
4. Dr. Sadia Khan
5. Dr. Riaz Ahmed
6. Dr. S. Mumtaz Danish Naqvi
7. Dr. Syed Adnan Hasan
8. Dr. Farah Inamullah
9. Dr. Zeeshan Akhter
10. Dr. Maryam Mudassir
11. Dr. Sadia Amber
12. Dr. Mehwish Mobeen

OBJECTIVES

1. Studying chemical properties for conducting research producing new materials and processes.
2. Engage with wide range of industrial perspective to make a contribution the field of chemical sciences.
3. Discover how literature is integrated with other emerging fields like chemical engineering
4. Encouraging professional development technical writing and publish research in reputed journals.
5. Revisiting chemistry to answer a specific question or solve a real-world problem finally to commercialization.

AVENUEUS & PROSPECTS

1. Academic careers in universities and institutes

2. Industrial and Scientific Research
3. Industrial Research and Development
4. Chemical consultation and advisory
5. Entrepreneurship and commercial start-ups

OUR MISSION

To attain quality in academics and research in turn formulating latest chemical technologies. We undertake to train highly skilled manpower with an entrepreneurial view. In turn help our university in becoming a globally recognized center of excellence in Higher Education and Research.

OUR VISION

Advance education and research for scientific application in most recent fields of chemical science and technology. Thus producing diversity of chemical substances and process technologies fostering industrial growth. In turn enriching lives in our society and to integrate with other economic stability measures.

AVAILABLE RESEARCH FACILITIES

1. Infrastructure includes specialized research laboratories, analytical equipment and workshop facilities.
2. Departmental Seminar Library, which contains most of the latest text and reference books on the related subjects. Students also have access to digital library provided by HEC.
3. Faculty believes in applied research in close co-ordination with the industries. Therefore, various field trips and colloquiums based on the latest chemical technology are also arranged.

POSTGRADUATE PROGRAMS

1. M.Phil
2. Ph.D.

FACULTY MEMBERS

SR.	Name & Designation	BPS / TTS / Adjunct / Visiting	Title & Subject Area [provide transcripts]	Basic Degree	Ph.D Degree Status
				Title & Subject Area [provide transcripts]	Research Area (based on key words given in published articles & books) [annex scatter plot of most frequently occurring key words]
1	Dr. Riaz Ahmed	BPS 20	Rheology Non-newtonian Fluids, Drip Irrigation Automation Neural network	M.Sc. 1986 Applied Chemistry and Chemical Technology, University of Karachi.	Ph.D. 1994 <i>Chemical Engineering</i> Cambridge University
2	Dr. M. Ashraf Kamal	BPS 21	Lubricants Edible oils Ores Chemo metrics	M.Sc. 1990 Applied Chemistry and Chemical Tech, University of Karachi.	Ph.D. 2011 Applied Chemistry & Chemical Technology, University of Karachi
3	Dr. S. Mumtaz Danish Naqvi	BPS 20	Chemical kinetics, Reaction Engineering Chemometric	M.Sc. 1994 Applied Chemistry and Chemical Tech, University of KHI	Ph.D. 2010 Applied Chemistry & Chemical Technology, University of Karachi

4	Dr. Mohib Reza Kazimi	BPS 21	Textiles, Polymers, Nanotech, Ionic liquids Biomaterials	M.Sc. 1996 Applied Chemistry and Chemical Technology, University of Karachi.	Ph.D. 2014 Chemical Engineering Universiti Malaysia Pahang, Al Sultan Abdullah
5	Dr. Sadia Khan	BPS 21	Biofuels Catalysis Analytical chemistry	M.Sc. 2003 Applied Chemistry and Chemical Technology, University of Karachi.	Ph.D. 2019 Applied Chemistry & Chemical Technology, University of Karachi
6	Dr. Syed Imran Ali	BPS 21	Polymer nanoparticles, nanocomposite drug release & encapsulation	M.Sc. 1998 Applied Chemistry and Chemical Technology, University of Karachi.	Ph.D. 2019 Polymer chemistry Eindhoven University of Technology, Netherlands
7	Dr. Syed Adnan Hasan	BPS 20	Chemical biotechnology, Environmental and Analytical chemistry	M.Sc. 1998 Applied Chemistry and Chemical Technology, University of Karachi.	Ph.D. 2010 Biodegradation of Fluorinated Environmental Pollutants University of Groningen, The Netherlands
8	Dr. Farah Inamullah	BPS 19	Production of base oil from waste lubricating oil. Applied Chemistry & Chemical Technology	M.Sc. 2000 Applied Chemistry and Chemical Technology, University of Karachi.	Ph.D. 2018 Applied Chemistry & Chemical Technology, University of Karachi
9	Dr. Zeeshan Akhter	BPS 19	Textile chemistry Synthesis of dyes, Technical textile and nanoparticles	M.Sc. 2005 Applied Chemistry & Chemical Tech, University of Karachi.	Ph.D. 2015 Applied Chemistry & Chemical Technology, University of Karachi
10	Dr. Maryam Mudassir	BPS 19	Polymer nano composite, Rheology, Electrocoagulation, Fiber composite	M.Sc. 2005 Applied Chemistry & Chemical Tech, University of Karachi.	Ph.D., 2018 Applied Chemistry & Chemical Technology, University of Karachi
11	Dr. Sadia Amber	BPS 19	Biofuels, Catalysis, Analytical chemistry	M.Sc. 2006 Applied Chemistry & Chemical Tech, University of Karachi.	Ph.D. 2019 Applied Chemistry & Chemical Technology, University of Karachi
12	Dr. Mehwish Mobeen	BPS 19		M.Sc. 2003 Applied Chemistry and Chemical Technology,	Ph.D., 2022 Applied Chemistry & Chemical Technology, University of Karachi

DEPARTMENT OF APPLIED PHYSICS

INTRODUCTION

Applied Physics is a post graduate department offering courses leading to BS. Degree in Applied Physics with specialization either in ELECTRONICS or SEMICONDUCTORS. The department also offers M.Phil degree program in Communication Electronics, Microprocessor/Microcontroller based system design and development of optical and electronic design software, Nanotechnology, and Renewable energy.

POSTGRADUATE PROGRAMS AVAILABLE

1. M.Phil.
2. Ph.D.

PROGRAM OBJECTIVES

The department offers excellent research facilities in Semiconductor Electronics, Thin Film Devices, Conducting Polymers, Specialized Glasses and Digital Signal Processing leading to M.Phil. and Ph.D. degree. The department can also offer M.Phil degree program in Communication Electronics, Microprocessor/Microcontroller based system design and development of optical and electronic design software, Nanotechnology and Renewable energy.

AIM AND SCOPE OF THE PROGRAM

Nanotechnology is the technology of the future hence keeping its importance in mind the department has established a well-equipped nanotech lab which caters the synthesis and characterization of nanoparticles. In addition renewable energy group is trying to meet the challenges of depleting traditional energy sources and using alternate energy source.

These programs are designed to prepare the students to meet the future challenges and get them well equipped with the advanced knowledge of the most experimental techniques.

REGULAR FACULTY MEMBERS WITH AREA OF SPECIALIZATION

S. No.	Name	Area of Specialization
1.	Dr. Zeeshan Alam Nayyar	Renewable energy (Wind Energy)
2.	Dr. Zia Ur Rehman	Nanotechnology
3.	Dr. Faisal Rafique	Electro-Optical and Electro-mechanical
4.	Dr. Ambreen Insaf	Renewable energy (Tidal Energy)

RESEARCH LABS & FACILITIES IN THE DEPARTMENT OF APPLIED PHYSICS

The department of Applied Physics owns three research laboratories, Nanoscale Synthesis & Research Laboratory (NSRL), Renewable Energy Research Laboratory.

1. **Nanoscale Synthesis & Research Laboratory (NSRL):** This lab consists of two sub sections
2. **KM Aleem Nanotech Laboratory:** This laboratory is named after KM Aleem (late), one of the alumni of this department. It was established with the generous funding and efforts of the family and friends (Alumni 1985) of late KM Aleem. This section caters the synthesis of nano materials in general and 2D materials in particular. Synthesis of 2D materials is done using Chemical Vapor Technique (CVT), Chemical Vapor Deposition (CVD) as well as hydrothermal method. Moreover, facility is also available to synthesize nanomaterials using wet chemical method. Synthesized materials can be used in electronics, optoelectronics, sensors, energy applications etc. We are actively working on the synthesis of TMDs, Mxenes, and other 2D materials in this laboratory.
3. **Ansari & Zaidi Thin Film Laboratory:** This lab is named after two of pioneer professors of this department to pay homage for their services to the department. This lab aims at growth of thin films for electronic, optoelectronics, sensor, and photovoltaic applications. Thin film growth is carried out

by thermal evaporation method using a vacuum coating unit. The lab is equipped with the facilities to study electrical transport and dielectric properties of thin films. Moreover, we also use spin coater and ultrasonic bath for the fabrication of thin films. We are planning to use these facilities to synthesis composite materials as well. These labs are supported by extensive publication track record in basic physics, applied physics and material physics journals of high repute.

4. **Renewable Energy Research Laboratory:** Renewable Energy lab caters the research related to wind energy, tidal energy, and their modeling applications.
5. **Modeling: electro-optical and electro-mechanical systems:** Modelling FEM simulation and developing applied electro-optical and electro-mechanical systems. EM Metamaterials designing in wide spectrum microwave band. Chemical and biomedical sensors and actuators based on ionic liquid and gels

DEPARTMENT OF BIOCHEMISTRY

INTRODUCTION

The Department of Biochemistry, established in 1962, is the pioneer independent department in Pakistan. Since its foundation, the department has contributed the University of Karachi's goals of excellence in research and education. Traditionally known for strength in major areas of research like Molecular Biology, Genetics, Cell Biology, Biotechnology and Bioinformatics, the current faculty is continuing to broaden its representation and reputation in other areas, most notably, Neuroscience, Clinical Biochemistry, Enzymology, Hematology, Immuno-chemistry, Genomics, Molecular Biophysics, Neurochemistry, Neuropharmacology, Biotechnology, Nutrition and Proteomics.

POST GRADUATE PROGRAMS AVAILABLE

1. M.Phil.
2. Ph.D.

OBJECTIVES OF PROGRAMS

M.Phil in Biochemistry Objectives

1. **Learn Advanced Biochemistry:** Understand advanced biochemical concepts and techniques. Specialize in specific areas of biochemistry.
2. **Develop Research Skills:** Conduct and analyze biochemical experiments. Use various biochemical tools and methods.
3. **Enhance Critical Thinking:** Improve problem-solving and analytical skills. Apply these skills to research questions.
4. **Improve Communication:** Write and present scientific information clearly. Share research findings effectively.
5. **Professional Growth:** Prepare for careers in research, industry, or teaching. Follow ethical research practices.

Ph.D. in Biochemistry Objectives

1. **Conduct Original Research:**
 - Perform unique and important research in biochemistry.
 - Become an expert in a specific research area.
2. **Master Advanced Techniques:**
 - Use advanced biochemical methods independently.
 - Design and carry out complex experiments.
3. **Think Critically and Innovate:**
 - Analyze scientific studies and data critically.
 - Develop new ideas and hypotheses.
4. **Communicate Scholarly Work:**
 - Write scientific papers and grant applications.
 - Present research at conferences.
5. **Lead and Collaborate:**
 - Lead research projects and teams.
 - Work well with other scientists.
6. **Maintain High Standards:**
 - Follow ethical research guidelines.
 - Prepare for leadership roles in various fields.

AIMS AND SCOPE OF THE PROGRAMS (M.Phil AND Ph.D.)

The M.Phil and Ph.D programs in the Department of Biochemistry are designed to offer students a deep dive into the world of Biochemistry, helping them to become experts in understanding the intricate details of living organisms at the molecular level. Here's what these programs aim to achieve and cover:

AIMS

1. **Deep Understanding:** Help students gain a thorough understanding of biochemistry, so they can explore and explain how life works at a molecular level.
2. **Research Expertise:** Train students to become skilled researchers, capable of designing and conducting their own experiments, analyzing data, and presenting their findings clearly.
3. **Critical Thinking:** Develop students' abilities to think critically and solve complex biochemical problems, contributing to scientific progress.
4. **Professional Growth:** Prepare students for various professional paths, whether in academia, industry, or other fields, by improving their communication, teamwork, and ethical research practices.
5. **Original Research:** Encourage students to create new knowledge through original research, making meaningful contributions to the field of biochemistry.

SCOPE

1. **Wide Research Areas:** Explore a broad range of topics such as enzyme function, genetic regulation, metabolism, structural biology, and the molecular basis of diseases.
2. **Interdisciplinary Work:** Foster collaboration with other fields like molecular biology, chemistry, physics, and medicine, to solve complex biochemical problems.
3. **Hands-On Lab Training:** Provide extensive hands-on experience with advanced lab techniques and technologies, like mass spectrometry, X-ray crystallography, and molecular cloning.
4. **Ongoing Learning:** Keep students engaged with the latest research through seminars, workshops, and discussions, ensuring they are up-to-date with current scientific advancements.

These programs are designed to mold students into well-rounded, knowledgeable biochemists who can make significant contributions to science and society.

REGULAR FACULTY MEMBERS WITH THEIR SPECIALIZATION / EXPERTISE

S. No	Name	Specialization / Expertise
1	Dr. Farhat Batool Professor & Chairperson	Neuroscience, Neuroendocrinology and Psychoneuroimmunology
2	Dr. Shamim A. Qureshi Professor	Phytopharmacology, Biotechnology, Clinical Biochemistry
3	Dr. Shah Ali ul Qader Professor	Enzymology, Biopolymer production and Immobilization
4	Dr. Hajra Naz Associate Professor	Neuropharmacology and Toxicology
5	Dr. Sadaf Naeem Associate Professor	Bioinformatics, Biophysics, Computational Molecular Biophysics and computational drug design
6	Dr Noman Syed Associate Professor (TTS)	Enzymology
7	Dr. Sadia Saleem Assistant Professor	Biochemical Neuropharmacology
8	Dr. Huma Ikram Assistant Professor	Biochemical Neuropharmacology
9	Dr. Faizan-Ul-Hassan Naqvi Assistant Professor	Molecular Biology
10	Dr. Yasmeen Rashid Assistant Professor	Bioinformatics, protein structure-function relationship studies and genomics
11	Dr. M. Farhan Kamali Assistant Professor	Biochemical Neuropharmacology
12	Dr Beena Hassan Assistant Professor	Neurochemistry
13	Dr. Fizza Naqvi Assistant Professor	Biochemical Neuropharmacology

14	Dr. Tooba Latif Assistant Professor	Natural Products chemistry and therapeutics
15	Dr. Ayesha Khan Assistant Professor	Neurochemistry
16	Dr. Uzma Urooj Assistant Professor	Protein Chemistry
17	Dr. Khawaja Ali Hasan Assistant Professor	Molecular Biology

RESEARCH FACILITIES AVAILABLE

1. **Dr. Zain memorial Seminar Library** of the department of biochemistry has the latest collection of books on Biochemistry with over 100 biochemistry-related journals in print and/or with on-line access. Computers with Internet facility are also available for the students and faculty.
2. **Computer Lab** maintained with Internet facility.
3. **Instruments Rooms:** The department is highly equipped with cutting edge techniques for research like: Atomic Absorption Spectroscopy (AAS), RT-PCR, High Performance Liquid Chromatography (HPLC) EC and UV detectors, Column Chromatography, ELISA, Spectrophotometer, Electrophoresis, Flame Photometry, Freeze Drying, Fluorimeter and Ultra centrifugation, Ultra Low Temperature Freezer, NanoDrop Spectrophotometer and Flame Photometer.

AVENUES OF RESEARCH

Biochemistry research covers many exciting areas. Here are some key ones:

Key Research Areas:

1. **Enzymes:** Studying how enzymes work and how they can be used in medicine and industry.
2. **Genes:** Exploring how genes are controlled and passed down, and how they influence life.
3. **Metabolism:** Understanding how cells convert food into energy and essential molecules.
4. **Protein Structures:** Figuring out the shapes of proteins and other molecules to see how they function.
5. **Neural Cell Communication:** Investigating how brain cells talk to each other and react to their environment.
6. **Disease Mechanisms:** Researching the molecular causes of diseases like cancer and diabetes to find new treatments.
7. **Proteins in Mass:** Studying all the proteins in a cell or organism to see how they work together.
8. **Genomics:** Analyzing entire genomes to understand genes and their roles.
9. **Chemical Biology:** Using chemistry to study and change biological processes.
10. **Biotechnology:** Applying biochemical knowledge to develop new technologies and products.
11. **Environmental Impact:** Studying how biochemical processes affect the environment and how to clean up pollutants.
12. **Nutrition:** Understanding how nutrients affect health at the molecular level.
13. **Systems Biology:** Combining data from various sources to understand how biological systems function as a whole.

These areas help us understand life better and find new ways to improve health, agriculture, industry, and the environment. Excellent career opportunities exist for the students acquiring a degree in Biochemistry. Our graduates are working in multinational industries, teaching and research organizations, Pharmaceutical industries, biomedical industries, Forensic science laboratories, hospitals and clinical laboratories and different public and private sector industries

DEPARTMENT OF BIOTECHNOLOGY

INTRODUCTION OF THE DEPARTMENT

The Department of Biotechnology at the University of Karachi is known for its excellent education and innovative research. Equipped with modern facilities and a seasoned faculty, the department specializes in vital biotechnological fields. Our programs prepare students to confront real-world issues, fostering progress in industry and society through collaborative efforts and scientific expertise. With rigorous academic training and research opportunities in our M.Phil. and Ph.D. programs, students are primed for leadership roles across academia, industry, and research institutions. Guided by esteemed faculty and empowered by state-of-the-art resources, students engage in interdisciplinary research, driving biotechnological advancements and addressing global challenges. Currently many of the graduates hold key positions in various national and multinational organizations as well as serving abroad and are making significant contributions in their field of specialization.

POSTGRADUATE PROGRAMS AVAILABLE

1. M.Phil.
2. Ph.D.

OBJECTIVES OF THE M.Phil. PROGRAM

1. To conduct scholarly activities in a professional and ethical manner.
2. To analyze case studies representatives of key areas of biotechnology.
3. To make student proficient in applying scientific principles and methods at an advanced level to diagnose and solve problems.
4. To produce a researcher who has in-depth knowledge in specialized areas of biotechnology and the skills to carry out original research.

AIMS OF THE M.Phil. PROGRAM

The M.Phil. Program aims to cultivate ethical and professional scholarly conduct, enhance students' analytical and problem-solving skills through advanced scientific principles, and develop in-depth knowledge and research capabilities in specialized areas of biotechnology.

SCOPE OF THE M.Phil. PROGRAM

The program covers advanced coursework, ethical training, and case study analysis, providing training in research methodologies and offering specialization through thesis research. It also emphasizes professional development, industry collaboration, and networking opportunities to prepare students for careers in biotechnology.

OBJECTIVES OF THE Ph.D. PROGRAM

1. To develop skills required to conduct independent research.
2. To prepare graduates for writing national & international projects aiming to solve challenges facing industry and society.
3. To make students able to communicate effectively, scientifically and professionally
4. To educate students so that they can exploit discoveries and can develop, design and synthesize safe and sustainable processes & products

AIMS OF THE Ph.D. PROGRAM

The Ph.D. Program aims to develop students' skills for conducting independent research, preparing them to write impactful national and international projects addressing industry and societal challenges, and ensuring they can communicate effectively and professionally.

SCOPE OF THE Ph.D. PROGRAM

The program encompasses advanced research training, project proposal writing, and effective scientific communication. It also focuses on educating students to leverage discoveries for the development, design, and synthesis of safe and sustainable biotechnological processes and products.

REGULAR FACULTY MEMBERS AND AREAS OF SPECIALIZATION

Highly qualified and experienced faculty are dedicated to providing quality education and conducting

research in key areas of biotechnology.

Faculty	Designation	Expertise Area
<i>Dr. Raheela Rahmat Zohra</i>	Associate Professor	Nanotechnology and Industrial Biotechnology
<i>Dr. Saifullah</i>	Associate Professor and Chairman	Biotransformation, Metabolomics, Plant Tissue Culture, Natural Product.
<i>Dr. Shafaq Aiyaz Hasan</i>	Assistant Professor.	Bioinformatics and Molecular biology
<i>Dr. Mahnaz Ahmad</i>	Assistant Professor	Industrial and Environmental Biotechnology
<i>Dr. Erum Hanif</i>	Assistant Professor	Immunology and Medical Biotechnology

INFRASTRUCTURE AND RESEARCH FACILITIES AVAILABLE

Classrooms are adequately equipped and have facility of multimedia projectors. Departmental seminar library has stock of about 500 books related to various fields of biotechnology and is updated periodically. The Department of Biotechnology has 09 research laboratories a part from four students laboratories. These laboratories are equipped with modern and fashionable equipment like RT-PCR, Enzyme Linked Immunosorbent Assay (ELISA), Nano-drop, High Performance Liquid Chromatography (HPLC) Unit, Thermocyclers, Electrophoresis Units, Vacuum Concentrator, Freeze Dryer, Rotary Evaporator, Spectrophotometer, Aspirators, Centrifuges, Shakerbators, various kinds of Microscopes with cameras, Clean Bench and BSL-2 Cabinets etc.

AVENUES FOR RESEARCH

Industrial biotechnology research aimed at developing sustainable processes for the production of chemicals, materials, or energy. Nanotechnology applications in various fields. Biotransformation of natural compounds for pharmaceutical or industrial purposes. Metabolomics studies to understand metabolic pathways in plants or microbial systems. Plant tissue culture techniques for the production of valuable secondary metabolites. Investigation of natural products for their therapeutic or industrial potential. Bioinformatics analysis of genomic or proteomic data related to diseases or biotechnological processes. Molecular biology studies focusing on gene expression, regulation, or genetic engineering of microorganisms. Environmental biotechnology projects focusing on waste treatment, bioremediation, or pollution control using biological systems. Immunological research on vaccine development, immune responses to pathogens, or autoimmune diseases. Medical biotechnology studies focusing on diagnostic tools, therapeutic agents, or personalized medicine approaches. These all faculty members are fields are working with industry in order to augment/solve community and country associated issues. The Department is facilitated with dedicated faculty labs; aiming towards the sustainable future prospects in the field of Biotechnology through producing well-trained postgraduate M. Phil. and Ph. D. students.

DEPARTMENT OF BOTANY

INTRODUCTION

The Department of Botany, University of Karachi was established in 1953 at the University of Karachi provides a diverse array of over ninety undergraduate and postgraduate courses aimed at offering valuable knowledge applicable to various sectors including education, agriculture, environmental NGOs, pharmaceuticals, and other industries. The department boasts a seminar library housing a vast collection of educational resources, guiding prospective students, alongside a computer lab equipped with internet facilities to aid students in honing their skills and staying abreast of the latest research trends of the twenty-first century. During the initial two years of study, students focus on botany as their major subject along with two subsidiary subjects from other university departments. In their final year, students have the opportunity to specialize in one of several branches of Botany, including Applied Botany, Mycology & Plant Pathology, Plant Ecology, Phycology & Marine Botany, Plant Physiology, and Plant Taxonomy.

The Department offers specialization in various branches of plant sciences, which proves beneficial for securing employment in agriculture-based organizations, teaching positions across colleges and universities, and contributing significantly to the country's education system. Graduates also find opportunities in non-governmental organizations like the World Wildlife Fund (WWF) and the International Union for Conservation of Nature (IUCN). Alumni from the Department are employed both nationally and internationally, reflecting the quality of education and training provided. With a strong emphasis on research and teaching, the Department has cultivated a large and productive group dedicated to advancing plant sciences. Faculty members are actively engaged in specialized fields, having successfully completed over fifty research projects supported by national and international funding agencies. The Department has an impressive publication record, with over two thousand research papers published in renowned journals, both nationally and internationally recognized by the Higher Education Commission (HEC) and with a notable impact factor. Presently, numerous research projects and publications are underway, showcasing the ongoing commitment to scientific advancement and knowledge dissemination.

THE DEPARTMENT OFFERS THE FOLLOWING POSTGRADUATES PROGRAMS:

1. M.Phil.
2. M.S.
3. Ph.D.

M.PHIL / MS AND Ph.D PROGRAM OBJECTIVES

The following objectives collectively aim to prepare graduate students to become proficient researchers and leaders in the field of Botany, capable of addressing the complex challenges facing the global environment and contributing to sustainable solutions for the benefit of society.

1. Advanced Research: To cultivate a culture of advanced botanical research among graduate students, fostering innovation and discovery in various sub fields of Botany.
2. Knowledge Enhancement: To deepen students' understanding of botanical principles, theories, methodologies, and applications through rigorous coursework, seminars, and workshops.
3. Specialization: To allow students to specialize in specific areas of botany such as plant physiology, taxonomy, ecology, genetics, or biotechnology, depending on their interests and career goals.
4. Critical Thinking: To develop students' critical thinking and analytical skills, enabling them to evaluate and solve complex botanical problems and contribute to the advancement of the field.
5. Collaboration: To promote interdisciplinary collaboration among students and faculty members within the department and across other departments or institutions, fostering a holistic approach to botanical research and applications.
6. Publication: To encourage students to publish their research findings in peer-reviewed journals and present them at national and international conferences, thereby contributing to the dissemination of knowledge and the advancement of the field.
7. Professional Development: To provide students with opportunities for professional development, including teaching experience, mentorship, and training in scientific writing, presentation skills, and grant writing, preparing them for successful careers in academia, industry, government, or non-profit organizations.

8. Ethical Conduct: To instill ethical values and principles in research practices, ensuring integrity, transparency, and responsibility in all aspects of botanical research and scholarship.

AIMS AND SCOPE OF M.Phil / MS AND Ph.D PROGRAM

The Scope of M.Phil. / MS and Ph.D. programs in the field of Botany typically encompass a wide range of objectives designed to equip students with advanced knowledge, skills, and expertise in various aspects of plant science. Here are some common aims and the scope of such programs:

The M.Phil./MS and Ph.D. programs in the Department of Botany at the University of Karachi offer an extensive scope for advanced study and research in various facets of plant science. With a rich tradition of academic excellence and a commitment to fostering innovative botanical research, these programs provide students with the opportunity to delve into diverse areas such as plant physiology, ecology, genetics, taxonomy, biotechnology, and conservation biology. Through rigorous coursework, hands-on laboratory experiences, and mentorship from esteemed faculty members, students gain a deep understanding of fundamental botanical principles and cutting-edge research methodologies. They are encouraged to pursue original research projects that address pressing challenges in agriculture, biodiversity conservation, environmental sustainability, and biotechnological innovation. Additionally, the programs emphasize interdisciplinary collaboration, ethical conduct, and professional development, preparing graduates for successful careers as researchers, educators, policymakers, and leaders in the field of Botany, both nationally and internationally.

FACULTY MEMBERS AND THEIR FIELD OF SPECIALIZATIONS

Professors

Dr. Rubina Abid (Chairperson)	(Plant Systematics, Biodiversity, Ethnobotany, Chemosystematics)
Dr. Zamin Shaheed Siddiqui	(Plant Physiology)
Dr. Sadaf Gul	(Phycology, seaweed ecology and physiology, Mangrove ecosystem)
Dr. Laila Shahnaz	(Phycology, Seaweed Biology, Phycochemistry)
Dr. M. Faheem Siddiqui	(Ecology and Dendrochronology)

Associate Professors

Dr. Seemi Aziz	(Plant Ecology)
Dr. Sharfun Nahar	(Plant Pathology)

Assistant Professors

Dr. Amna Ahmed	(Plant Anatomy)
Dr. Maria Hamid	(Plant Pathology)
Dr. Muhammad Azeem	(Plant Physiology)
Dr. Naeem Ahmed	(Plant Physiology)
Dr. Afshan Rahman	(Plant Physiology)
Dr. Hafiza Asma	(Plant Pathology, Plant molecular biology)
Dr. M. Waseem Abbasi	(Plant Pathology)
Dr. M. Yousuf Adnan	(Plant Ecophysiology)
Dr. Asma Hanif	(Plant Pathology)
Dr. Sana	(Plant Systematics, Biodiversity, Ethnobotany)

Lecturer

Mrs. Hira Ejaz	(Plant Pathology)
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AVAILABLE RESEARCH FACILITIES

The Department boasts well-equipped laboratories catering to the needs of undergraduate, graduate, postgraduate, and Ph.D studies alongside an extensive collection of books and research papers housed in the Seminar library. Additionally, it has established a computer lab facilitating rapid access to study materials, thus enabling students to delve into various aspects of plant sciences research. Moreover, the Department has inaugurated the Shameel Memorial Library, the Stress Physiology and Phenomics Center (funded by South Korea), and the Biosaline Research Lab and Experimental Field, all aimed at advancing research and

productivity. At the forefront of these facilities is the state-of-the-art Central Lab, equipped with cutting-edge instruments including HPLC, PCR, and high-resolution microscopes, which form the cornerstone for conducting high-quality research endeavors.

INFRASTRUCTURE

1. The Department has spacious lecture hall and audio-visual rooms. Further, an auditorium is present to conduct seminars and other program.
2. To deal with the queries and to facilitate students an administrative desk is also available besides Chairperson room.
3. All faculty members have offices alongside their own research laboratories providing a conducive environment for M.Phil. and Ph.D. research students.
4. Centralized laboratory is functioning along with the state of the art and sophisticated Scientific Equipment's and Instruments for post graduate students.
5. Girls Common is also available in the Department.
6. The department is surrounded by beautiful and well maintained lawns having wide variety of ornamental, medicinal and fruit plants.

AVENUES FOR RESEARCH

Research in Botany encompasses a diverse range of avenues, reflecting the broad scope of plant science and its interdisciplinary connections with other fields. Here are some key avenues of research in botany:

1. **Plant Taxonomy / Systematics:** Classifying, identifying, and naming plants based on their evolutionary relationships and morphological, anatomical, and molecular characteristics, contributing to our understanding of plant diversity and investigating the evolutionary history, adaptation, speciation, and diversification of plants, using phylogenetic analyses, comparative genomics, and experimental approaches to unravel the patterns and processes of plant evolution.
2. **Plant Physiology:** Investigating the physiological processes that govern plant growth, development, metabolism, and responses to environmental stimuli such as light, temperature, water, and nutrients. Studying plant-environment interactions, plant responses to biotic and abiotic stresses, and mechanisms of stress tolerance and adaptation, with implications for agricultural productivity, ecosystem resilience, and environmental sustainability.
3. **Plant Ecology:** Studying the interactions between plants and their biotic and abiotic environments, including topics such as community ecology, ecosystem dynamics, biodiversity, conservation biology, and the impacts of human activities on natural ecosystems. **Dendrochronology:** Scientific method of dating tree rings to determine the age of a tree or the timing of events in the environment. By examining the pattern of rings formed in the trunk of a tree, scientists can infer information about past climates, environmental conditions, and ecological changes. **Plant-Soil Interactions:** Exploring the role of plants in soil formation, nutrient cycling, and ecosystem functioning, as well as the effects of soil properties and microbial communities on plant growth and health.
4. **Plant Genetics and Genomics:** Exploring the genetic basis of plant traits, genetic diversity, gene expression, and genome organization, using molecular techniques and genomic tools to improve crop traits, develop genetically modified organisms, and conserve endangered species.
5. **Plant Molecular Biology and Biotechnology:** Investigating the molecular mechanisms underlying plant processes such as gene regulation, signal transduction, and stress responses, and applying biotechnological approaches to improve crop productivity, disease resistance, and tolerance to environmental stresses.
6. **Plant-Pathogen Interactions:** Understanding the mechanisms of plant-pathogen interactions, including plant defense mechanisms, pathogen virulence factors, host-pathogen co-evolution, and the development of sustainable strategies for plant disease management.
7. **Plant Developmental Biology:** Studying the molecular and cellular mechanisms that control plant

development, including embryogenesis, organogenesis, pattern formation, and senescence, to understand how plants achieve their diverse forms and functions.

8. **Ethnobotany and Plant Conservation:** Documenting the traditional knowledge and uses of plants by indigenous peoples, studying the medicinal, cultural, and ecological significance of plant species, and developing conservation strategies to protect endangered plants and their habitats.
9. **Plant Biogeography and Climate Change:** Investigating the distribution patterns of plant species across geographic regions and their responses to climate change, including shifts in species ranges, phenology, and ecosystem dynamics.

DEPARTMENT OF CHEMISTRY

INTRODUCTION

Chemistry is a novel discipline of science. It helps to understand the origin of life and also suggests ways to use non-living things for giving comfort to life. To train the young generation to enable them to meet this task easier it is necessary to make the students understand the principles underlying the composition, structure and properties of substances along with the transformation that they undergo. It is therefore; obvious that chemical orientation cannot be imparted in isolation and that it has to be developed through a cocktail of physics, biology on one hand and engineering and applied sciences on the other hand. The Department of Chemistry, University of Karachi is well equipped for pursuing this goal. It has a competent faculty, well-equipped laboratories and other infrastructure.

The Postgraduate Program in Chemistry is intended to meet the growing needs of competent manpower of Pakistan and that of the international community. Department of Chemistry imparts professional and academic excellence to young people of Pakistan. It caters for all the disciplines of chemistry i.e. inorganic, organic, physical and analytical chemistry. In recent years many candidates have completed M.Phil. and Ph.D. work. Every year the researchers of the department publish several research papers and chapters in books.

The department faculty is highly qualified holding Ph.D. degree many of our teachers have presented their research in international conferences/seminars. Most of the teachers are pursuing a vigorous research program within the limited resources. The department also hosted a number of national and international conferences and published articles in pre reviewed international journals.

POSTGRADUATE PROGRAMS AVAILABLE

1. MS
2. M.Phil
3. Ph.D.

PROGRAMS OBJECTIVES / AIMS AND SCOPE OF THE PROGRAM

At the Post graduate level, the department offers Master of Science degree program (MS/M.Phil) which is Academic and research program to become professional chemists teachers or further pursue chemistry Doctoral program. In addition, the department offers Chemistry Doctoral Program (Ph.D.) to highly motivated chemistry majors, interested in conducting a faculty with a wide variety of professional interests representing all areas of chemistry. This supports classes and research opportunities, which span the entire chemical spectrum. The department seeks to remain equipped with modern chemical instrumentation, so that students learn to use the tools of the chemist's trade.

LIBRARY

The Department of Chemistry has Digital Library facility with large number of recent and classic text and reference books of chemistry as well as subscription to leading journals. The library has a stock of more than 10,000 books and research journals on various fields of Chemistry; Membership to this library entitles students to borrow books for a specified period of time. The department library has local area network facility from which most scientific journals can be accessed.

LABORATORY AND RESEARCH RESOURCES

The Department of Chemistry is equipped with many modern instruments for training of students for challenging careers and research. These include, pH/Ion meters, conductometers, polarimeters, refractometers, potentiometers, electrochemical analyzers, HPLC, atomic absorption spectrometers, UV-Visible spectrophotometers, FTIR Spectrometers, Guoy Balance for magnetic susceptibility measurements, flame photometers, gas chromatograph-mass spectrometer nuclear magnetic resonance spectrometer as well as some other modern instruments. The Department of Chemistry is constantly trying to upgrade its laboratory and research resources so that its scholars can maintain a competitive edge in the research and job market.

AVENUES FOR RESEARCH

Chemistry Scholar has not only gain a vast research experience but also a wide choice of employment opportunities in many industries and research institutions that are emerging in the country such as Textile industry, Pharmaceutical Industry, Clinical laboratories, Oil refineries, Chemical industry, Paint industry, Cement industry, Beverages & Food Product Industry, Analytical & Instrumental Services Agencies, Polymer Industry

REGULAR FACULTY MEMBERS WITH AREAS OF SPECIALIZATION

Teacher's Name	Designation	Areas of Specialization
Dr. Hajira Tahir	Professor	Physical Chemistry
Dr. Rehana Saeed	Professor & Chairperson	Physical Chemistry
Dr. Nasreen Fatima	Professor	Inorganic Chemistry
Dr. Firdous Imran Ali	Professor	Organic Chemistry
Dr. Itrat Anis	Professor	Organic Chemistry
Dr. Uzma Ashiq	Professor	Inorganic Chemistry
Dr. Sheikh Mohiuddin	Professor	Analytical Chemistry
Dr. Erum Zahir	Professor	Physical Chemistry
Dr. Imran Ali Hashmi	Professor	Organic Chemistry
Dr. Rifat Ara Jamal	Associate Professor	Inorganic Chemistry
Dr. Shazia Nisar	Associate Professor	Inorganic Chemistry
Dr. Raheela Naz	Associate Professor	Inorganic Chemistry
Dr. Zahid Hussain	Associate Professor	Organic Chemistry
Dr. Syed Tariq Ali	Associate Professor	Organic Chemistry
Dr. Syed Kashif Ali	Associate Professor	Organic Chemistry
Dr. Agha Arslan Wasim	Associate Professor	Analytical Chemistry
Dr. Mohsin Ali	Associate Professor	Analytical Chemistry
Dr. Saba Arslan Wasim	Associate Professor	Analytical Chemistry
Dr. Humaira Bano	Assistant Professor	Inorganic Chemistry
Dr. Naushaba Zafar	Assistant Professor	Inorganic Chemistry
Dr. Tehseen Ahmed	Assistant Professor	Physical Chemistry
Dr. Summyia Masood	Assistant Professor	Physical Chemistry
Dr. Syed Nawazish Ali	Assistant Professor	Organic Chemistry
Dr. Qurat-ul-Ain	Assistant Professor	Inorganic Chemistry
Dr. Hamida Sultan Muhammad	Assistant Professor	Analytical Chemistry
Dr. M. Asad Khan Tanoli	Assistant Professor	Inorganic Chemistry
Dr. Perveen Fazil	Assistant Professor	Organic Chemistry
Dr. Safi Asim Bin Asif	Assistant Professor	Analytical Chemistry
Dr. Nizam Uddin	Assistant Professor	Organic Chemistry
Dr. Syed Muhammad Saad	Assistant Professor	Organic Chemistry
Dr. Maria Ashfaq Lodhi	Assistant Professor	Inorganic Chemistry
Dr. Noshab Qamar	Assistant Professor	Inorganic Chemistry
Dr. Sadaf Iqbal	Assistant Professor	Inorganic Chemistry
Dr. Nida Ali	Assistant Professor	Inorganic Chemistry
Dr. Fakhrun Nisa	Assistant Professor	Analytical Chemistry
Dr. Muhammad Balal	Assistant Professor	Analytical Chemistry
Dr. Muhammad Saad	Assistant Professor	Analytical Chemistry

DEPARTMENT OF COMPUTER SCIENCE

INTRODUCTION

University of Karachi's illustrious Department of Computer Science was inaugurated in 1984 and instigated its scholarly pursuits in the academic year of 1985-86, and stands as a beacon of academic excellence and innovation among the academia and the industry.

The Department of Computer Science offers a magnanimous assortment of courses across multiple tiers, punctiliously designed to endow prospects for advanced studies and innovative research in the realms of Computer Science and Software Engineering. The Department of Computer Science is dedicated to nurturing substantially proficient computer professionals, adept in addressing the evolving demands of the industry both domestically and internationally.

The mission is to foster a comprehensive educational environment that supports the advancement of computer science, software engineering and IT. The vision is to be a leading department in research and education, contributing to the IT industry and academic community through innovation and excellence in education and research

POST GRADUATE PROGRAMS

A considerate assortment of distinguished postgraduate programs are offered at the Department of Computer Science, encompassing MS, M.Phil, and Ph.D. Degrees. These esteemed programs have been pedantically engendered with the deep-seated aim of imparting exceptional education and fostering ground-breaking research in the fields of Computer Science, Software Engineering and Information Technology.

OBJECTIVES OF THE POST GRADUATE PROGRAMS

The primary objectives of the postgraduate programs at the Computer Science Department include the development of advanced knowledge and skills in computer science, preparation of students for research and professional careers, and contribution to the technological advancement of society. These programs aim to advance knowledge through original research, preparing students for careers in national and international academia/industry and research institutions, while fostering a solid foundation in various computing fields.

The scope of the postgraduate research programs encompasses deep research in areas such as theoretical computer science, data science, artificial intelligence, and beyond.

PRIMARY GOALS

1. To provide advanced knowledge and expertise in computer science.
2. To prepare Students for careers in academia, industry, and research.
3. To foster Innovation and research in cutting-edge technologies.

INTENDED OUTCOMES

1. Mastery of advanced topics in computer science.
2. Development of research and analytical skills.
3. The ability to design and implement complex software systems.
4. Evolve the research culture to facilitate the tech industry.

INFRASTRUCTURE

The Department of Computer Science is currently housed in UBIT building, the UBIT building by itself is a modern architectural marvel comprising two blocks. These blocks encompass state-of-the-art lecture rooms, discussion rooms, computer laboratories, and faculty offices. The infrastructure is further enhanced with facilities for teleconferencing, a well-stocked library, administrative offices, separate prayer area, and common rooms. This complex has been continually upgraded to support the academic and research endeavors of the Computer Science Department.

Classrooms and Lecture Halls

1. The classrooms are equipped with modern technology, including projectors, smartboards, and high-speed internet, facilitating an advanced learning environment.

2. The lecture halls boast the capacity to host guest lectures and accommodate large classes, ensuring ample space for intellectual discourse.

Libraries

1. The libraries house extensive collections of computer science textbooks, research journals, and digital resources, providing a comprehensive repository of knowledge.
2. Access to online databases and digital libraries is provided, ensuring students and faculty have the latest information at their fingertips.

OTHER FACILITIES

1. **Collaborative Workspaces:** Designated areas for students to engage in group projects and collaborative work, fostering a spirit of teamwork and innovation.
2. **Seminar Rooms:** Specifically designed for hosting workshops, seminars, and academic discussions, these rooms are equipped to facilitate high-level intellectual engagement.
3. **Student Lounges and Common Areas:** Comfortable spaces for informal study and relaxation, offering a balanced environment for academic pursuits and leisure.

The infrastructure at Computer Science Department stands as a testament to the commitment to excellence, providing an environment conducive to both scholarly achievements and groundbreaking research.

RESEARCH FACILITIES AVAILABLE AT THE DEPARTMENT OF COMPUTER SCIENCE

Specialized Labs

1. **Artificial Intelligence and Machine Learning Lab:** This lab is furnished with high-performance computing systems and advanced software, enabling extensive research in the fields of artificial intelligence and machine learning.
2. **Cybersecurity Lab:** Secure environments are provided for the testing and development of cutting-edge security protocols and systems, fostering advancements in cybersecurity.
3. **Robotics Lab:** This lab features state-of-the-art equipment, including robotic arms, drones, and other hardware essential for pioneering research in robotics and automation.

Equipment and Resources

1. **High-Performance Computing Clusters:** These clusters are available for undertaking large-scale data analysis and complex computational tasks, facilitating high-level research activities.
2. **Virtual Reality/Augmented Reality Facilities:** Advanced facilities are provided for research in immersive technologies, supporting explorations in virtual and augmented reality.
3. **Data Centers:** Equipped to support big data research and cloud computing experiments, these centers are integral to conducting expansive and innovative research projects.

The research facilities at the Department of Computer Science are meticulously designed to support and enhance the scholarly and investigative pursuits of its faculty and students, ensuring a conducive environment for groundbreaking discoveries and technological advancements.

AVENUES OF RESEARCH

The dynamic landscape of Computer Science and technology presents numerous trending avenues of research, each contributing to the continual advancement of the field. The Computer Science Department's primary focus lies in the development of human research in the field of Computational Sciences. The department is dedicated to exploring several pivotal research areas. Research endeavors are methodically designed to encompass a wide array of domains, some of which are as Follows,

Quantum Computing,

Edge Computing, Blockchain and Distributed Ledger Technologies, Autonomous Systems and Robotics

Cyber-Physical Systems (CPS), Explainable AI (XAI), Augmented Reality (AR) and Virtual Reality (VR)

Natural Language Processing (NLP) and Speech Recognition, Smart Agriculture, and Smart Cities and

Urban Computing, Big Data Analytics, Data Communication, Network security, Software defined network.

Professional Linkages

1. Close links are maintained with professional organizations like IEEE and the ACM at both national and international levels to elevate the professional and academic standards of the faculty and

students.

Events and Seminars

1. International conferences, symposiums, and workshops on emerging technologies are frequently organized by the department.
2. Regular seminars conducted by industry experts and visiting scholars contribute to the intellectual enrichment of students and faculty alike.
3. Hackathons, coding competitions, and tech fairs are hosted, providing platforms for innovation and skill development.

Additional Opportunities

1. Internship programs with leading technology companies are facilitated, offering students invaluable industry experience.
2. Collaborations with industry for applied research projects ensure that students engage in practical, real-world problem-solving.

These initiatives reflect the department's unwavering commitment to fostering an environment of continuous learning, professional growth, and international collaboration, thereby ensuring that its members remain at the forefront of technological advancements and academic excellence.

REGULAR FACULTY MEMBERS

Dr. M. Sadiq Ali Khan; Ph.D (KU), MS (SSUET), BS (SSUET)

Chairman & Associate Professor (msakhan@uok.edu.pk)

Areas of Interest: Network security; Cyber Security; Adhoc Vehicular Networks; SDN; Data Communication, Network security, Software defined network.

Dr. Nadeem Mahmood; Post. Doc. (IIU Malaysia), Ph.D (KU), MCS (KU)

Associate Professor (nmahmood@uok.edu.pk)

Areas of Interest : Temporal and Spatial Systems, Artificial Intelligence, Health Information Systems, Software Engineering

Dr. S. M. Khalid Jamal; Ph.D, M.Phil(C/Fi), M.B.A (KU), BS (KU)

Assistant Professor (s.m.khalid@uok.edu.pk)

Areas of Interest : Data Science, Database & Data-warehouse Systems, Grid/Cloud Computing, Data / Text / Web Mining & Knowledge Discovery, MIS, Information Extraction / Integration, Big Data Analytics, Social Media / Network Analysis

Dr. Muhammad Saeed; Ph.D (KU), MS (LUMS), BS (KU)

Assistant Professor (saeed@uok.edu.pk)

Areas of Interest : Operating Systems, Networking, Parallel Computing

Dr. Syed Asim Ali; Ph.D (KU), M.A. (KU), BS (KU)

Assistant Professor (asim@uok.edu.pk)

Areas of Interest : Human Computer Interaction, Multimedia Communication

Dr. Farhan Ahmed Siddiqui; Ph.D (Comp. Sc.), MBA (KU), BS (SSUET), MCP, MPEC, MIEEE

Assistant Professor (farhan@uok.edu.pk)

Areas of Interest: Computer Networks & Communication, WSN, Programming & OOP, Financial Accounting / Management

Dr. Humera Bashir

Assistant Professor (humera.azam@uok.edu.pk)

Computer Graphics, Image Processing, Software Engineering, Artificial Intelligence, Health Information Systems

Dr. Humera Tariq; Postdoc (UoM-USA) , MCS-MS/PhD (UoK-Pakistan), B.E (NED-Pakistan)

Assistant Professor (humera@uok.edu.pk)

Areas of Interest: Deep Learning, Machine Learning, Bio-Medical Image Processing, Computer Vision, Computer Graphics, Simulation

Dr. Shaista Rais

Lecturer (shaista.rais@uok.edu.pk)

Areas of Interest: Modeling and Simulation, Artificial Intelligence, Health Information Systems, Software Engineering.

DEPARTMENT OF FOOD SCIENCE & TECHNOLOGY

INTRODUCTION OF DEPARTMENT

The department of Food Science & Technology was established in 1994 to cater the needs of the fast growing food industries in the country and also to meet the ever growing demand for food scientists and technologists in government, research and educational establishments. Its mission is to provide high-quality education and training to students and prepare them for careers in the food industry, academia, or government, to conduct basic and applied research for the ultimate benefit of the food industry and consumers and to provide assistance and services to the food industry. Over the years, the department has developed strong links with the industry. Recently the department has established an Industrial Advisory Board comprising members from reputed Food industries to help us shape our curriculum from an industrial perspective to keep it relevant and to ensure that our curriculum reflect current developments and industry skills requirements. The department has always been actively engaged in providing advisory services and consultancy to food industries and also organizing seminars, conferences and training courses in collaboration with industries. In addition to food industries, the department also works in collaboration with government organizations such as PSQCA, PCSIR, PARC, SFA and different NGOs. Realizing the importance of human nutrition in daily life and emergencies like earthquake, drought, famine and man-made disasters, department offered another Four year BS Program entitled Human Nutrition & Dietetics from 2023. The Curriculum for the program has been developed in view of local as well as global health needs.

POSTGRADUATE PROGRAMS AVAILABLE

The Department of Food Science and Technology offers a variety of postgraduate programs designed to advance knowledge and professional skills in specialized areas of Food Science and Technology. The available programs include:

1. M.Phil in Food Science and Technology
2. Ph.D. in Food Science and Technology

PROGRAMS' OBJECTIVE

1. **To Develop Advanced Expertise in Food Science and Technology:** Equip students with advanced knowledge and specialized skills necessary for addressing complex challenges in food science, enabling them to become leaders and innovators in the food industry, academia, and research institutions.
2. **To Foster Cutting-Edge Research and Innovation:** Encourage and support postgraduate students in conducting pioneering research that contributes to scientific advancements in food technology, food safety, and nutrition, leading to peer-reviewed publications and patents.
3. **To Enhance Interdisciplinary Collaboration:** Promote interdisciplinary research and collaboration among various departments, institutions, and industries, facilitating a holistic approach to solving global food-related issues and advancing sustainable food systems.
4. **To Address Global Food Security and Sustainability:** Develop strategies and technologies aimed at improving food security, minimizing post-harvest losses, and enhancing the sustainability of food production and processing systems.
5. **To Advance Food Safety and Quality Management Systems:** Train students in advanced methodologies and technologies for ensuring food safety, quality assurance, and compliance with international standards and regulations.
6. **To Develop Innovative and Nutrient-Rich Food Products:** Engage in the development and commercialization of novel, health-promoting food products with enhanced nutritional profiles and functional benefits for diverse consumer needs.
7. **To Promote Sustainable Utilization of Food Industry By-Products:** Investigate and implement sustainable practices for recycling and repurposing food industry by-products and waste into valuable new products, contributing to a circular economy.
8. **To Provide Expert Consultancy and Advanced Analytical Services:** Offer high-level consultancy and advisory services to the food industry, utilizing state-of-the-art analytical techniques for the assessment, improvement, and development of food products.

9. **To Foster International Collaboration and Exchange:** Establish and maintain strategic partnerships with leading international food science departments, research centers, and industries to facilitate the exchange of knowledge, students, and staff, and to participate in global research initiatives.
10. **To Promote Public Awareness and Education on Food Safety and Nutrition:** Organize advanced educational programs, seminars, and workshops aimed at raising awareness among professionals and the public about the importance of safe, nutritious, and sustainable food practices.

AIMS AND SCOPE OF PROGRAMS

Aim

The aim of the postgraduate program in Food Science and Technology is to develop highly skilled professionals and researchers who are equipped with advanced knowledge and practical expertise to drive innovation, enhance food safety and quality, and address global food challenges. The program aims to prepare graduates for leadership roles in academia, industry, and research institutions, fostering a deep understanding of food systems and their impact on health, sustainability, and the economy.

Scope

The postgraduate program in Food Science and Technology encompasses a comprehensive and interdisciplinary curriculum designed to integrate advanced theoretical concepts with practical applications in the field. The scope of the program includes:

1. **Advanced Food Science and Technology**
 - In-depth study of the chemical, physical, microbiological, and nutritional aspects of food.
 - Exploration of advanced food processing and preservation techniques to enhance food quality and safety.
2. **Research and Innovation**
 - Encouragement of cutting-edge research that addresses critical issues such as food security, post-harvest losses, and sustainable food systems.
 - Development of innovative food products with improved nutritional profiles and functional benefits.
3. **Sustainable Practices and Environmental Impact**
 - Examination of sustainable practices in food production, processing, and distribution.
 - Strategies for minimizing the environmental impact of food industry operations and reducing food waste.
4. **Food Safety and Quality Assurance**
 - Advanced training in food safety protocols, quality control measures, and compliance with international food standards and regulations.
 - Utilization of state-of-the-art analytical techniques for food product diagnostics and improvement.
5. **Interdisciplinary Collaboration**
 - Promotion of interdisciplinary projects and partnerships with other academic departments, research institutions, and industry stakeholders.
 - Facilitation of international collaborations and exchange programs to broaden the scope and impact of research and education.
6. **Professional Development and Career Preparation**
 - Preparation of graduates for diverse career opportunities in academia, research, industry, and government sectors.
 - Emphasis on leadership, innovation, and problem-solving skills to address complex food-related challenges.
7. **Public Health and Nutrition**
 - Focus on the development and promotion of safe, nutritious, and health-promoting food products.
 - Public education initiatives to raise awareness about the importance of food safety, nutrition, and sustainable consumption practices.

The program is designed to produce research scholars who are not only knowledgeable and skilled but also innovative thinkers and leaders capable of making significant contributions to the field of food science and technology on a global scale.

REGULAR FACULTY MEMBERS AND THEIR AREA OF EXPERTISE

The Department of Food Science and Technology boasts a diverse team of experienced and dedicated faculty members who are experts in various fields of food sciences and advanced technologies.

S. No.	Faculty Members	Area of Expertise
1	Dr. Abdul Haq Associate Professor & Chairman	Food Science, Biopolymer, Green Chemistry
2	Dr. S. M. Ghufraan Saeed Associate Professor	Oils and Fats, Nutraceuticals
3	Dr. Tahira Mohsin Associate Professor	Starch Technology, Cereal Technology, Biodegradable films and composite flour
4	Dr. Feroz Alam Assistant Professor	Biodegradable film, Complex carbohydrates, Bioactive compounds
5	Dr. Zubala Lutfi Assistant Professor	Waste, Cereals, Hydrocolloid, Nutrition
6	Dr. Anjum Nawab Assistant Professor	Biodegradable packaging film, Modified starch
7	Dr. Lubna Mobin Assistant Professor	Bioactive components, encapsulation, micronutrients fortification
8	Dr. Mariam Sheikh Assistant Professor	Biodegradable and active food packaging, chemical modification of starch, cereal based product development
9	Dr. Abeera Moin Assistant Professor	Food Waste valorization, Biopolymers, Biodegradable packaging, Consumer acceptance

RESEARCH FACILITIES AVAILABLE

The department is equipped with state-of-the-art research facilities that support a wide range of academic and industrial research activities:

- Seminar and Digital Library:** The department's library houses an extensive collection of books, journals, and digital resources in chemical engineering and related fields. The digital library provides access to recent research publications.
- Laboratories:** We have well-equipped laboratories for research in areas such as edible lipids, food processing, food engineering, food Waste valorization, biopolymers, biodegradable packaging. These include pilot-scale reactors, analytical instruments, and specialized equipment for characterization and product development.
- Infrastructure**
The Department of Food Science and Technology is a modern facility designed to support both teaching and research. The infrastructure includes:
- Air conditioned Lecture Halls:** Equipped with multimedia facilities for effective teaching and presentations. Along with regular classes these halls are also used for workshops, conferences, and collaborative research activities.
- Research Labs:** Dedicated spaces for conducting experimental research and project work.
- Faculty Offices:** Designed to facilitate interactions between students and faculty for academic guidance and research discussions.
- Student Clubs and Leisure activities:** Basketball court, Sports society, Rotract Club, Competitions and Industrial visits.

INFRASTRUCTURE

The Department has the infrastructure to support new trends in learning. All lecture rooms, labs, and classrooms provide support for e-learning, and faculty members frequently use these facilities.

The infrastructure and facilities that support new trends in learning, along with their adequacy are given below:

1. Modern computer Labs with latest technologies are available in the department
2. Advanced Visual Aids and tools.

AVENUES FOR RESEARCH

These research avenues offer diverse opportunities for advancing knowledge and innovation in the field of food science and technology, addressing critical challenges, and contributing to the development of sustainable and healthy food systems. Some of the research avenues (not limited to) are enlisted below;

1. Food Safety and Microbiology
2. Food Processing and Preservation
3. Nutritional Science and Functional Foods
4. Sustainable Food Systems
5. Food Chemistry and Biochemistry
6. Food Engineering and Technology
7. Food Policy and Economics
8. Sensory Science and Consumer Behavior
9. Food Allergies and Intolerances
10. Food Industry By-Products and Waste Utilization
11. Advanced Analytical Techniques
12. Food-Environment Interactions
13. Biodegradable packaging

DEPARTMENT OF GENETICS

INTRODUCTION OF THE DEPARTMENT

The Department of Genetics was established in 1969 at the University of Karachi that offers the degrees of BS, MS, M.Phil. and Ph.D. in Genetics. All of these programs are conducted by a faculty consisting of highly qualified individuals, many having degrees and research experience from technologically advanced countries.

The Department of Genetics comprises a diverse community of graduate and postgraduate scholars dedicated to exploring research avenues within microbial, plant, animal, and human genetics. A sizeable number of research students are working on problems of national importance that may have a bearing on health, agriculture, environmental, biotechnology, chemical industrial sectors to name a few.

Our department's mission is to foster an environment where genetics, as a foundational science, lays the groundwork for nurturing human potential, benefiting individuals, society at large, and the environment. The department takes great pride in maintaining an intellectual and cognitive atmosphere in order to promote learning and to provide to its students an experience they will remember fondly for the rest of their lives.

From deciphering the genetic roots of diseases to exploring the mechanisms driving diversity in nature, our curriculum provides a comprehensive grasp of genetics across various organisms and systems. Through practical experiments, fieldwork, and interactive lectures, students develop a profound appreciation for the role genetics plays in shaping the natural world. Many of the department's graduates are serving the country both in Pakistan and abroad at positions important with respect to the scientific, technological, and economic development of Pakistan.

Graduate students learn in a conducive and comfortable environment with modern teaching aids in classrooms in the state of the art building. Recently the departmental labs are equipped with the latest instruments worth millions of rupees funded by the Higher Education Commission of Pakistan (HEC). Assessments are done as per university policies.

FACULTY MEMBERS

Professors

Dr. Maqsood A. Ansari M.Sc., (Karachi), MS, PBC Biotech (USA), Ph.D. (UK)
(Chairman)

Dr. Uzma Badar M.Sc., Ph.D. (Karachi)

Associate Professors

Dr. Fouad M. Qureshi M.Sc., Ph.D. (Karachi)

Dr. Simeen Mansoor M.Sc., Ph.D. (Karachi)

Dr. Erum Shoaib M.Sc., Ph.D. (Karachi)

Assistant Professors

Dr. Nadia Khan M.Sc., Ph.D. (Karachi)

Dr. Ayesha Riaz M.Sc., (Karachi), MS, (USA), Ph.D. (UK)

Dr. Shumaiza Anis M.Sc., Ph.D. (Karachi)

Dr. Rifat Z. Ahmed M.Sc., Ph.D. (Karachi)

Dr. Sarwat Afshan M.Sc., Ph.D. (Karachi)

Dr. Aribah Naz M.Sc., Ph.D. (Karachi)

Adjunct Professor

Dr. Obaid Y. Khan M.Sc., (Karachi), Ph.D. (UK)

DEPARTMENT OF GEOGRAPHY

INTRODUCTION OF THE DEPARTMENT

Department of Geography is one of the pioneer departments of University of Karachi. It was established in 1952 under the chairmanship of Professor Dr. Manick B. Pithawala. Professor Oscar Schmider was one of the leading German Geographers who served as Chairman in the department. Department has been providing teaching and research in various fields of Geography for the last sixty years. Department has been served by leading Pakistani Geographers who were educated in USA, UK, Germany, India etc. The Department has an excellent academic history. The Department of Geography is the only Department among Universities of Pakistan which hosted International Geography Union (IGU) Conference. More than twenty Professors have served the Department as Chairpersons. Presently Prof. Dr. Sheeba Afsar is Chairperson. The Department has qualified academic staff which includes Professors, Assistant Professors and Visiting Faculty.

Department of Geography organizes teaching for the Graduate and Post Graduate courses including M. Phil and Ph.D. research programs. The main objective of the department is to train the people for geographical research in both aspects of pure and social sciences. In this way research becomes more helpful for planning, management, allocation, distribution and evaluation of resources and their problems.

POSTGRADUATE PROGRAMS AVAILABLE

The Department of Geography at the University of Karachi, established in 1952, has a rich history of leadership from renowned geographers. It offers comprehensive teaching and research in various geographic fields, producing a large number of graduates working globally. The department, which has the highest number of Ph.D. / M.Phil. Graduates in Pakistan, offers graduate and post-graduate programs and has hosted several geography conferences. It emphasizes practical geographical research skills and maintains strong international academic partnerships.

1. M.Phil
2. Ph.D

PROGRAM OBJECTIVES

The program's objectives align with the university mission and are measured through comprehensive assessments. Strategies include practical field trainings, integrating advanced technologies, industry collaboration, and regular program updates to achieve these objectives.

AIMS AND SCOPE OF THE PROGRAM

The Department of Geography at the University of Karachi aims to make students highly professional in the field of Geography, GIS technology, and Remote Sensing technology. They provide field survey skills, professional knowledge, and training opportunities for teachers. The program's objectives align with the university mission and are measured through comprehensive assessments. Strategies include practical field trainings, integrating advanced technologies, industry collaboration, and regular program updates to achieve these objectives.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION

Name of Teacher	Areas
Prof. Dr. Shamshad Akhtar	Physical Geography, Quantitative Methods, Population and Settlement, Muslim World
Prof. Dr. Sheeba Afsar	Natural Hazard Management, Advance Cartography, Spatial Analysis, Map and Air Photo Analysis, Physical Geography and Human Geography Specialization: Appraisal of Urbanization and Quality of Life by Employing Remote Sensing and GIS Techniques: A Case Study of Karachi
Dr. Saima Shaikh	Urban Ecology, Diseases Ecology, Cartography, RS, GIS and Geography of Pakistan
Dr. Anila Kausar	Agro Climatology, Urban Geography, Regional Geography and Urban Land use Planning

Dr. Yasmeen Anis	Climatology, Oceanography, Coastal Morphology, Human Geography and Political Geography Specialization: Impact of Urbanization along coastal areas, through remote sensing techniques Course Supervisor: Coastal Geomorphology # 763 and Fluvial Geomorphology # 760 Coastal Geomorphology #763: Physical features along the coast and their associated environment, ecosystems, habitats, estuarine systems, creeks, Mangroves: their importance. Impact of natural hazards over the coast and role of particular topography for protection against these hazards, impact of urbanization over the coast and its modification accordingly. Fluvial Geomorphology # 760: The Physical features associated with the river system: drainage patterns, the head works, valleys and interfluvies, old and active floodplains, channel Geometry the deltas and estuaries. River and it types: seasonal and perennial rivers. Role of river system in supporting a particular habitat, in the modification of a landscape, the floods and its modification in particular landscape, causes, major river systems and associated irrigation or agriculture and Impact of urbanization over a river system, Urban flooding.
Dr. Salman Zubair	Metropolitan Transportation, Locational Analysis and Economic Geography, Geography of Pakistan and Culture Geography
Dr. Lubna Ghazal	Water Resources and Management, Geomorphology, Geography of Pakistan, Economic Geography, Ecology of Natural Resources, Climatology, Arid-land: Desertification, EIA and Management
Dr. Imran Ahmed Khan	GIS and Public Health, Biogeography, Diseases Ecology, Human Geography, Medical Geography and Economic Geography

RESEARCH FACILITIES AVAILABLE

Computer Laboratory

Computer laboratory of the Department of Geography was established in March, 1988. Initially it consisted of three PCs and a printer. Currently this laboratory has fifteen computers, Intel Core 2 duo. Department has purchased MapInfo Professional Software, which is one of the state-of-art packages for GIS and Remote Sensing application. Presently, it is being used for the purpose of teaching and research, in the fields of computer aided cartography, GIS and Remote Sensing.

Museum

On the ground floor of the Department there is a Museum containing relic of prehistoric cultures of Pakistan. These have been collected from an area extending to 120 miles around Karachi. The material ranges from early Paleolithic age through Middle and upper Paleolithic, Mesolithic and Neolithic to Harappan Age. The museum also displays vertebrate fossils of Miocene Age derived from Sindh Kohistan and coastal area of Sindh.

Seminar library

The department contains a fully-fledged seminar library which offers books, Atlases, different types of Maps and topographic sheets to students. A section donated by Dr. I. H. Zaidi and Dr. M. I. Siddiqui is also available having their valuable collection of books on various fields of Geography. Different precious research journals and published material are also available.

INFRASTRUCTURE

The Department is housed in a two storied building where lecture rooms, laboratories, seminar library and staff rooms are provided. The seminar library caters for the immediate reading needs of the students and is provided with 1000 books form donations. The Map Library of the Department is well equipped with topographical maps of various areas on various scales. Total number of such maps is about 2000, aerial photographs and satellite imageries are available in the Department.

AVENUES FOR RESEARCH

Urban Landscape Ecology, Hydro-Geography, Mobile technology for Geographic Data Collection, Global Positioning System (GPS), Spatial Database Programming in Geography, Foundations of Drone Mapping, Urban Agriculture and Regional Food Systems, Climate Change Studies, Digital Cartography, Natural Hazard Management and Road Safety in Geography.

ANY OTHER RELEVANT INFORMATION

Geography Students generally find useful employment in some major fields, i.e. education, administration, and private organization & other cellular companies, K-Electric, SSGC, Archeological Department, Planning Commission, and provincial departments of planning and development have employed geographers for their Geo-Spatial skills. Private firms, and Non-Governmental Organizations i.e. ECIL Consulting Services, Cabernet, Fiber optics, (NGOs) also utilize the expertise of geographers. There are various fields of application in Pakistan where geographer can explore job opportunities successfully to survive and Empowering Communities.

RESEARCH ASSISTANT

1. **Land surveyor:** In SOP, KDA, LDA, CDA and private surveying organizations
2. **Transport Planners:** In TEB (Traffic Engineering Bureau) Karachi Planners for Katchi abadies, KMC (Karachi Metropolitan Corporation) Sindh Kachi Abadies authority
3. **Cartographer:** In survey of Pakistan, in Geological survey of Pakistan, in Civil Aviation, Meteorological department, SUPARCO, in Oil Companies, Geodetic Surveying
4. **Consultants:** In engineering firms like ECIL, EA/ES, BlueChip Intl., Proplan etc., Soil Survey of Pakistan, a number of NGOs, Atlas publishing companies (like OUP, FEP etc.)
5. **Others:** Population census organization, statistical bureau, AKRSP, (Agha Khan Rural Support Program)

DEPARTMENT OF GEOLOGY

INTRODUCTION OF THE DEPARTMENT

Department of Geology was established in 1956 in old building of the campus and shifted to the present building in beginning; most of the students were foreigners from different countries like Iran, Jordan, Iraq and Somalia. Department of Geology has good relationship with the oil exploration companies like PPL, OGDCL, UTP, ENI, UEPL, POL and OMV etc. as well as services companies like Weatherford, Schlumberger, Baker Hughes, Geoservices and Exlog etc. The department has an impressive geology museum which houses a large collection of valuable rocks fossils and mineral specimen. Since a large number of employees and key position holders in oil and gas sector are graduates of Department of Geology, University of Karachi. Being alumni of the department, all pass out have established cordial relationship with department of Geology for the promotion of subject and Department. In recent past, Department has signed MoUs with PPL, SECMEC Engro, HDIP and Goldensun Petroleum to strengthen the department.

Postgraduate Programs Available:

1. M.S
2. M.Phil
3. Ph.D

PROGRAMS OBJECTIVES

M.Phil/ Ph.D. program is designed to:

1. Produce a research worker who has acquired good all round knowledge of geology, in depth knowledge in specialized areas, and the skills to carry out original research.
2. Provide persons who have excellence to carry out research work and give rise new endeavors either in exploration of natural resources or remediation of natural hazards for serving relevant fields and humanity.
3. Establish a cadre of specialists and professionals in various fields of geology who can provide effective leadership in conducting and supervising high level research in various fields of geology.
4. Provide opportunities of professional growth and development to scholars in general and in-service personnel in particular.

A large number of Ph. D. and M. Phil degrees have so far been awarded in the last few years. At present scholars are registered for M. Phil and Ph. D. degrees in the fields of economic geology, geochemistry, sedimentology, engineering geology, hydrogeology, petroleum geology and marine geology. In spite of heavy load of teaching, the senior teachers undertake funded research project from different agencies. The staff members of the department have published a large number of research articles in international scientific journals within country and abroad.

AIMS AND SCOPE OF THE PROGRAM

Department aims to produce effective and responsible leaders by creating insightful knowledge and inspiring minds in dialogue with the world around. We focus on providing competency-driven education, leveraging our resources to provide experiential learning, committing to continuous improvement and linking our research and outreach efforts to the driving forces of our economy.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION/EXPERTISE

<u>Name of Faculty Member</u>	<u>Designation</u>	<u>Area of specialization</u>
Dr. Erum Bashir	Professor & Chairperson	Geochemistry/ Mineralogy
Dr. Sadaf aseem	Assistant Professor	Hydrogeology
Dr. Adnan Khan	Assistant Professor	Hydrogeology/Environment Geology
Dr. Sadaf Ismail	Assistant Professor	Petroleum Geology
Dr. Shella Bano	Assistant Professor	Soil & Groundwater
Dr. Bushra Shabab	Assistant Professor	Geochemistry, Stratigraphy
Dr. Maria Kaleem	Assistant Professor	Hydrogeochemistry, Petrography
Mr. Muhammad Arsalan	Lecturer	Remote Sensing & GIS

RESEARCH FACILITIES AVAILABLE (LIBRARY, LABORATORY, COMPUTER LAB)

1. **Library:** A large numbers of books available in our Seminar Library. The Department has a separate Reference Library for Post Graduate Students where a large number of books for post graduate students are also available. Students also avail Dr. Mahmud Husain Library. University Library Books Grant is available to acquire and maintain libraries.
2. **Laboratory Facilities:** Following subjects laboratories are available for research purposes. Geochemistry, Sedimentology, Optical Mineralogy, Stratigraphy and Paleontology, Petrology & Mineralogy, Marine Geology, Engineering Geology, Thin section Preparation Lab, Photomicrography Lab, Computer Laboratory, Environmental Geology, Geophysics (VES Resistivity /Terra meter), Central Analytical Lab (AAS, Spectrophotometer, Flame Photometer).
3. **Computing Facilities:** Department of Geology has adequate networking computer facilities.
4. **Multimedia / Projectors:** The department has 06 multimedia and 02 Laptops (obsolete version)
5. **Website:** The university website <http://www.uok.edu.pk> has a link for Department of Geology and each faculty member has its own official e-mail address.
6. **Internet:** The department has Internet facility and all computers are interconnected with main communication.

INFRASTRUCTURE

The department has three spacious buildings in which teacher's offices, class rooms, laboratories and libraries are present. Department also has an impressive geology museum which houses a large collection of valuable rocks, fossils and minerals specimen. Around one hundred seating capacity Auditorium is also functioning in the Department. Department have three gardens with large numbers of plantation and flowers.

AVENUES OF RESEARCH

Mineral, Oil & Gas, Energy and Coal sectors. Beside these, other research includes fields of Hydrogeology, Geochemistry, Engineering Geology, Geophysics, Sedimentology, Oceanography, Soil and Rock etc.

DEPARTMENT OF HEALTH, PHYSICAL EDUCATION & SPORTS SCIENCES

INTRODUCTION

Physical Education is a way of education by means of physical movement. From Molecular Biology to Material Sciences nothing of value can be expected from a society if the teaching and research in Health and physical education is ignored or if it is relegated to a secondary status. Being a natural phenomenon, movement is the essence of life and the entire mechanism of human body is geared to it. Experts believe that exercise is also as important as food. While food provides necessary energy for the body, exercise tones up the human organism so that it can utilize this energy in a better way. Since a healthy body is necessary for a healthy mind, physical education as a discipline is more intimately concerned with human development than educational disciplines.

Physical fitness is not only a concept but it reflects a practical approach to human betterment in terms of health. The survival of a nation is undoubtedly dependent on physical fitness which ensures general proficiency in all walks of life, and is inevitably linked with better economic growth. It is needless to emphasize the importance of physical education as a discipline as it already stands recognized at the university level. All the advanced countries, realizing its virtues and benefits, have included this subject as an essential part of curriculum at all levels of education.

The Department has been working strenuously towards a program of teaching and research to meet the needs of the country in Physical Education & Sports Sciences.

MISSION STATEMENT OF DEPARTMENT HPES

To remain committed to achieve the highest standards in teaching, learning & research in accordance with the international standard.

POST GRADUATE PROGRAM AVAILABLE

1. M.Phil
2. Ph.D

OBJECTIVES

1. To create opportunities for higher learning, in the subject of Health and Physical Education at post graduate level.
2. To inculcate the observance of law and dignity of labor in the students through health and physical education.
3. To integrate health and physical education with the general educational set-up of the university.
4. To promote physical, mental, social, emotional and spiritual growth of the youth in a suitable environment.
5. To inculcate a sense of self-awareness, self-confidence and self-reliance in the youth.
6. To encourage the students to make purposeful beneficial use of their surplus energies and leisure.
7. To develop such qualities as to make an individual a patriotic, responsible, disciplined, competent and productive citizen.
8. To develop in the students the capacity for intelligent self-direction in matters of health and recreation.
9. To encourage personal hygiene and good living for fitness and wellbeing.
10. To offer activities leading to character building, development of cooperative attitudes, behavioral adjustments and qualities of leadership and sportsmanship.
11. To offer programs leading to the learning of skills and techniques of various games and sports.
12. To highlight the significance and importance of Islamic rituals with reference to movement.

RESEARCH FACILITIES

The Department is equipped with all the required practical facilities. A variety of sports facilities including playing fields for cricket, hockey, football and athletics, a squash court and indoor games facilities in the gymnasium are also available on the campus. Competitive sports for both male and female students are organized at inter-departmental, inter-collegiate and inter-university levels through a variety of sports like hockey, football, cricket, badminton, athletics, etc, every year. The Department have a fully equipped Human Performance Laboratory to incorporate the latest techniques and performance analysis methodologies.

SPECIFIC RESEARCH AREAS (AVENUES OF RESEARCH)

The Department of Health, Physical Education & Sports Sciences has a broad scope for qualitative as well as quantitative research in the field of Kinesiology, Exercise Physiology, Socio economic impact of Physical Education & Sports Sciences

AIMS & SCOPE

1. Physical education develops the alertness of mind.
2. Physical education provides knowledge about health and its hazards and communicable and non communicable diseases.
3. Through physical activities leisure time can be utilized properly.
4. Through physical education human body can be developed in good proportion.

CAREER OPTIONS IN PHYSICAL EDUCATION

Career Options

Physical Education Teacher
Personal Sports Trainer
Fitness Instructor
Aerobics Instructor

Employment Avenues

Schools, Colleges, Universities
Freelancing, recreation centers
Fitness Studios, Gyms, Community centers
Health clubs, Pilates studios

FACULTY MEMBERS / DRC

FACULTY					
DETAILS OF M.Phil / Ph.D FACULTY					
S.No.	Name	Designation	Permanent / Visiting	Title of degree	Expertise / Area of Specialization
1.	Prof. Dr. Basit Ansari	Professor	Permanent	Ph.D	Exercise Physiology
2.	Prof. Dr. Shah Ali-ul-Qadar	Professor	Permanent	Ph.D	Muscular Protein Enzyme Technology Chemical Biochemistry
3.	Prof. Dr. Anila Amber Malik	Professor	Permanent	Ph.D	Sports Psychology, Psychometrics.
4.	Prof. Dr. Haris Shoaib	Professor	Permanent	Ph.D	Pharmaceutics
5.	Dr. Mazahir Zia Shaikh-PT	Assistant Professor	Permanent	M.Phil	Physical Therapy
6.	Ms. Shazia Tabussum	Assistant Professor	Permanent	M.Phil	Physical Fitness
7.	Dr. Humayun Imran Azeemi-PT	Assistant Professor	Permanent	M.Phil	Physical Therapy

DEPARTMENT OF HEALTH, PHYSICAL EDUCATION & SPORTS SCIENCES **(Doctor of Physical Therapy Program)**

INTRODUCTION

Physical Education is a way of education by means of physical movement. From Molecular Biology to Material Sciences nothing of value can be expected from a society if the teaching and research in Health and physical education is ignored or if it is relegated to a secondary status. Being a natural phenomenon, movement is the essence of life and the entire mechanism of human body is geared to it. Experts believe that exercise is also as important as food. While food provides necessary energy for the body, exercise tones up the human organism so that it can utilize this energy in a better way. Since a healthy body is necessary for a healthy mind, physical education as a discipline is more intimately concerned with human development than educational disciplines.

Physical fitness is not only a concept but it reflects a practical approach to human betterment in terms of health. The survival of a nation is undoubtedly dependent on physical fitness which ensures general proficiency in all walks of life, and is inevitably linked with better economic growth. It is needless to emphasize the importance of physical education as a discipline as it already stands recognized at the university level. All the advanced countries, realizing its virtues and benefits, have included this subject as an essential part of curriculum at all levels of education.

The Department has been working strenuously towards a program of teaching and research to meet the needs of the country in Physical Education & Sports Sciences.

AN OVERVIEW OF THE PROGRAM

This is a time of innovation in medical education when changes in health care delivery systems have profound implications for treatment. There is public and professional demand for more relevance in educational programs with funding bodies and government requiring accountability and quality in education. Throughout the world there is an increasing recognition of the need for Physical therapist who work in the health care system to obtain training for better treatment. The Ph.D Physical therapy has been designed to meet the country's need for the Research oriented Physical therapists who are better equipped with updated knowledge and advance skills.

MISSION OF DEPARTMENT

Department of Health, Physical Education and Sports Sciences is committed to produce competent physical therapist by providing knowledge and skilled based educational so inculcating the element of research with intent to improve patient care.

MISSION OF POSTGRADUATE PROGRAM

The post graduate program aims to inculcate the element of research among physical therapy professionals thus promoting evidence based physical therapy practices for the better health care through clinical and academic services introduction.

SCOPE OF PROGRAM

Physical therapy services are worldwide considered as an integral part of health care services, starting from the emergency or intensive care unit to community-based rehabilitation, health and wellness and fitness programs. Physical Therapist is now a part of health care teams in primary, secondary and tertiary care units thus inculcating research and evidence based physical therapy practices with the intent to improve quality of education and patient care. Student education and training at the university is directed towards producing graduates who will contribute towards enhancing health education and the promotion of healthy life styles in populations from all socio-economic strata of Pakistan.

POST GRADUATE PROGRAM AVAILABLE

1. M.Phil
2. Ph.D in Physical Therapy

OBJECTIVES OF THE Ph.D DEGREE PROGRAM***Ph.D Physical therapy***

1. Ph.D in Physiotherapy has been planned to furnish qualified candidates with the mandatory skills needed to operate in the field of Medicinal Chemistry of Various places and associated enterprises along with government bodies.
2. Ph.D in Physical Therapy is a doctorate program.
3. The duration of this program is of five (05) years.
4. Ph.D in Physical Therapy Program provides training and experiences that prop up interdisciplinary subjects learning, encourage critical and theory-based problem-solving skills to address to help peoples.
5. Candidates who are under this program are taught basic psychology to enhance understanding of people's mental state.
6. A candidate who graduates Ph.D in Physical Therapy can join research programs, Lecturer, Therapist, etc.
7. The candidate earns handsome amount in the initial years.

WHY STUDY PH.D PHYSICAL THERAPY?

1. Ph.D in Physical Therapy program is advantageous for candidates who want to achieve leadership roles in the field of Physical therapy. During a degree in Physical Therapy, Candidates are allowed to work in various organizations and practices.
2. The main purpose of the Ph.D in Physical Therapy program is to make candidates responsive and achieve skillful knowledge obtainable on the subject and distribute their theses and earn accordingly.
3. Ph.D in Physical Therapy facilitates the integration of practical fieldwork with academic subjects.

PH.D PHYSICAL THERAPIST JOB PROFILES

After the successful completion of the course, one can opt to work either in the public sector or the private. Both of these sectors provide good opportunities.

Physical Therapy is a fruitful and rewarding career, after the candidates clear the license examination of a particular country, higher opportunities open up for a fresher to seek work in the healthcare industry.

A Ph.D physical therapist is responsible to prescribe remedial exercises for easing pain in muscles and bones. They also help in strengthening muscles after major accidents to regain lost control. A physical therapist also ensures that the patient is balanced neurologically. Some of the Job Positions where the graduate's work are as mentioned below:

Job Profiles	Job Description
Physical Therapist	A physical therapist is responsible for analyzing patients. They are also Responsible for the psychoanalysis and understanding of statistics for preparation and front actions and agenda initiative.
Lecturer	A Lecturer works with colleges and universities and guides future candidates towards learning and implementing the basics of physical therapy. They also facilities to help improve the well-being of people as a community.
Researcher	Researcher Physical Therapists show analysis and then administer statistics beginning early days teaching disability, and physical condition projects. They are also accountable for adding professional status to the institute.
Consultant	They provide direct services to patients who seek treatment under them.

PH.D PHYSICAL THERAPIST FUTURE SCOPE

1. A candidate after successful completion of a degree in Ph.D in Physical Therapy is eligible to take Job Opportunities.

2. Some of the job roles that are available after the booming achievement of this Program will be including programs, Lecturer, Therapist, etc. The candidate earns good in the initial years after some experience and practical knowledge the salary can hike up.
3. The prospect of Ph.D in Physical Therapy program is high and estimated to increase in the upcoming time.
4. Ph.D is the highest degree a person can obtain in Ph.D in the Physical Therapy field, individual can do research work in the area.
5. To enable the post graduate students to upgrade their knowledge and skills in the field of Physical Therapy
6. To acquire the understanding of advance clinical skills across the field of physical therapy; critically evaluate and contribute to research in their clinical area.
7. To incorporate evidence based physical therapy practice by learning clinical decision making, patient preference and practice knowledge
8. To enable students to enhance the research and development in the field of Physical Therapy
9. To improve the student's abilities to develop and carry researches that shall lead to establish evidence-based practices for the prevention and treatment of impairments, functional limitations and disability.
10. To provide the students with relevant knowledge and skills to assume positions of responsibility and leadership in the field of Physical Therapy.

Conclusively Ph.D in Physical Therapy Program provides training and experiences that prop up interdisciplinary learning, encourage critical and theory-based problem-solving skills to address psychology, and foster the growth of people's minds.

SCOPE OF THE DEGREE PROGRAM (MARKET, SOCIAL AND EMPLOYABILITY)

1. The Ph.D is a Philosophical Doctorate. Ph.D in Physical Therapy is a doctorate program of 3 years. This is the degree required to become a professor at a University. The Ph.D is the highest degree offered by a University.
2. Physical Therapy Ph.Ds are given many positions like Researcher, Therapy Manager, Research Assistant Customer Care Manager, Sports Physio Rehabilitator, Self Employed Private Physical Therapist and more.
3. Most PTs with a Ph.D working full time can expect to earn in the top quarter of salaries, which is over \$100,000.
4. Earning a Ph.D gives credibility in the workplace as well as highly valuable transferable skills, having this degree opens opportunities in fields or positions where having an advanced degree is recommended or required.
5. Completing a Ph.D in physical therapy program is a respected accomplishment and for good reason. Those who earn a Ph.D in Physical Therapy have invested considerable time and effort in completing their degree program. Completion is often associated with prestige and credibility along with a personal sense of accomplishment that comes from reaching the highest level of academic achievement.
6. Not only will this accomplishment impact personally, but it will also influence professional life. Many employers value the knowledge and skills that come with earning a Ph.D. This will give credibility in the job market, showing both employers and coworkers that individual is a hardworking and dedicated professional.
7. While earning Ph.D individuals will not only gain specialized knowledge within physical therapy field but also transferable skills that they can use, no matter what career path they ultimately follow. Learning proper research methods and how best to take notes is an invaluable skill that a Ph.D physical therapist can use in most workplace settings.
8. Ph.D physical therapist will also learn important communication skills, including presenting and writing. Employers value job candidates who have good writing skills and who are comfortable presenting information to other employees. Communication skills are critical in every workplace and are constantly ranked among the most important skills for successful professionals.
9. Ph.D physical therapist gains credibility in the workplace as well as highly valuable transferable skills, they will find that having this degree opens opportunities in fields or positions where having an advanced degree is recommended or required.

10. It's a common misconception that a Ph.D only prepares professionals to enter the world of academia. A Ph.D physical therapist can prepare individuals to enter a profession in any aspect of physical therapy, depending on the focus on the degree. Although a Ph.D physical therapist can still choose to pursue a career as a professor or researcher at a university, but he can also look into a career in growing Hospitals industry or a higher-level position in his current profession.
11. For example, with a Ph.D in physical therapy would be able to pursue careers in a variety of professions, such as publishing or grant writing. A physical therapy Ph.D would prepare individuals for a variety of data analytic positions or even consulting. These are just a few examples of jobs outside academia that a physical therapist could pursue with a Ph.D.
12. So, earning a Ph.D in physical therapy can open higher-level management positions or a better salary. Given that physical therapist will now have a highly desired skill set, they might also be able to advance more quickly in their current career and gain more benefits after earning this advanced degree.
13. Throughout the process of earning Ph.D, a physical therapist will most likely be collaborating with professors on research efforts and writing a dissertation. Getting to know professors in the current or desired field is a great opportunity, whether physical therapist looking for a job or not. Networking with professionals is often how people find job opportunities that they wouldn't have found just looking online.
14. Ph.D physical therapist will also be building connections with other graduate students. They can be a great source of knowledge and support if they ever need it while earning their Ph.D and beyond. These connections can become part of professional network and valued lifelong relationships.

PH.D PHYSICAL THERAPIST FUTURE SCOPE

1. A candidate after successful completion of a degree in Ph.D in Physical therapy is eligible to take Job Opportunities.
2. Some of the job roles that are available after the booming achievement of this Program will be including programs, Lecturer, Therapist, etc. The candidate earns good in the initial years after some experience and practical knowledge the salary can hike up.
3. The prospect of Ph.D in Physical Therapy program is high and estimated to increase in the upcoming time.
4. Ph.D is the highest degree a person can obtain in Ph.D in the Physical Therapy field, individual can do research work in the area.
5. Conclusively Ph.D in Physical Therapy Program provides training and experiences that prop up interdisciplinary learning, encourage critical and theory-based problem-solving skills to address psychology, and foster the growth of people's minds.

OBJECTIVES OF THE M.PHIL DEGREE PROGRAM

M.Phil Physical Therapy

1. M.Phil in Physical Therapy has been planned to furnish qualified candidates with the mandatory skills needed to operate in the field of Medicinal Chemistry of Various places and associated enterprises along with government bodies.
2. M.Phil in Physical Therapy is a post graduate program.
3. The duration of this program is of three (03) years.
4. M.Phil in Physical Therapy Program provides training and experiences that prop up interdisciplinary subjects learning, encourage critical and theory-based problem-solving skills to address to help peoples.
5. Candidates who are under this program are taught basic psychology to enhance understanding of people's mental state.
6. A candidate who graduates M.Phil in Physical Therapy can join research programs, Lecturer, Therapist, etc.
7. The candidate earns handsome amount in the initial years.

Why Study M.Phil Physical Therapy?

1. M.Phil in Physical Therapy program is advantageous for candidates who want to achieve leadership roles in the field of Physical Therapy.
2. During a degree in Physical Therapy, Candidates are allowed to work in various organizations and practices.

3. The main purpose of the M.Phil in Physical Therapy program is to make candidates responsive and achieve skilful knowledge obtainable on the subject and distribute their theses and earn accordingly.
4. M.Phil in Physical Therapy facilitates the integration of practical fieldwork with academic subjects.

M.PHIL PHYSICAL THERAPIST JOB PROFILES

After the successful completion of the course, one can opt to work either in the public sector or the private. Both of these sectors provide good opportunities.

Job Profiles	Job Description
Physical Therapist	A physical therapist is responsible for analyzing patients. They are also Responsible for the psychoanalysis and understanding of statistics for preparation and front actions and agenda initiative.
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Researcher	Researcher Physical Therapists show analysis and then administer statistics beginning early days teaching disability, and physical condition projects. They are also accountable for adding professional status to the institute.
Consultant	They provide direct services to patients who seek treatment under them.

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2. Physical Therapy M.Phil are given many positions like Researcher, Therapy Manager, Research Assistant Customer Care Manager, Sports Physio Rehabilitator, Self Employed Private Physical Therapist and more.
3. Most PTs with a M.Phil working full time can expect to earn in good salaries.

- Earning a M.Phil gives credibility in the workplace as well as highly valuable transferable skills, having this degree opens opportunities in fields or positions where having an advanced degree is recommended or required.
- Completing a M.Phil in physical therapy program is a respected accomplishment and for good reason. Those who earn an M.Phil in Physical Therapy have invested considerable time and effort in completing their degree program. Completion is often associated with prestige and credibility along with a personal sense of accomplishment that comes from reaching the highest level of academic achievement.
- Not only will this accomplishment impact personally, but it will also influence professional life. Many employers value the knowledge and skills that come with earning a M.Phil. This will give credibility in the job market, showing both employers and coworkers that individual is a hardworking and dedicated professional.
- While earning M.Phil individuals will not only gain specialized knowledge within physical therapy field but also transferable skills that they can use, no matter what career path they ultimately follow. Learning proper research methods and how best to take notes is an invaluable skill that a M.Phil physical therapist can use in most workplace settings.
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- Throughout the process of earning M.Phil, a physical therapist will most likely be collaborating with professors on research efforts and writing a dissertation. Getting to know professors in the current or desired field is a great opportunity, whether physical therapist looking for a job or not. Networking with professionals is often how people find job opportunities that they wouldn't have found just looking online.
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- A candidate after successful completion of a degree in M.Phil in Physical Therapy is eligible to take Job Opportunities.
- Some of the job roles that are available after the booming achievement of this Program will be including programs, Lecturer, Therapist, etc. The candidate earns good in the initial years after some experience and practical knowledge the salary can hike up.
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FACULTY MEMBERS / DRC

FACULTY					
DETAILS OF M.Phil / Ph.D FACULTY					
S.No.	Name	Designation	Permanent / Visiting	Title of degree	Expertise / Area of Specialization
1.	Prof. Dr. Basit Ansari	Professor	Permanent	Ph.D	Exercise Physiology

2.	Dr. Sadaf Ahmed	Assistant Professor	Permanent	Ph.D	Pain Management & Psycho Physiology
3.	Dr. Faizan Mirza	Assistant Professor	Permanent	Ph.D	Neuromuscular Screening & Rehabilitation
4.	Dr. Mazahir Zia Shaikh-PT	Assistant Professor	Permanent	M.Phil	Physical Therapy
5.	Ms. Shazia Tabussum	Assistant Professor	Permanent	M.Phil	Physical Fitness
6.	Dr. Humayun Imran Azeemi-PT	Assistant Professor	Permanent	M.Phil	Physical Therapy

RESEARCH FACILITIES

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SPECIFIC RESEARCH AREAS (AVENUES OF RESEARCH)

The Department of Health, Physical Education & Sports Sciences has a broad scope for qualitative as well as quantitative research in the field of Kinesiology, Exercise Physiology, Socio economic impact of Physical Therapy.

DEPARTMENT OF MATHEMATICS

INTRODUCTION

The Department of Mathematics at the University of Karachi has a rich history dating back to its establishment in 1956, with Prof. Dr. Suleman Kerawala serving as its first chairman. Renowned as one of the largest departments within the Faculty of Science, it upholds a mission to infuse vitality into mathematics and reciprocally enrich life through its teachings and research endeavors. Embracing both pure and applied mathematics and interdisciplinary pursuits, the department offers undergraduate, graduate, and postgraduate programs. Students are nurtured in mathematical reasoning, equipped with essential tools, and empowered with the skills to apply mathematical principles effectively. Central to its objectives is fostering a deep understanding and appreciation of mathematics as a dynamic and creative discipline. To support this, students can access state-of-the-art facilities, including computer labs and seminar libraries, supplemented by a newly established conference hall for academic and research activities.

POSTGRADUATE PROGRAMS AVAILABLE

1. M.Phil. (Mathematics)
2. Ph.D. (Mathematics)

PROGRAM OBJECTIVES

The M. Phil. and Ph.D. program directly contributes to the departmental mission by nurturing future mathematicians and preparing them for impactful careers. The program offers a rigorous curriculum designed to achieve the following objectives:

1. **Solid Mathematical Foundation:** Equip students with a strong foundation in core mathematical concepts, including analysis, algebra, topology, and applied mathematics, preparing them for advanced research or professional applications.
2. **Research Skills Development:** Introduce students to research methodologies, literature review techniques, effective communication strategies, and responsible data analysis practices, fostering independent research skills.
3. **Exploration of Specialized Fields:** Offer a range of elective courses to allow students to explore specific areas of mathematics that align with their interests, such as mathematical modeling, computational mathematics, or financial mathematics.
4. **Preparation for Further Studies:** Prepare students for doctoral programs or research careers by developing critical thinking, problem-solving, analytical abilities, and the ability to formulate and investigate original research questions.
5. **Interdisciplinary collaboration:** Recognizing the importance of mathematics across disciplines, we actively collaborate with other departments to offer qualified instructors and contribute to the mathematical needs of various fields.

AIMS AND SCOPE OF THE PROGRAMS

1. M.Phil. in Mathematics provides students with an opportunity to deepen their understanding of mathematical theory, develop advanced research skills, and prepare for both academic and non-academic careers in a wide range of fields.
2. Ph.D. in Mathematics is designed to equip students with the knowledge, skills, and expertise needed to conduct original research at the highest level and to make significant contributions to the field of mathematics. It is a rigorous and intellectually demanding program that prepares graduates for leadership roles in academia, industry, and beyond.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION / EXPERTIES

Sr. #.	Name of Ph. D. Faculty Members	Specialization / Expertise
01.	Dr. Najeeb Alam Khan	Applied Mathematics, Mathematical Modeling, Dynamical Systems, Decision Sciences etc.
02.	Dr. Syeeda Sadia Zia	Climate Analysis, Fluid dynamics
03.	Dr. Saba Naz	Data Analysis, climate and optimization
04.	Dr. Tanveer Ahmed Siddiqi	Applied Mathematics, Stochastic Processes, Optimization, Artificial Intelligence (decision making mode), Mathematical Statistics
05.	Dr. Muhammad Imtiaz	Operations Research, Analysis
06.	Dr. Syed Inayatullah	Operations Research and Optimization
07.	Dr. Muhammad Ayaz	Mathematical Modeling, Numerics, Differentiation, Artificial Neural Networks
08.	Dr. Asif Raza Khan	Mathematical Inequalities and Applications
09.	Dr. Saqib Ur Rehman	Climate Dynamics
10.	Dr. Fozia Hanif	Optimization, Cryptography, Wireless Sensors Networks, Underwater Sensors, Block chain, Machine Learning, Deep Learning, Quantum Cryptography, Mathematical Modeling and Simulations
11.	Dr. Syed Ahmed Hassan	Applied Mathematics, Mathematical Modeling, Climatic Influenced River Flow Modelling and Forecasting, Solar influenced Hydroclimate Variations
12.	Dr. Hina Zaheer	Optimizations Research
13.	Dr. Asma Rani	Optimization
14.	Dr. Noor Fatima Siddiqui	Applied Mathematics
15.	Dr. Samreen Ahmed	Applied Mathematics
16.	Dr. Ateeq Razzak	Solar and Renewable Energy
17.	Dr. Salman Safdar	Mathematical Biology, Epidemiology and Non-linear Dynamical Systems
18.	Dr. Samira Sahar Jamil	Applied Topology
19.	Dr. Sumayyah Saadi	Mathematical Inequalities
20.	Dr. Mehresh Shafi Khan	Hydroclimatic Modeling

RESEARCH FACILITIES AVAILABLE (LIBRARY, LABORATORY, COMPUTER LAB)

Two thousand and two hundred books are available at seminar Library and a computer lab is available with multimedia facility.

AVENUES FOR RESEARCH

1. Applied Topology
2. Dynamical Systems
3. Hydroclimatic Modeling
4. Mathematical Epidemiology
5. Mathematical Inequalities
6. Operations Research
7. Optimization
8. Solar and Renewable Energy

ANY OTHER RELEVANT INFORMATION YOU WOULD LIKE TO MENTION

The published research work of the departments fulfills the United Nations Sustainable Development Goals SDGs #3, #8, and #9. The department published more than hundred article in peer reviewed Journals in the last five years.

DEPARTMENT OF MICROBIOLOGY

INTRODUCTION

Microbiology is a broad discipline which has both basic and applied aspects. Therefore, microbiologists must have good understanding about microbial taxonomy, genetics, immunology and physiology in order to identify these agents, utilize their metabolic potential on commercial levels and to manipulate these organisms for the betterment of humanity.

The Department of Microbiology at the University of Karachi was established in 1957 with only 22 students and one full-time teacher (late Prof. Dr. A. A. Anwar) and a part time teacher cum student (Late Prof. Dr. Pirzada Misbahuddin Ahmed Siddiqui). Since this was the first and only Department of Microbiology for many years, in this country, offering a degree, Prof. Dr. Anwar could be truly regarded as the Father of Microbiology in Pakistan.

Because of the importance of the subject and its applicability, this department is now regarded as one of the best departments within the faculty of Science. As a result this department has become popular and a large number of students opt to study Microbiology as their major. Furthermore, this department caters the needs of more than 500 students who opt to study Microbiology as one of their minors each year.

Besides offering B.S. / Honors and Masters degrees, this department has a reputation of its M.S. / Ph.D. and Ph.D. programs. At present, more than 150 scholars are enrolled in MS and Ph.D programs.

POST GRADUATE PROGRAMS

1. M.Phil/M.S.
2. Ph.D.

PROGRAM OBJECTIVES

The doctoral program in Microbiology trains students at the highest level to assume research and leadership roles in the biological sciences. The objectives of Post Graduate Programs in Microbiology programme are;

1. To produce qualified microbiologists with the competencies to provide services to the private and public sectors including environment, agriculture, medicine and industries.
2. To build national and regional technological and infrastructural capacities for imparting terminal education, training and research on contemporary microbiology.
3. To develop and foster collaborations with developed countries and institutes of global excellence, facilitating the exchange of knowledge and development of cooperation in related industries.
4. To strategically prepare the Karachi, Pakistan region to actively interact with the developing world in Microbiology services, research and development.

AIMS AND SCOPE

The Graduate of M.Phil/ Ph.D. in Microbiology program will:

1. Engage in and conduct original research in the field of microbiology.
2. Have the attributes of a Subject specialist, Scientific researcher, Educator, Effective communicator and Collaborator
3. Have the attributes of a Subject specialist, scientific researcher achieved by developing trained personnel, in research skills and methodology.
4. Serve as Educators who are capable of teaching various sub discipline of microbiology
5. Serve as Scientific Researcher to conduct quality and credible research.
6. Compete successfully for productive employment or postdoctoral training in industry or academic institutions.

RESEARCH FACILITIES

A computer lab, seminar library and dedicated, well-equipped M.Phil/ Ph.D. research lab are available at the department. Following instruments/ facilities are present:

1. Biosafety cabinets with incinerators
2. Laminar Flow
3. Thermocycler and electrophoresis unit

4. Liquid Nitrogen storage tanks, Refrigerators, ice makers
5. Spectrophotometer
6. CO₂ incubator
7. Orbital shaker
8. Water baths
9. Incubators
10. Microscopes
11. Animal house

REGULAR FACULTY MEMBERS

S #	FACULTY	DESIGNATION	SPECIALIZATION
01.	Dr. Tanveer Abbas	Professor & Chairman	Food Microbiology and Antimicrobial Agents
02.	Dr. Syed Abdus Subhan	Professor	Molecular Biology, Virology and Bioinformatics
03.	Dr. Abdul Wahab	Professor	Isolation of Antimicrobial Compounds from Actinomycetes and Endophytes
04.	Dr. Zaid Ahmed Pirzada	Professor	Streptococcal Pathogenesis, Clinical Microbiology
05.	Dr. Asma Naim	Professor	Clinical Bacteriology, Pathogenesis and Antimicrobial resistance; Bioinformatics
06.	Dr. Muhammad Sohail	Professor	Isolation of microbial enzymes; Antimicrobial resistance
07.	Dr. Muneera Naz Baloch	Professor	Bioactive Agents of Lactic Acid Bacteria
08.	Dr. Nusrat Jabeen	Professor	Molecular Biology, Virology and Bioinformatics
09.	Dr. Sehar Afshan Naz	Professor	Clinical Mycology
10.	Dr. Adnan Khan	Associate Professor	Study on Mode of transmission and Control of Helicobacter and Campylobacter infections
11.	Dr. Asma Saeed	Associate Professor	Bacterial genetics; Plasmid interactions and incompatibility
12.	Dr. Farheen Ansari	Associate Professor	Antibacterial and antifungal activities of medicinal and wild plant extracts
13.	Dr. Maryam Shafique	Associate Professor	Applied Microbiology and Biotechnology
14.	Dr. Nayyar Mehmood	Assistant Professor	Enzymology; Bacteriocins
15.	Dr. Fouzia Haider	Assistant Professor	Molecular Biology and Biotechnology
16.	Dr. Sana Ejaz	Assistant Professor	Molecular Medicine and Microbiology
17.	Dr. Wajeeha Binte Asad	Assistant Professor (T)	Industrial & Microbial enzyme
18.	Dr. Anjum Zehra Naqvi	Assistant Professor	Molecular Biology; Antimicrobial Agents
19.	Dr. Urooj Zafar	Assistant Professor	Environmental Microbiology; Antimicrobial Resistance
20.	Dr. Yasir Raza	Assistant Professor	Molecular Microbiology; Cell and Tissue Culture
21.	Dr. Sadiya Muhammad Saleemullah Khan	Assistant Professor (T)	Applied and Environmental Microbiology/ Biotechnology
22.	Dr. Amna Alam	Assistant Professor (T)	Antimicrobial Agents; Viral Diagnostics; Molecular Epidemiology

DEPARTMENT OF PHYSICS

INTRODUCTION

The department of Physics was established in 1952 with Professor Dr. Mujtaba Karim as the founding Chairman. Since then the department has played an important role in the development of Science and Technology in the country. A considerable number of students have obtained their degree in Physics and are now rendering their services in reputable organizations within and outside the country. The department has twenty two (22) regular faculty members, out of which eleven (11) have Ph.D qualification and rest of them all have M.Phil. Degrees. The department of Physics post-graduate research programs leading to the award of MS/M.Phil. and Ph.D. degrees. These post-graduate programs are strictly monitored as per the University of Karachi rules and procedures and are administratively governed by Advanced Studies and Research Board (ASRB), the University of Karachi. Enrolled students and scholars are required to choose a research advisor of their choice of research discipline from the faculty. Under the guidance of the research advisor(s), students submit their research proposal as Synopsis to ASRB for approval. After their course and completion of approved research work, students are required to submit their research thesis which is refereed by local and foreign referees who are eminent scientist and researchers in their respective fields. To keep students abreast of the recent developments in the field of Physics in general and in their respective disciplines in particular, research advisors regularly give opportunities to their research students to participate in local and internal conferences and seminars / workshops.

POST-GRADUATE PROGRAMS

1. MS Program 30 Credit Course Work (Eligibility for admission 16 years of education) In this stream, as per ASRB and HEC policy enrolled students are required to complete a prescribed course work (Cr 24). Students have the option to take two more courses worth 3+3 credit to complete their degree requirement of 30 credits. Provided all other necessary requirements are fulfilled, MS degree is awarded by the University of Karachi. MS qualification is 18 years of educational.
2. M.Phil Program 24+06 Credits (Eligibility for admission 16 years of education) In this stream, as per ASRB and HEC policy enrolled research scholars to complete their degree by completing total required credits in terms of courses (Cr 24) and have to complete their theoretical or experimental research thesis worth 6 credit to fulfill M.Phil. Degree requirement. After all necessary requirements are fulfilled, M.Phil. Degree is awarded by the University of Karachi. M.Phil qualification is 18 years of educational.
3. Ph.D Program 18+12 Credits (Eligibility for admission 18 years of education) Research scholars are required to complete 18 credits of course work as per ASRB and HEC policy. During the course of Ph.D program, research scholars are required to complete their approved research thesis and must have at least one W-Category research publication. After completion of all necessary requirements, as per policy, Ph.D Scholar is required to give a Ph.D defense presentation to be eligible for an award of Ph.D degree.

PROGRAMS OBJECTIVES

After successful completion of the post-graduate programs, following are the long-term and in multifaceted scenario:

1. Have the understanding of the implications and responsibilities of good and credible research practices in pursuing quality research.
2. Have developed a broad understanding of the subject that can be applied in dissemination and creation of knowledge.
3. Achieved a level of competency and have developed a capacity to work independently and as a team member.
4. Be able to identify the research question and be able gather and assimilate relevant the data for the application of mathematical and research tools.
5. Is proficient in translating the research problem in to a viable and practical solution as demanded by the private and public sector organization.
6. Is proficient in writing research proposals, research publications and is able to communicate in written and verbal forms with the members of the community in particular and with broader audience in general.
7. Have developed a threshold level of being able to supervise research.

AIMS AND SCOPE

Aim

Consistent with the aims of the University of Karachi, Department of Physics seeks to train its research students and scholars so that they have a sound research skills, have acquired latest theoretical and experimental understanding and foremost to develop critical thinking in achieving aims and objectives; and to have developed a mechanism of enquiry which is ethical and professional in an expert manner.

Scope

In addition to the coverage of broad areas of Physics research, programs aims to develop broader understanding of research challenges of the society. To promote a research atmosphere / thinking which is both intellectual and conducive to searching knowledge to answer questions.

PH.D FACULTY MEMBERS AND DEPARTMENTAL RESEARCH COMMITTEE (DRC)

All Ph.D faculty members are members of the Departmental Research Committee (DRC). As per post-graduate policy, DRC is tasked with carrying out all administrative activities related to research in the department.

Following are DRC faculty members:

1. Dr. Shabana Rizvi, **Professor & Chairperson**
Specialization: Solid State Physics, Biophysics & Medical Physics
2. Dr. Imran Ahmad Siddiqui, **Professor**
Specialization: Laser Physics, Quantum & Atom Optics, Quantum Field Theory and Gravitation, Wind and Tidal Energy, Brain-Computer Interfacing (BCI)
3. Dr. Intikhab Ulfat, **Professor**
Specialization: Solid State Physics, Synchrotron Spectroscopy, Medical Physics, Renewable Energy Resources.
4. Dr. Syed Rizwan Ali, **Professor**
Specialization: Solid State Physics, Spintronics, Thin Films, Magnetic Nanostructures & Vacuum Technology
5. Dr. Muhammad Naeem Khan, **Professor**
Specialization: Magnetic Semiconductor, Molecular Magnets, Nanomagnetism, Spintronics.
6. Dr. Syed Zeeshan Abbas, **Associate Professor**
Specialization: Renewable Energy Resources, Electronics
7. Dr. Sadaf Qaseem, **Associate Professor**
Specialization: Solid State of Physics, Dilute Magnetic Semiconductor Nanoparticles, Band gap Engineering
8. Dr. Saif Uddin Jilani, **Assistant Professor**
Specialization: Nuclear Physics and Space Science
9. Dr. Sadaf Fatima, **Assistant Professor**
Specialization: Solid State Physics, Synchrotron based X-ray absorption spectroscopy for Nanomaterials.
10. Dr. Sheikh Zeeshan Iqbal, **Assistant Professor**
Specialization: Laser Spectroscopy, Laser Induced Break Down Spectroscopy (LIBS)
11. Dr. Muhammad Khalid, **Assistant Professor**
Specialization: Nano Technology and Electronics

AVENUES OF RESEARCH

The most significant aspect of learning Physics is that it is an experimental science, not an abstract intellectual discipline. At the departmental level, currently M.Phil. & Ph.D. programs are running and research studies are being done in wide ranging fields of theoretical and experimental Physics. In these studies, collaboration from other organizations & institutes is also sought when needed. Qualified faculty members are actively engaged in research areas such as:

1. Nanotechnology
2. Atomic and Molecular Spectroscopy & Optics
3. Laser
4. Space Physics
5. Medical Physics and Biophysics
6. Neurophysics and Brain-Computer Interfacing

7. Renewable Energy Resources and Environmental Science
8. Material Science
9. Electronics
10. Theoretical Physics

FACILITIES

Beside teaching laboratories, department has a number of other laboratories and computing facilities facilitating in theoretical and experimental research:

1. Spectroscopy Research Laboratory
2. Solid State Research Laboratory
3. Condensed Matter Research Laboratory
4. Brain Computer Interface Laboratory
5. Electronics Laboratory
6. Computer Laboratory
7. Seminar Library
8. Mechanical Workshop

CAREER OPPORTUNITIES

Students enrolled in M.Phil and Ph.D programs are offered Teaching Assistantships and Teaching Associateships. In such capacities students / research scholars are required to work full-time in the department and are assigned laboratory and teaching courses. This not only has monetary benefits but also serves as improvements / training of students. Depending on the teaching load requirements of the department, at the start of each semester applications are invited and based on interview candidates are selected.

After completing and getting M.Phil and Ph.D degrees, opportunities are available both in public and private sector Higher Education Institutions (HEIs) and research organizations where they can serve in their respective disciplines. Graduates also have the opportunity to apply in international research organizations and foreign universities for Post-Doctoral studies. M.Phil and Ph.D Physics graduates develop critical thinking and objective approaches, virtue of which appears as leadership qualities. This gives them opportunity to use their expertise to serve as team leads and consultants in public and private sector organizations and in business and corporate environments.

DEPARTMENT OF PHYSIOLOGY

INTRODUCTION TO THE DEPARTMENT

As the backbone of every medical breakthrough, the Department of Physiology occupies a prominent and exalted position within the Faculty of Science at the University of Karachi. Founded by Prof. Dr. Mahmood Hussain Qazi in 1964, the department holds the distinction of being the first institution outside medical colleges to introduce teaching and research programs in Basic and Experimental Physiology at both graduate and postgraduate levels. It proudly stands as the only institute in Pakistan to have produced M.Phil and Ph.D scholars in various disciplines of Physiology over the years. Currently, the department is actively engaged in teaching and research, offering advanced studies in core areas such as neurophysiology, endocrine physiology, reproductive physiology, poultry physiology, pathophysiology, and hematology. Recognizing that Physiology forms the foundational base of biological science, resonating from the cellular level to the organism as a whole, the department is committed to nurturing young minds to become future scientists and lifelong learners. The mission of the Department of Physiology is to produce experts who can contribute to cutting-edge research, generating endless possibilities for the development of novel therapeutic drugs and technologies. By providing a rigorous and comprehensive education, the department aims to equip its students with the skills and knowledge necessary to excel in the modern age of Biosciences, Clinical/Biomedical Sciences, and other Applied Biological Sciences.

POSTGRADUATE PROGRAMS AVAILABLE

1. M.Phil in Physiology
2. Ph.D in Physiology

OBJECTIVES FOR Ph.D AND M.Phil RESEARCH PROGRAMS IN PHYSIOLOGY

The Ph.D and M.Phil post-graduate research programs in Physiology are meticulously designed to deliver quality education in Physiology through research, fundamentally aligned with the university's mission. By focusing on advanced research, critical skills development, and comprehensive education, the program supports the university's goal of fostering academic excellence and innovative research. The objectives of the program are as follows:

1. Provide a stimulating approach to the major forces instrumental in shaping basic, clinical and applied skills in physiology.
2. Foster the development of professional skills through a well-designed curriculum based on experiments, training, and research.
3. Provide quality education and training that embody the aspirations of the Pakistani population and the global community.
4. Facilitate the creation, preservation, integration, transmission, and utilization of knowledge in physiology.
5. Encourage critical analysis, interpretation, and discussion of factual information and data related to physiology.

PROGRAM SCOPE

These objectives ensure that students are well-prepared to contribute to the field of Physiology, advancing both their personal careers and the broader scientific community. The program aims to produce graduates who are not only knowledgeable but also skilled in research and practical applications, meeting the needs of both national and international communities. Our M.Phil. and Ph.D programs aim to produce graduates who are not only knowledgeable in the theoretical aspects of Physiology but also skilled in practical and research applications. By equipping students with advanced knowledge, professional skills, and a strong research foundation, the M.Phil and Ph.D programs in Physiology ensure that graduates are well-prepared to contribute to the scientific community and address global challenges in medical and biological sciences.

INFRASTRUCTURE AND FACILITIES OF THE DEPARTMENT OF PHYSIOLOGY

The Department of Physiology at the University of Karachi boasts necessary infrastructure that supports both teaching and research at the highest levels. The department is equipped with modern laboratories furnished with advanced apparatus such as PCR machines, spectrophotometers, Power Labs and electrophoresis units, enabling comprehensive experimental and applied research in various physiological disciplines. The well-equipped labs cater to techniques in electrophysiology, microscopy, physiological

chemistry, hematology, metabolic physiology, general physiology, and clinical physiology. Classrooms are designed for interactive learning, well-organized seminar library provides the latest textbooks and references on various physiology disciplines, while dedicated computer facilities, including specialized research units and a separate computer laboratory, support computational needs for academic and research activities. The department also includes an animal house, which is essential for conducting advanced physiological experiments and research involving animal models.

REGULAR FACULTY MEMBERS WITH THEIR AREA OF SPECIALIZATION AND EXPERTISE

Professor

1. Dr. Taseer Ahmed Khan: Molecular & Poultry Physiology

Associate Professors

1. Dr. Lubna Naz: Pathophysiology and Herbal therapeutics
2. Dr. Sadaf Ahmed: Behavioural and Psychophysiology
3. Dr. Anwar Ali: Cell & Molecular Physiology
4. Dr. Mehru Nisa: Molecular Physiology & Clinical Endocrinology

Assistant Professors

1. Dr. Arifa Savanur: Neuromuscular Physiology
2. Dr. Zulfiqar Ahmed: Poultry Science
3. Dr. Nazish Iqbal Khan: Integrative & Pathophysiology
4. Dr. Ghazala Yasmeen: Renovascular Physiology
5. Dr. Sumera Sohail: Pathophysiology
6. Dr. Syed Mudassir Haider Rizvi: Neuromuscular Physiology
7. Ms. Ambreen Niazi (M.Phil) Reproductive and pathophysiology
8. Ms. Samia Mushtaq (M.Phil) Hematology and Natural therapeutics

Lecturers

1. Dr. Faizan Mirza: Physiology of Aging & Psychophysiology
2. Ms. Uzma Firdous (M.Phil) Reproductive Endocrinology
3. Dr. Muhammad Nisar: Endocrinology

DEPARTMENT OF STATISTICS

INTRODUCTION OF THE DEPARTMENT

Department of Statistics was established in 1963. Prof. Muhammad Iqbal was the first Chairman. It is one of the Significant Departments in the faculty of Science. The proliferation of quantitative information in society demands that we help to develop the next generation of statistics professionals while increasing the statistical literacy of our students and the frequency of collaboration with our colleagues. The aim of the department of statistics is to develop concrete concepts of the subject in the students to transform it to a research oriented department. Students are taught advanced statistical softwares to enhance their understanding of data and sharpen research skills.

POSTGRADUATE PROGRAMS AVAILABLE AT STATISTICS

1. M.Phil
2. M.S.
3. Ph.D

PROGRAM'S OBJECTIVES

The main objectives of M.Phil/ MS/ Ph.D. Statistics are aligned with the objectives of the university and the program mission statements which focuses, interactive teaching, class presentation and research methodology. These are as follows:

1. To guide students about acquisition of real data from various sources.
2. To enhance the data handling and data smoothing skills so as to prepare data for statistical analysis.
3. To impart and enrich the computing skills of the students by making them familiar with various softwares; Eviews, STATA, Minitab, R, Python, SPSS and SAS and other advanced software's.
4. To improve and enhance the writing skills of the students enabling them for report compilation and thesis/article write up.

AIMS AND SCOPE OF THE PROGRAM

The aim of the program is to impart statistical knowledge and skills in the students, enabling them to handle manage and analyze all type of data from field to forecasting, necessary for better planning development and growth of the country. Statistics has great scope in all fields which include medical, biological, business, finance, economics, and computer science etc.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION/ EXPERTISE

Highly qualified and competent teachers are part of the faculty. They are very dedicated and loyal with their job and rigorously involved in quality teaching and research. Following are the Department Research Committee members with their area of expertise

- | | |
|---|--|
| 1. Dr. Farah Yasmeen (Assistant Professor): | Demography, Time Series |
| 2. Dr. Arfa Maqsood (Assistant Professor): | Time Series, Econometrics, Sampling Distributions |
| 3. Dr. Suboohi Safdar (Assistant Professor): | Design of Experiment & Quality Control, Regression, Survival |
| 4. Dr. Bushra Shamshad (Assistant Professor): | Multivariate Analysis and Categorical data |
| 5. Dr. Samreen Fatima (Assistant Professor): | Analysis, Sampling Distributions |
| 6. Dr. Tayyab Raza Fraz (Lecturer): | Time Series Multivariate Analysis |
| | Econometrics, Time Series forecasting |

RESEARCH FACILITIES AVAILABLE

The department has a spacious seminar library enriched with subject related books. Students can utilize the library for reading, discussion and for completing their assignments. The students can also get membership of LEJ digital library. Computer lab with updated PC's is also available for the research students.

INFRASTRUCTURE

The department has comfortable infrastructure to support conclusive learning environment.

AVENUES OF RESEARCH

Students graduating from this department have ample opportunities of getting employment in various public/private R&D organizations and educational sectors.

DEPARTMENT OF ZOOLOGY

INTRODUCTION

Department of Zoology was established in 1951 and now it holds a pivotal position in the faculty and can also be compared with any department of Zoology elsewhere in and outside Pakistan. The Department right from its inception has been imparting instruction at B.Sc., BS, and M.Sc. levels, besides providing facilities for post-graduate research leading to M.Phil, and Ph.D. degrees. More than two thousand (2000) research papers have been published by the faculty members in National and International research journals. Various monographs and thirty two books have been published by the faculty members. About 300 students are enrolled each year at various levels in the department including various specializations. Several are being enrolled each year in M.Phil and Ph.D programs.

The graduates of the department while catering the needs of the education department, where they are readily absorbed as teachers of Zoology in schools, colleges and University also have job opportunities in the department of Agriculture, Plant protection, Fisheries, Zoological Survey, Zoological Gardens Safari parks, Pakistan Council of Scientific and Industrial Research, National Institute of Oceanography, Pakistan Agricultural Research Council, Wild life conservation and management of WWF, IUCN, Public and Private sectors pathological laboratories

POSTGRADUATE PROGRAMS AVAILABLE AT ZOOLOGY

1. MS
2. M.Phil
3. Ph.D

PROGRAMS OBJECTIVES

The main objectives of M.Phil/ MS/ Ph.D. Zoology are aligned with the objectives of the university and the program which focuses, interactive teaching, class presentation and research methodology. These are as follows:

1. To enhance the data skills to prepare data for zoological analysis
2. To improve the students enabling them for report compilation and thesis/article write up different specializations of Zoology, Marine Zoology, Wildlife and Fisheries, Entomology, Limnology, Parasitology, Research Methodology

AIMS AND SCOPE OF THE PROGRAMS

The aim of the program is to impart zoological knowledge and skills in the students, enabling them to handle manage and analyze all type of specimens from field .Zoology has great scope in all fields which include medical, biological, teaching ,field research, Agriculture and computer science etc. Understanding the aspects of different specialization is very important for providing sound scientific advice in research.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION/ EXPERTISE

Qualified teachers are part of the faculty. They are very dedicated and loyal with their job and involved in quality teaching and research. Following are the Department Research Committee members with their area of expertise

Professors

1. Prof. Dr. Masarrat Jahan Yousuf (Dean of Science)
Specialisation Entomology
2. Prof. Dr. Farzana Yousuf (Chairperson)
Specialisation Marine Zoology
3. Prof. Dr. Shoaib
Specialisation Limnology
4. Prof. Dr. Soleha Rahman
Specialisation Marine Zoology

Associate Professors

1. Dr. Karim Gabol Specialisation Wildlife and Fisheries
Assistant Professors

Assistant Professor

1. Dr. Sumera Farooq
Specialisation Marine Zoology
2. Dr. Shahina
Specialisation Limnology
3. Dr. Shakila Parveen
Specialisation Parasitology
4. Dr. Ghazala Yasmeen
Specialisation Wildlife and Fisheries
5. Dr. Rizwana Ghaffar
Specialisation Parasitology
6. Dr. Tariq Rajput
Specialisation Entomology
7. Dr. Roohi Kanwal
Specialisation Wild life and Fisheries
8. Dr. Samreen
Specialisation Parasitology
9. Dr. Atiya Batoool
Specialization Limnology

M.Phil / Ph.D Courses Assigned

1. Prof. Dr. Masarrat Jahan Yousuf (Dean of Science)
Specialisation Entomology
ZOE -ZOE-801 Toxicological techniques
ZOE-812 Enzymology
2. Prof. Dr. Farzana Yousuf
Marine Zoology
ZOM-712
Aquatic Toxicology
3. Prof. Dr. Shoaib
Limnology
ZOL 712
ZOL -811
4. Prof. Dr. Soleha Rahman
Marine Zoology
Ph.D title: Molluscan fauna of rocky shores of Karachi-A comparative ecological study
5. Dr. Karim Gabol
Wild life and Fisheries
First semester .M.Phil
ZWF.731 . Conservation and management of Wild lands and protected areas
ZWF.711.Conservation of endangered species

Second Semester

ZWF 732. Advance Wild life Research Design

ZWF.712. Biodiversity Assessment and monitoring

1. Dr. Shakila Parveen
Parasitology
ZOP 712 MS
ZOP -711 , Advanced Helminthology
2. Dr. Rizwana Ghaffar
Parasitology
ZOP - 811 Methods in Parasitology (1st Semester)
ZOP - 712 Infectious Diseases (2nd Semester)
3. Dr. Shahina
Limnology
ASR 701, Research Methodology
ZOO-722, Aquaculture Management and Fisheries
4. Dr. Roohi
Wildlife and Fisheries
Wildlife radiotelemetry ZWF.811
5. Dr. Samreen
Pataditology
ZOP-712 Infectious disease
6. Dr. Atiya Batoool
Limnology
ASR -702
Research Methodology

RESEARCH FACILITIES AVAILABLE

Zoology department has a seminar library available with subject related books and many Journals for researchers. Students can utilize the library for reading, discussion and for completing their assignments. The students can also get membership of L EJ digital library. Computer lab with updated PC's is also available for the research students.

INFRASTRUCTURE

The department has comfortable infrastructure to support conclusive learning environment.

AVENUES OF RESEARCH

Students after completion their studies opportunities of getting employment in various public/private R&D organizations and educational sectors. e.g. Fisheries, National Institute of Oceanography, WWF, Korangi Fisheries etc.

DR. A. Q. KHAN INSTITUTE OF BIOTECHNOLOGY & GENETIC ENGINEERING (BIOTECHNOLOGY) (KIBGE)

INTRODUCTION

Dr. A. Q. Khan Institute of Biotechnology and Genetic Engineering (KIBGE) is a research institute with state-of-the-art facilities to conduct research on modern biological sciences and to impart higher education at the University of Karachi. The institute was established with an aim to foster research-based knowledge relevant to the developing world in the scientific domain of biotechnology and genetic engineering. There are four interdisciplinary key areas of research at KIBGE that include agriculture biotechnology, environmental biotechnology, industrial biotechnology and medical biotechnology.

Agriculture Biotechnology focuses research on food safety and security issues such as antimicrobial resistance, pesticides and heavy metal contamination in food chain and sustainable crop production using proteomic and genomic approaches.

Environmental Biotechnology focuses research on developing biopolymeric nanofibers, nano- composites, nanohybrid systems using sustainable approaches for controlling environmental pollutants widespread in the environment.

Industrial Biotechnology focuses research on production and structural elucidation of various industrial metabolites that include enzymes, antimicrobial peptides, biopolymers and nanomaterials that are synthesized under *in-vivo* / *in-vitro* conditions using microbial or synthetic biology approaches.

Medical Biotechnology focuses research on hereditary and acquired disorders prevalent in the community, cancer genomics, cardiovascular complications, metabolic and inflammatory disorders for disease biomarker discovery and validation.

KIBGE has established a sound scientific research environment that promotes quality research in aforementioned scientific fields.

RESEARCH AT KIBGE

KIBGE has well-equipped laboratory facilities in the field of biotechnology, molecular biology and genetic engineering. The infrastructure of the institute has been developed according to the needs and requirements of innovative and advanced research. The key objective of the institute is to train scholars by imparting rigorous training along with preparing them to be a productive member of the scientific community. Scholars are provided supportive research intensive environment that enhances leadership qualities which motivate them to make discoveries in the field of biotechnology and genetic engineering. Scholars also have the benefit of understanding contemporary techniques from national and international subject-experts in the field of biological sciences and related fields.

NETWORKING PARTNERS

The institute is part of several international research networks and encourages students and researchers to pursue training abroad. KIBGE is one of the principal partners in the network of collaborative research involving partners from all over the world, including Australian National University, Australia; University of Buea, Cameroon; University of Zagreb, Croatia; INSERM, University of Paris, France; Max Planck Institute, Germany; Pasteur Institute of Iran, Iran; University College Dublin, Ireland; University Sains Malaysia, Malaysia; University of Auckland, New Zealand; Ohio University, USA; Bangor University, United Kingdom; University of Malaya, Malaysia and many other international institutes.

SPECIFIC RESEARCH AREAS

Students are provided with a scientific platform to achieve outstanding research level with comprehensive biotechnological skills. The current curriculum guarantees biotechnological specialization ranging from research communication skills to research and development. Other avenues include bioinformatics, biostatistics, cytogenetics, plant biotechnology, industrial and environmental biotechnology, human molecular genetics, biosafety, ethics and regulatory affairs etc.

CAREER OPPORTUNITIES

Graduate scholars from KIBGE are trained to develop subject-specific transferable skills that are currently valued by all employers in academia, industry and government organizations. Scholars may find employment in specialized biotechnology and biotech related industrial sectors such as pharmaceutical, health care, food manufacturing, aquaculture and agricultural industries. KIBGE also fosters hands-on research opportunities to the students and alumni towards securing places at various international higher education institutions and networking partners abroad.

FACULTY MEMBERS***Director General (Acting)***

1. Prof. Dr. Khalid Mahmood Iraqi

Professors

1. Prof. Dr. Saddia Galani
2. Prof. Dr. Afsheen Aman
3. Prof. Dr. Sitwat Zehra
4. Prof. Dr. Saima Saleem

Assistant Professors

1. Dr. Sehrish Fatima
2. Dr. Afsheen Arif
3. Dr. Asma Ansari
4. Dr. Syeda Nuzhat Nawab
5. Dr. Nadir Naveed Siddiqui
6. Dr. Ishrat Jamil

POSTGRADUATE PROGRAMS OFFERED

1. M.Phil. in Biotechnology (KIBGE)
2. Ph.D. in Biotechnology (KIBGE)

DR. PANJWANI CENTER FOR MOLECULAR MEDICINE & DRUG RESEARCH (MOLECULAR MEDICINE) (PMDC)

INTRODUCTION

The Dr. Panjwani Center for Molecular Medicine and Drug Research (PCMD), was established in 2004 through a generous donation of Dr. Panjwani Memorial Trust in the memory of Dr. Mohammad Hussain Panjwani, a leading scholar and philanthropist. Since then the PCMD has initiated several academic programs at the M. Phil / Ph. D. levels manpower through, focusing on prevalent diseases of Pakistan. The core mission of the center is to train highly qualified manpower in the emerging fields of molecular medicine and drug development. The efforts are focused on developing greater understanding of the pattern and causes of most common diseases of Pakistan in order to develop effective diagnostic tools and affordable treatments.

The PCMD has several public service units, including DRAP (Drug Regulatory Authority of Pakistan) certified center for bioequivalence studies and clinical research (CBSCR), diagnostic laboratory, genome center, National Virology Institute, Sindh Forensic DNA and Serology Laboratories, etc. All these units are also ISO 9001 and 17025 accredited facilities. It also a UNESCO category II center.

SPECIFIC RESEARCH AREAS WITH WELL EQUIPPED LABS INCLUDING

1. Bioinformatics and Computational Biology
2. Cancer Research
3. Cell and Molecular Biology
4. Clinical Pharmacology
5. Drug Discovery, Design and Synthetic Medicinal Chemistry
6. Electrophysiology
7. Epidemiology
8. Genomics
9. Infectious Diseases
10. Metabolic Disorders
11. Virology
12. Human Genetics and Gene therapy
13. Infectious Agents and Diseases
14. Metabolic Disorders
15. Molecular Enzymology
16. Molecular Immunology
17. Molecular Pharmacology
18. Neurosciences and Neuropharmacology
19. Proteomics and Metabolomics
20. Structural Biology
21. Tissue Engineering and Stem Cell Research
22. Tropical Disease Research
23. Virology

FACILITIES

1. Biobank
2. Diagnostic and Clinical Research
3. Imaging
4. Molecular, Microbial and Drug Banks

SPECIFIC RESEARCH AREAS

Visit our website <http://iccs.edu>

FACULTY MEMBERS

Visit our website:

<http://iccs.edu/page-panjwani-faculty>

PROGRAMS AVAILABLE

M.Phil. and Ph.D. in various disciplines of Molecular Medicine.

HEJ RESEARCH INSTITUTE OF CHEMISTRY (CHEMISTRY)

INTRODUCTION

The H.E.J. Research Institute of Chemistry, International Center for Chemical and Biological Sciences (ICCBS), is one of the finest academic research institutions of developing world in the field of chemical and bioorganic chemical sciences. Its large complex is comprised of 10 research buildings, equipped with some of the region's most sophisticated laboratory equipment. The complex also includes a pilot plant, electrical powerhouses, green houses, animal house facility, auditorium, and a residential, apartment buildings, and several international guesthouses.

The H.E.J. Research Institute of Chemistry carries out research, training, product development, and service delivery various branches of chemical and bioorganic sciences. The center also provides analytical testing for a broad range of clients for both the public and private sectors, nationally and internationally.

Over the past 55 years, over 1,500 students have earned M.Phil. and Ph.D. degrees from the institute. These degree-granting programs are based on world-class training to young researchers from developing countries, including those belonging to the Organization of Islamic Cooperation (OIC), and Non-Aligned Movement (NAM), and UNESCO member states.

The institution has won the Islamic Development Banks (IDB) Prize for best Science and Technology Institution twice in 2004 and 2010. The ICCBS is a UNESCO category II Center and WHO collaborating center.

CAREER OPPORTUNITIES

Graduates of the H.E.J. Research Institute are serving the nation in various capacities. The graduating scholars receive offers of post-doctoral and faculty positions in foreign, as well as in the top universities of Pakistan. The scholars who are not interested in academic positions are absorbed in local industries as R & D or production scientists.

FACILITIES

The center has a large doctoral program in organic chemistry, analytical chemistry, physical chemistry, inorganic chemistry, biochemistry, and textile chemistry with over 400 Ph.D. students. The main areas of research, and training of students include natural product chemistry, protein chemistry, pharmacology, computational chemistry, nano-chemistry, and plant biotechnology. The analytical, spectroscopic, computational, clinical and other facilities of the institute are *at par* to any good institution in the world. The center also has ISO 17025 / 9001 certified Industrial Analytical Center is providing services to over 600 industries in Pakistan. The L.E.J. Nanotechnology Center of the Institute is a state-of-the art facility in this field.

A number of goal oriented projects relating to the chemistry of natural products, protein chemistry, plant biotechnology, and pharmacology of herbal medicine are being vigorously pursued which has led to the award of over 1,500 M. Phil. and doctorate degrees. The scientific contributions of over 10,000 research papers, over 400 books, and over 350 international patents have earned international recognition.

SPECIFIC RESEARCH AREAS

Visit our website <http://iccs.edu>

FACULTY MEMBERS

Visit our website <http://iccs.edu/page-hej-faculty>

PROGRAMS AVAILABLE

M. Phil., and Ph.D. Program in various disciplines of Chemistry, Bio-organic Chemistry, Biotechnology and Pharmacology.

DR. M. AJMAL KHAN INSTITUTE **OF SUSTAINABLE HALOPHYTE** **UTILIZATION (BOTANY)**

INTRODUCTION

The Dr. Muhammad Ajmal Khan Institute of Sustainable Halophyte Utilization (MAK-ISHU), aptly re-named after its founder Director, was established in 2006 to serve humanity through a better understanding of salt-tolerant plants (halophytes) and their sustainable use for food, medicine, fodder, forage, biofuel and climate change mitigation. The Institute has trained several students and researchers through international and national collaborative projects and has made significant breakthroughs in saline agriculture by using saline resources. The quality of the research at MAK-ISHU can easily be judged by more than 500 international peer-reviewed articles. The institute is proud of its post-graduate students who completed their research successfully and are now actively serving in prestigious public and private, teaching and research institutes, colleges, and Universities throughout Pakistan.

POSTGRADUATE PROGRAMS AVAILABLE

1. M.Phil. in Botany (ISHU)
2. Ph.D. in Botany (ISHU)

PROGRAMS OBJECTIVES

The main objective of the postgraduate programs at MAK-ISHU is to train students to 1) Understand the mechanistic basis of natural high-stress tolerance of halophytes through eco-physiological studies, supported by biochemical and molecular tools and 2) Investigate ways to use halophytes for various purposes in saline and water-scarce regions in harmony with nature.

AIMS AND SCOPE OF THE PROGRAMS

The emphasis of postgraduate programs at MAK-ISHU is to train students on cutting-edge research techniques and avenues for a better understanding of stress tolerance mechanisms leading ultimately towards sustainable halophyte utilization. These postgraduate programs are in line with several United Nations Sustainable Development Goals (SDGs), particularly SDG 2 (Zero Hunger), SDG 7 (Affordable and Clean Energy), SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible Consumption and Production), SDG 13 (Climate Action), SDG 14 (Life Below Water), and SDG 15 (Life on Land).

REGULAR FACULTY MEMBERS AND THEIR SPECIALIZATION

Professors

1. Dr. Bilquees Gul (Ecophysiology; Proteomics; Halophytes Utilization)
2. Dr. Salman Gulzar, Director (Halophyte Biology; Photosynthesis & Photobiology; Climate Change)
3. Dr. Irfan Aziz (Wetland and Mangrove Ecosystems; Phytoremediation)

Associate Professors

4. Dr. Muhammad Zaheer Ahmed (*Ion Homeostasis; Transcriptomics; Intercropping Systems*)
5. Dr. Abdul Hameed (*Redox Homeostasis; Seed Biology; Biotechnology*)

Research Officers

1. Dr. Zain Ul Abideen (*Biofuels; Biochar; Nano-technology; Saline Agro-ecosystem*)
2. Dr. Tabassum Hussain (*Photosynthesis; Proteomics; Microscopy*)
3. Dr. Muhammad Qasim (*Phytochemistry; Metabolomics; Medicinal Xero-halophytes*)

RESEARCH FACILITIES

MAK-ISHU has state-of-the-art facilities for carrying out research in lab, greenhouse and field conditions. Instruments such as Atomic Absorption Spectrometer, CNS analyzer, GC-MS, HPLC, Ion Chromatograph, Gel Doc System, PCRs, Li-COR Photosynthesis System, Walz-PAM Fluorometer, Osmometer and Plant Water Status Console are available to carry out state of the art research in Halophyte Biology.

INFRASTRUCTURE

MAK-ISHU has good infrastructure to cater to research in laboratories, greenhouses, and field conditions. There are ten well-equipped working laboratories, a centralized instrument Lab, a tissue culture Lab. and three greenhouses.

CAREER OPPORTUNITIES

A number of MAK-ISHU graduates have been successful in acquiring teaching and research positions in public and private colleges, universities and R&D organizations related to plant and environmental sciences.

AVENUES FOR RESEARCH

- Plant Ecology and Ecophysiology
- Biochemistry and Biotechnology
- Phytochemistry of medicinal halophytes
- Transcriptomics and Proteomics
- Ecotoxicology and Phytoremediation
- Halo-fuel and Saline Agriculture

MARINE REFERENCE COLLECTION AND RESOURCE CENTER

INTRODUCTION OF THE DEPARTMENT

The Marine Reference Collection and Resource Center is the oldest marine biodiversity research center in Pakistan. It was founded in 1967 as an Invertebrate Reference Museum by renowned marine scientist of the country Prof. Dr. Nasima M. Tirmizi (late), who was also founder director of the first Institute of Marine Science in Pakistan. MRC&RC has been providing excellent facilities to the marine science community of Pakistan and regional countries. It supports marine biodiversity research through comprehensive taxonomic data on marine fauna and flora and is in fact one of the unique research and training facilities in the region with a mission emphasizing taxonomic research, conservation and education. The center, which has only been a feature of invertebrate taxonomic studies, has demonstrated an ability and agility in responding to the management and conservation of marine biodiversity in a national, regional and international context. MRC&RC has a preeminent museum dedicated to inspiring curiosity and learning about the invertebrate fauna through its unparalleled collections, exhibits and research.

So far 44 research projects have been accomplished through national and international fundings/grants. The faculty and research students of the center have produced 800 publications including research papers, articles, monographs, books and illustrated keys. MRC&RC has been publishing Pakistan Journal of Marine Sciences since the last three decades. This is the only research journal from Pakistan on Marine Sciences which is HEC recognized, indexed at WEB OF SCIENCE Master Journal List. The center is linked with a number of local and foreign institutions for collaborative research including: *University of Réunion, France.*, University of Kwa Zulu-Natal, South Africa., Sinop University, Turkey., Gazi University, Turkey., University of Baghdad, Iraq., University of Chittagong, Bangladesh., Sudan University of Science & Technology., University De Sao Paulo, Brazil., University of Kelaniya, Sri Lanka., Sindh Fisheries Department, Marine Fisheries Department, National Institute of Oceanography, SUPARCO, PCSIR, Coastal Development Authorities etc.

POSTGRADUATE PROGRAMS AVAILABLE

4. M.Phil
5. Ph.D
6. Post graduate training and research supervision.

PROGRAMS' OBJECTIVES

1. Clearly articulate the specific research goals and outcomes.
2. Familiarize with existing research. This will help identify gaps in the current knowledge.
3. Keep updated with the latest developments, methodologies, and technologies relevant to research work. Attend conferences, workshops, and seminars, and subscribe to reputable journals and publications.

AIMS AND SCOPE OF THE PROGRAM

Exploration of coastal areas natural resources: The Center supports marine biodiversity research through comprehensive taxonomic data on marine fauna & flora and thus offers research and training facilities in the region with a mission emphasizing on conservation and education.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION/EXPERTISE

S. No.	Names	Area of Specialization
1.	Dr. Farhan Shafiq Ghory	Larval rearing of Crustacea-Decapoda, Systematic of Brachyura and Sipuncula.
2.	Dr. Muhammad Uzair Khan	Taxonomy of Fish and Coastal Biodiversity.

3.	Ms. Wafra Matanat Zhaeen	Systematic of sea anemone and Stomatopod, Ecological and Environmental Impact and Assessment.
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RESEARCH FACILITIES AVAILABLE

The center has a carefully maintained Museum (repository) with more than 10,000 specimen, Preservation room, Wet laboratory to conduct research on live specimen, Ichthyology and Pollution lab, Mangrove lab, Molluscan lab, Plankton lab, SEM (Scanning Electron Microscope) room, Photographic lab, an Auditorium and Museum library with more than 3000 books, journals, reports, proceedings etc. A shore laboratory/Marine Research Station situated at Sandspit beach is also part of Marine Reference Collection and Resource Center.

INFRASTRUCTURE

1. Wet laboratory to conduct research on live specimen
2. Labs (All the working laboratories are provided with necessary equipment)
3. Museum (repository)
4. Museum library
5. Internet Access

AVENUES FOR RESEARCH

1. Explore marine biodiversity research through comprehensive taxonomic data on marine fauna & flora along the northern Arabian Sea.

NATIONAL NEMATOLOGICAL RESEARCH CENTER (NEMATOLOGY)

INTRODUCTION

National Nematological Research Center (NNRC) is a leading research and educational institute dedicated to the generation, acquisition of knowledge about Nematology in Pakistan. Nematology is a highly specialized field of biological sciences of agriculture. NNRC was established in 1974 and now has come to symbolize as an efficient, dynamic and forward looking research Center in Pakistan.

NNRC is ISO Certification 9001 – 2015 for the compliance of its diagnostic services and education program. The Center has been engaged in publishing bi-annual scientific journal, *Pakistan Journal of Nematology* (PJN) since 1983 and PSN Newsletter since 2002, covering the significant research in fundamental and applied Nematology within and outside the country. It provides diagnostic services related to nematode diseases. Beside this, several books, booklets, manual, proceedings and many national and international patents are the publications of the Center.

The Center has collaborations in research with many international and national agricultural research institutions and universities as well as various other departments of the University of Karachi. Numerous research projects have been awarded to NNRC by various funding agencies such as PSF and HEC. Diagnostic, advisory and consultancy services have also been provided to the farmers, scientists and extension workers from all over the country and abroad.

The NNRC collaborates with the government, international research bodies, research centers and non-governmental organizations to create awareness on vital issues of nematode problems and provides forum for all the Nematologists, researchers, agriculturists, extension workers and students of the country to exchange information about nematode diseases by organizing national and international conferences, seminars and workshops regularly.

Research and teaching jobs are available for the post graduates in agriculture related organizations/ universities/ institutions in the country and abroad. Wide range of career opportunities are also accessible in various organizations such as PARC, NARC, Plant Protection, Pakistan Atomic Energy Commission, Institutes of Biotechnology & Genetic Engineering, Environmental Protection Agency, Integrated Pest Management (IPM) services, Quarantine, Seed Certification, Pesticide formulations, Food Industries and Research and Development private sector and marketing. The Nematologists are also hired for diagnostic and consultancy services for the field and laboratory services.

POSTGRADUATE PROGRAMS AVAILABLE

1. M. Phil. In Nematology
2. Ph.D. in Nematology

PROGRAMS OBJECTIVES

The programs are solely entitled to educate and train the young students of biological sciences by providing them necessary skills and experiences to take sound decisions in research field.

AIM AND SCOPE OF THE PROGRAM

The aim is to promote the subject of Nematology to develop highly trained professionals through M. Phil / Ph. D. degree program.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION

Director

Dr. Saboohi Raza (Molecular Plant Pathology/ Biotechnology)

Research Officers

Dr. Tabassum Ara Khanam (Entomopathogenic Nematodes)
Dr. Salma Javed (Molecular /bioinformatics)

Teaching Associate

Dr. Erum Adeel (Taxonomy, Disease Management)

Visiting Faculty

Dr. Nasira Kazi (Taxonomy/ Marine nematodes)
Dr. Firoza Kazi (Plant Parasitic Nematodes)

RESEARCH FACILITIES AVAILABLE

NNRC has well equipped laboratories for research work on fundamental and applied Nematology of plant parasitic, free- living soil, fresh water, marine and entomopathogenic nematodes. Over 19,000 nematode specimens are preserved in the Nematode Collection Unit. Computer facilities and internet access are available for researchers. The seminar Library has a collection of more than 1000 books, journals and periodicals related to nematology.

INFRASTRUCTURE

Well maintained class rooms, seminar library, labs and large experimental field and two green houses are also available for the research purpose.

AVENUES FOR RESEARCH

The research programme includes work on the molecular taxonomy, biology, pathogenicity, population dynamics, control of plant parasitic nematodes, use of marine nematodes as pollution indicators, entomopathogenic nematodes. Short and long term training programs in Nematology are also arranged for researchers, students, extension workers and farmers of the country.

DR. ZAFAR H. ZAIDI CENTER FOR PROTEOMICS (BIOCHEMISTRY)

INTRODUCTION

Proteomics refers to the study of all proteins being expressed in a tissue, cell or an organ at a particular point and time. It reflects a snapshot of expressed proteins in the sample being analyzed. Proteomics is being utilized as a core technology globally to understand the role of protein molecules involved in health and disease. The cutting edge technologies used in proteomics have much greater impact in terms of early diagnosis and offers tremendous potential in prevention and therapeutics. Proteomics facilitates fundamental understanding of diseases by discovery of biomarkers to improve clinical diagnostics. This technology uses a combination of chromatography, electrophoresis, mass spectrometry and bioinformatics tools. Various projects on different types of cancers such as oral, liver and breast carcinoma, diabetes, osteoarthritis, role of antioxidants in disease conditions, stem cells therapeutics, venomomics, plant proteomics and salivary proteomics are in progress at the institute. During the last 14 years the institute has produced 13 Ph.D and 11 M.Phil.

DEGREE PROGRAMS OFFERED

- M.Phil in Biochemistry
- Ph.D in Biochemistry

PROGRAMS OBJECTIVE

M.Phil and Ph.D. programs at Dr. Zafar H. Zaidi Center for Proteomics offer training in Biochemistry analysis techniques used in research that cover study areas such as structure function relationship, biological properties and activities, bioinformatics and molecular analysis. The programs are designed for students with a background in biological sciences. The programs focus on investigation of proteins samples coming from variety of sources including human samples, animals, microbial and plants. The M.Phil program has a particular emphasis on basic training in a range of skills based subjects and provides a professional training for aspiring candidates. The Ph.D. program involves the advanced level of research with improving the skills further, addressing the interpreting capabilities and developing an individual as an independent researcher.

AIMS AND SCOPE OF THE PROGRAM

Deciding to pursue for an M.Phil degree in Biochemistry is the beginning to the road of learning. Students understand the basic principles and applications of the techniques used in biochemistry and its applications in various biological domains such as health and diseases. Students can enroll in academic positions after completion of degree as well as pursue their studies further towards a Ph.D degree.

Opting for Ph.D in Biochemistry at our institute provides opportunities for our students to enroll for jobs in industries as well as academia. We aim is to supervise and train our young forthcoming scientists and build a strong foundation for them to take up responsible positions in society.

LIST OF FACULTY MEMBERS WITH THEIR AREA OF SPECIALIZATION/EXPERTISE

Associate Professor

Dr. Zehra Manzoor (Director)

Research Interests:

Recent research interests include comparative proteomics, functional proteomics, biomarker identification and drug-induced modifications in breast, liver and oral cancer tissues, serum and cell lines. Recently, Dr. Manzoor is investigating the involvement of PDGFR pathway in cancer progression. She has published her work in international journals of the relevant field.

Dr. Mehtab Alam

Research Interests:

Research interest of Dr. Mehtab Alam is based on Peptide synthesis, Toxinology, Venom Chemistry, Ion channel modulators, Isolation and purification of proteins, Protein Bioinformatics, and Therapeutic peptides.

Assistant Professor

Dr. Syed Faraz Moin

Research Interests:

Current research interest of Dr. Syed Faraz Moin is Proteomics/Protein Biochemistry that includes protein characterization from snake venom, saliva, dental material, plant, and microbial origin. Currently, Dr. Moin is working on dental material specifically characterization of nanocomposites and proteins extracted from the dentine. He has authored international publications in the field.

Dr. Amber Noman

Research Interests:

Dr. Amber Noman is working on protein identification, quantification, and protein-protein interactions having aim to identify signaling networks of cancer cells that can be employed for therapeutic interpolation. Current research interest of Dr. Noman focuses on ASS1 gene regulation in hepatocellular carcinoma and the identification of carcinogenesis in liver cancer through EGFR pathway.

Dr. Kanwal Haneef

Research Interests:

Current research interest includes stem cell differentiation in response to hypoxia, genetic modifications, and small molecule treatments. This includes stem cell isolation, propagation, characterization, and their therapeutic applications. Dr. Haneef is also investigating the molecular mechanisms regulating the differentiation of stem cells using proteomics techniques.

FACILITIES

ZCP has state-of-the-art facilities in the field of Proteomics. The Institute is equipped with instruments such as vertical and horizontal electrophoresis, 2D gel electrophoresis, Western blotting, gel documentation system with Quantity One software, gel spot cutter with 2D gel analysis software (PD Quest), chromatographic instruments e.g. FPLC, HPLC, ice flaker, ultracentrifuge, PCR, qPCR, cell culture, fluorescence microscope, Cryostat and mass spectrometer. The institute has a computer lab with complete *in-silico* solutions for data analysis, protein structure prediction, web based searches and protein docking studies. ZCP also has a library with different reference books and journals related to genomics, proteomics and bioinformatics.

SPECIFIC RESEARCH AREAS

1. Clinical Proteomics
2. In Silico Studies in Drug Designing And Discovery
3. Oxidative Stress in Breast Cancer, Hepatocellular Carcinoma, Diabetes, and Osteoarthritis.
4. Plant Protein Biochemistry
5. Regulation of Micro RNA in Clinical Conditions.
6. Salivary Proteomics
7. Stem Cells Proteomics
8. Study of Differentiation Mechanisms in 2D and 3D Culture
9. Synthesis and Characterization of Dental Material
10. Venom Proteomics

INSTITUTE OF SPACE SCIENCE & TECHNOLOGY

INTRODUCTION

Space Science encompasses all scientific disciplines involving space exploration and the study of natural phenomena and physical bodies. From investigating the Planetary Boundary Layer to probing the depths of deep space, Space Science offers unparalleled opportunities to explore the mysteries of the universe. Advancements in Space Science and Technology not only expand our understanding of the cosmos but also drive scientific capacity building, technical human resource development, and the creation of spin-off technologies that contribute to economic and social development. For instance, Space Technology is crucial in urban planning, natural disaster monitoring, and environmental management through satellite and remote sensing technologies. Furthermore, space provides a strategic vantage point for military applications, with space militarization becoming increasingly significant among major spacefaring nations.

The Institute of Space Science and Technology (ISST) was established as an autonomous research institute of the University of Karachi in 1994 to produce skilled professionals in Space Science and Technology through comprehensive teaching and research programs. The principal aim of the Institute is to provide top-tier education in Astrophysics, Atmospheric Physics, Remote Sensing and GIS, Satellite Communication, and Aerospace Studies, fostering indigenous research and R&D capabilities. ISST has already produced groundbreaking research in diverse areas such as Computational Astronomy, Ionospheric Physics, Ozone Layer Depletion, Climate Change, Aerospace Science, Satellite Communication, and Remote Sensing & GIS. ISST collaborates with prestigious institutions such as the Pakistan Space Agency SUPARCO and the School of Marine and Atmospheric Sciences, Stony Brook University, NY, USA. Additionally, ISST maintains close ties with Pakistan's largest privately-owned Taqwa Observatory in Balochistan and collaborates with various industries, including UAS-Global, to provide hands-on experiences for students.

POSTGRADUATE PROGRAMS AVAILABLE

1. M.Phil.
2. Ph.D.

INNOVATIVE CURRICULUM

An updated and comprehensive curriculum designed to keep pace with the rapid advancements in Space Science and Technology.

CAREER DEVELOPMENT

Programs focused on developing the technical and professional skills required to excel in various space-related careers.

PROGRAM OBJECTIVES

1. To comprehend the latest advancements in Space Science and Technology
2. To generate new knowledge in Space Science
3. To conduct cutting-edge research in the domain of Space Science
4. To develop innovative space technologies
5. To engage students in semester projects for practical knowledge application

AIMS AND SCOPE OF THE PROGRAMS

Our M.Phil. and Ph.D. programs aim to cultivate research expertise to address indigenous problems related to Space Science and Technology. The updated curriculum is designed to provide students with in-depth theories and knowledge of the latest scientific developments in Space Science. The programs also aim to integrate students into the field of research and development through a diverse range of space-oriented elective subjects offered in the research degrees. By participating in these programs, students will gain the skills and knowledge necessary to contribute to scientific advancements and technological innovations, positioning themselves as leaders in the field of Space Science and Technology.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION / EXPERTISE

Learn from highly qualified faculty members who are experts in their respective fields, offering mentorship and guidance in research endeavors.

1. **Prof. Dr. Muhammad Jawed Iqbal** (Gravitation Physics, Atmospheric Physics Computational Astronomy)
2. **Dr. Mirza Jawwad Baig** (Oceanography, Climate Change Modelling)
3. **Dr. Naeem Sadiq** (Environmental Science, Ionospheric Physics, Solar Radiation, GIS)
4. **Dr. Syed Tanveer Iqbal** (Laser Spectroscopy, Space Weather, Astrophysics, Computational Cosmology)

RESEARCH FACILITIES AVAILABLE

1. **Cutting-Edge Research Facilities:** Access to GIS, Computer, Aerospace, Physics and Electronics laboratories.
2. **Astronomical Observatory:** Access to students to use the astronomical observatory, providing students with hands-on experience for deep space observation.

AVENUE FOR RESEARCH

1. Computational Astronomy
2. Computational Cosmology
3. Space Weather
4. Climate Change
5. Applications of Remote Sensing & GIS
6. Drone Technology
7. Aerodynamics
8. Propulsions (Jet and Rockets)
9. Satellite Technology

INTERNATIONAL COLLABORATIONS

Opportunities for collaborative research with renowned institutions and organizations globally, broadening academic and professional horizons.

INDUSTRY PARTNERSHIPS

Strong ties with industry leaders to ensure practical training and exposure to the latest technological advancements. Join ISST to embark on a journey of discovery, innovation, and excellence in the fascinating field of Space Science and Technology. Become part of a community dedicated to pushing the boundaries of human knowledge and contributing to significant scientific and technological breakthroughs.

CENTER OF EXCELLENCE FOR WOMEN'S STUDIES

INTRODUCTION

Since its inception, the Center of Excellence for Women's Studies (CEWS) University of Karachi has been engaged in offering teaching and research facilities in a friendly environment to prepare students to act as agents of change. We are the first to offer B.A (Hons.), M.A and M.Phil. / Ph. D. degree programs in Pakistan. Currently the Center offers admission in BS 4 Years Program (morning & evening), BS 3rd Year (morning & evening), and M.Phil. / Ph. D. programs. Women's Studies, an interdisciplinary subject, aims to make visible women's role and contributions towards building a gender-bias free society for a holistic development of the nation. We offer a variety of courses to address the issues of women and gender, both nationally and internationally. The major focus of our teaching and research is to bridge-up the gap between the academia and women at the grass-root level. Thus, our students move beyond the Campus and interact with women to document their every-day lived experiences.

CAREER OPPORTUNITIES

As Women's Studies prepares students to act as agents of social change, therefore, our students are well equipped and trained to take up jobs that a person qualified in any other discipline can undertake. Majority of CEWS graduates are now hired in various positions by private organizations, government and non-government agencies. As one of the requirements of our teaching is practicum, our students work as interns with various GOs / NGOs and thus gain practical training and gain experience. To help our students in their future careers, we remain engage with the corporate sector.

FACILITIES

CEWS has well-qualified faculty to help and assist students in their quest for knowledge. We also invite scholars, academics and activists from various fields to share their scholarship and experiences with our students.

Our library with its holding of more than 4000 books, journals, reports on women and gender is the only library of its kind in the country. We have a collection of press-clippings on almost all aspects of women's lives. Our library is accessed not only by our students but it also caters to the academic search needs of scholars, activists and journalist who wish to avail this facility. Our library catalogue is accessible through electronic portal.

SPECIFIC RESEARCH AREAS

1. Introduction to Women and Gender Studies
2. Women Technology & Entrepreneurship
3. Women & Work: Child Labour, Girl Child, Self-Employed Women
4. Women, Religion & Religious Traditions
5. Health Concerns of Women & Gender
6. Women's History
7. Women & Law
8. Women & Politics
9. Women & Media
10. Women & Health
11. Women Sports & Physical Education
12. Social Construct of Gender
13. Women in Islam
14. Women in World Religion
15. Psychology of Women
16. Development Planning (nationally & globally) and Women

POSTGRADUATE PROGRAMS OFFERED

1. M.Phil.
2. Ph.D.

PROGRAM AIM, MISSION, OBJECTIVES AND OUTCOMES***Vision of the Department***

To engage Women's Studies students, teaching and non-teaching faculty in gender oriented productive and research activities in order to bridge the gap between academic knowledge about women and gender.

Aim

To help develop critical insight about social realities, lived experiences and systemic oppressions.

M.Phil. / Ph.D. Program Mission

Students in Women's Studies are expected to adopt Women's Studies as a change-making exercise. It also seeks to engage students in the production of knowledge that emerges from feminist critique of social, cultural and institutional structures that promote and strengthen patriarchy, for the elimination of gender biases.

Mission Statement of the Department

The mission of the Center is to play a role as a catalyst for change, bridging the gap between academic knowledge about women and their everyday life experiences, along with to prepare a human resource for institutions of women and gender development. To train students in such a way that they learn to critically analyze patriarchal structures that hampers the growth of an egalitarian society — socio-economically, culturally, politically, and religiously. Also, engage students in an on-going academic activity through class-discussions and research in the production and dissemination of knowledge through a feminist critique of the historic and current social, cultural, and institutional structures that hamper social justice and create gender-based oppressive society. Moreover, the aim is to equip students to develop a research culture that encourages and contributes to gender equality and equity.

Objectives of the Program

The program focuses on extending the knowledge and research expertise of students through scientific inquiry. It is designed for individuals who have the intellectual capacity for advanced research and who wish to contribute to future development of Women's Studies as a discipline.

To understand multiplicity of feminist theories and the major feminist texts associated with them.

1. To understand the need and purpose of conducting research.
2. To examine how interdisciplinary feminist paradigms generate research for social change and policy intervention.
3. To understand how the dynamics of feminist research methods give identities to the 'objects' of research.

M. PHIL. / PH. D PROGRAM OUTCOMES

1. Students write and disseminate researches conducted about women and gender issues.
2. Students will demonstrate an analytical understanding of some major feminist theories by drawing out dominant themes considering the ways in which class, ethnicity, and sexual orientation have interacted with women's gendered identity and roles.
3. Student will develop thesis in consultation with the research supervisor.
4. Student will conduct six seminars of one hour each during one year of course work.

SCOPE OF THE PROGRAM

The M.Phil. / Ph.D. Women's Studies program opens up numerous career prospects for graduates and prepares them for various professional fields

FACULTY MEMBERS

Name	Designation	BPS	Degree	Title of the Ph.D. Thesis
Dr. Asma Manzoor	Professor/ Director	21	Ph. D.	A Study of Socio-Economic Conditions for the Establishment of Export Processing Zones and Reasons of Employment Attraction for Women in the Karachi Export Processing Zone
Dr. Shagufta Nasreen	Associate Professor	20	Ph. D.	Impact of Globalization Policies (fiscal Austerity, privatization and trade liberalization) conducted on the women workers of urban industrial sector of Karachi
Dr. Aliyah Ali	Associate Professor	20	Ph. D.	Causes of Crimes among female prisoners in Karachi Jail, a study of the socio-economic and environmental problems and effects on personality
Dr. Seema Manzoor	Assistant Professor	19	Ph. D.	Women Sufi's: A Veiled Paragon in South Asia

INSTITUTE OF MARINE SCIENCE

INTRODUCTION OF THE INSTITUTE

Marine Science is a branch of Earth Science that deals with all aspects related to oceans, seas, their coasts and seabed. It covers a wide range of topics including Ocean, currents, waves, and tides; marine organisms and ecosystem dynamics; geophysical fluid dynamics; plate tectonics and geology of the sea floor including their minerals and hydrocarbon potential; and fluxes of various chemical substances and physical properties within the ocean and across its boundaries. These diverse topics relate to a multitude of disciplines like chemistry, physics, biology, geology, meteorology, and geography that oceanographers blend together to comprehend knowledge of the world oceans and processes within it. Marine science is a contemporary field of education having emerged as a hybrid of traditional fields such as physical attributes of oceans, biology, chemistry and geology.

POSTGRADUATE PROGRAM AVAILABLE

1. M.Phil.
2. Ph.D.

PROGRAM OBJECTIVES

M. Phil. Marine science offers a wide range of multi-disciplinary areas of study comprising four core areas; Biological, chemical, geological and physical oceanography. Programme is designed for students with these diverse backgrounds. This programme focuses on marine science with particular emphasis on marine environment, coastal development, sustainability and includes basic training in a range of skills based subjects and provides a professional training for aspiring candidates.

AIMS AND SCOPE OF THE PROGRAM

Marine Science is aimed over preparing graduates for a variety of interesting careers and opportunities. A marine graduate/ scientist can be employed in federal, state and local government agencies to manage and monitor the use of marine resources, solve problems and conduct research. They can be employed by private industries such as seafood, fisheries, aquaculture, exploratory marine geology, satellite imagery, and ecological modeling, including environmental agencies and numerous non-government organizations.

Regular Faculty Members with their area of specialization/expertise

Name	Qualification	Designation	Area of Expertise
Dr. Nuzhat Afsar	Ph.D. (Marine Biology)	Director/ Associate Professor	Marine Biology/Marine Pollution / Marine Environment
Dr. M. Shoaib Kiani	Ph.D. (Zoology)	Assistant Professor (T)	Marine Zoology/ Marine mammal and large marine vertebrates
Dr. Nazia Arshad	Ph.D. (Zoology)	Assistant Professor (T)	Marine Zoology/ Benthic Ecology/Mesofauna
Mr. Kaiser Abbas	Ph.D. (in progress)	Taxonomist	Marine Pollution/ Sea Weeds, Mangroves/ Biocemical Work

RESEARCH FACILITIES AVAILABLE

The Institute of Marine has limited facilities but it is doing its best to make efficient use of available facilities and build on them consistently. For instance recent developments towards upgrading the facilities are present in the form of Sieve Shaker, HPLC, and Atomic Absorption Spectroscopy etc. for analysis of Biological, Geological, Chemical and Physical samples.

INFRASTRUCTURE

Ambitious and progressive get-up-and-go towards infrastructure of the Institute of Marine Science continued to upgrade its infrastructure.

AVENUES FOR RESEARCH

1. **Physical Oceanography**, or **marine physics**, studies deal with physical attributes of the oceans including temperature-salinity structure, mixing, waves, tides and currents, light and sound transmission, etc.
2. **Chemical Oceanography** or **marine chemistry** is the study of chemistry of the ocean and its chemical interaction with the atmosphere.
3. **Biological Oceanography** or **marine biology** is the study of plants, animals and microbes of the ocean and their ecological interaction with the habitat. The sustainable use of living resources may be provided through three separate modules with specific electives designed for fisheries, aquaculture and conservation biology.
4. **Geological Oceanography** or **marine geology** studies the structure and morphology of the ocean floor, tectonic activity and volcanism associated with plate margins, continental margins, beaches and coastal areas, sediment transport and deposition regimes, and offshore mineral and hydrocarbon deposits etc.

High-quality education expertise of the faculty that will prepare graduates and future scientists to accept the challenges of leadership, positions within the seafloor mining, seafood export, blue economy/ fisheries promulgation, and aquaculture and associated industries to strengthen the country's economy. The institute shall promote and encourage the pursuit for knowledge in new areas of marine science, innovation in skills and expertise for sustainable usage of renewable and non-renewable resources from sea i.e., natural product research and patent product usage in pharmacy, biomedical sciences & biotechnology.

ANY OTHER RELEVANT INFORMATION

Mission of these Marine Science Programmes is to equip students with an understanding of the world's oceans and a positive reception of the importance of marine ecosystems in a global perspective. This mission could be consummated through a combination of hands-on laboratory and field experience, supported by a broad background in the marine sciences, including basic to advanced knowledge of the natural science disciplines of biology, chemistry, physics, geology, statistics and mathematics.

INSTITUTE OF ENVIRONMENTAL STUDIES

INTRODUCTION

Established in 1982, this prestigious institute boasts the title of being the forerunner in environmental studies. Their curriculum offers a range of programs, from undergraduate degrees to Ph.D. Modern facilities and WHO recognition solidify their commitment to environmental education and research. Graduates leave empowered to tackle environmental challenges and promote public health.

OBJECTIVES OF THE POSTGRADUATE PROGRAMS

1. Enhance the learning and teaching environment to foster academic excellence.
2. Cultivate high-quality research in environmental and related fields, culminating in the awarding of M.Phil. and Ph.D. degrees.
3. Develop and upgrade laboratory facilities to facilitate cutting-edge research and practical training.
4. Provide and fortify consultative services in environmental matters and related disciplines.
5. Foster collaborations and partnerships with institutions both nationally and internationally to promote knowledge exchange and cooperation in environmental studies and allied fields.

AIM AND SCOPE OF M.PHIL. AND PH.D. PROGRAMS IN THE INSTITUTE

Aim

The M.Phil. and Ph.D. programs offered by the Institute of Environmental Studies aim to foster advanced research and scholarship in the field of environmental studies. These programs are designed to equip students with the theoretical knowledge, methodological skills, and critical thinking abilities necessary to address complex environmental challenges effectively. Through rigorous coursework, independent research, and interdisciplinary collaboration, students are prepared to become leading experts, scholars, and practitioners in environmental science, policy, and management.

Scope

This program isn't just about memorizing facts. You'll tackle environmental issues from all sides (Science, Law, Economics) through cutting-edge research and practical experience. Explore specific areas like climate change or sustainable development. Gain hands-on skills through fieldwork and projects with real-world partners. We emphasize a global perspective, fostering solutions that consider diverse viewpoints. Learn from leading experts and contribute to environmental knowledge through your research, ultimately informing policy and creating a more sustainable future.

REGULAR FACULTY MEMBERS

Dr. Muhammad Farrakh Nawaz

Dr. Farrakh, Director and Associate Professor, holds a Ph.D. in Environmental Sciences and post-docs from Germany and Australia. His research focuses on sustainable forestry, including carbon sequestration and phytoremediation. With numerous accolades, he has supervised over 30 post-graduate students and authored 100+ publications.

Prof. Dr. Moazzam Ali Khan

Professor Dr. Moazzam Ali Khan, a veteran faculty member since 1990, specializes in Climate Change, low-cost wastewater treatment and recycling, focusing on water-scarce areas. He's engaged in international projects, published over 120 papers.

Prof. Dr. Omm-e-Hany

Prof. Dr. Omm-e-Hany has done her Ph.D. in Microbiology from University of Karachi. Her research is focused on biological and biochemical effects of environmental contaminants. Her key areas of study included environmental issues pertaining to public health and hygiene.

Dr. Suraiya Jabeen

Dr. Suraiya Jabeen is working on Bioremediation of Hazardous Waste, Water and wastewater, Industrial Pollution management, Public Health & Hygiene. She is working as an Assistant Professor in the Institute. She did Masters and Ph.D from this institute.

Dr. Waqar Ahmed

Dr. Waqar Ahmed is Assistant Professor in the Institute. He has done his Masters and Ph.D in Environmental Studies on ecology of mangroves. His field of expertise is ecology of mangroves, coastal pollution, wildlife, air pollution and noise.

Dr. Aamir Alamgir

Dr. Aamir Alamgir, an Assistant Professor, with a Ph.D. in Climate Change issues. He is working on coastal ecology and climate response. He has published over 75 research articles.

Dr. Noor Fatima

Dr. Noor Fatima is a Lecturer in the institute with a Ph.D in climate change in 2022 from the Institute of Environmental Studies. She focuses on environmental microbiology, waste management, and biotechnology, with prior experience at Environmental Management Consultants.

Ms. Humaira Kanwal

Ms. Humaira Kanwal, working as Lecturer, with a M.Phil degree in 2022. She is enrolled in Ph.D. With expertise in Environment, Health, and Safety, she contributed significantly to projects like the Thar Coal Fields Project.

RESEARCH FACILITIES AVAILABLE

1. Environmental Microbiology and Environmental Chemistry Lab
2. Instrument Room
3. Seminar Library
4. Sampling instruments and Field Measurements devices

AVENUES OF RESEARCH

Climate change, Biodiversity Conservation, Environmental Pollution, Environmental Health, Sustainable Resource Management, Ecosystem Services, Environmental Policy and Governance, Urban Environmental Studies, Environmental Justice,

CENTER FOR PLANT CONSERVATION

INTRODUCTION

Center for Plant Conservation was established in Karachi University with the funding of HEC. It is the hub of academic and research activities. The Center is divided into two major sub-departments. (I) **Karachi University Herbarium** is a fundamental source of identification of plants and permanent reference Center via dried plant specimen and Flora of Pakistan. It serves the primary need of identification for basic and applied research in Botanical Sciences, Agriculture, Pharmacy, Genetics, Pharmaceutical Chemistry, Aerobiology, and Biotechnology. (II) **Prof. Dr. S.I. Ali Botanical Garden** is a conservation center spread over an area of 35 acres divided into 13 regions, holds well documented approximately 3000 plant species, in living form for the purposes of conservation, research, display and education. Seed bank in the botanical garden, containing seeds of more than 500 variety of plant species of Pakistan. A wide range of plants are also propagated by different methods in nursery.

The aim of the Center is to provide and promote facilities for advanced study and research in various aspects of plant conservation and ethno-botany. It enhances the knowledge of plant conservation and sustainable utilization of plant biodiversity of rare and endemic plants of Pakistan and also provides opportunities for collaborative research with related departments/ institutes at national and international levels.

POST GRADUATE DEGREE PROGRAMS AVAILABLE

1. M.Phil
2. Ph.D

PROGRAM OBJECTIVES

The objectives of an M.Phil/Ph.D program in a Center for Plant Conservation are typically designed to align with the overarching goals of plant conservation, sustainable utilization and building capacity in the area of Plant biodiversity to serve the community.

AIMS AND SCOPE OF THE PROGRAM

The primary aim of the M.Phil/Ph.D program in a Center for Plant Conservation is to cultivate a new generation of skilled researchers and professionals equipped with advanced knowledge and expertise to address the critical challenges in Plant Systematics, Conservation and Biodiversity preservation. The program strives to foster innovative research, interdisciplinary collaboration, and practical solutions to conserve plant species and their habitats, ensuring the sustainability of ecosystems for future generations.

The scope of the M.Phil/Ph.D program encompasses a wide range of activities and areas of study, including Plant Systematics, Plant Biodiversity, Seed Collection activities, Inventory of Rare and Endemic Plants, Botanical Surveys, Plant Conservation, Tissue Culture, Germ plasma storage, Ethnobotany, Horticulture; Propagation & Cultivation of Plants; and Phytochemistry.

REGULAR FACULTY MEMBERS

Dr. Roohi Bano (In-Charge and Associate Professor)

Dr. Roohi Bano, is the Incharge of Center for Plant Conservation. She has been associated with the field of teaching and research for more than 18 years. Her fields of expertise are Plant systematics, Seed and Pollen morphology, Phytogeography, Biodiversity, Chemical studies, basic and applied techniques.

Dr. Shaukat Ali (Senior Taxonomist)

Dr. Shaukat Ali is the Senior Taxonomist in this Center. He has been doing Research and Teaching since 17 years. His main field of research are Plant Taxonomy, Phytogeography, Palynology, Horticulture and Ecology.

Dr. Muneeba Khan (Taxonomist)

Dr. Muneeba Khan is working as Plant Taxonomist in Center for Plant Conservation. She has a experience of more than 15 years in the field of Research and Teaching. Her main areas of interest are Palynology, Aerobiology, and Plant Taxonomy.

RESEARCH FACILITIES AVAILABLE

Flora of Pakistan is a research based series of plants of Pakistan comprise of 225 volumes, providing baseline information for research scholars/ scientists at National and International level and also for exchange research program. This Flora has been written and edited by the world renowned Botanist Prof. Dr. Syed Irtefaq Ali and Prof. Dr. Muhammad Qaiser. Flora of Pakistan has put Pakistan on the botanical map of the world. This Flora has been widely used by the environmentalists, pharmacists, phytochemists, and foresters for their research related to Plants.

Karachi University Herbarium (KUH) has more than 150,000 plants species (mostly flowering plants), including those which are several decades old. Scientific collection of Plants documentation and preservation of naturally occurring flora in Karachi University Herbarium is the basic practice of all the researchers which have been associated with Plant Systematics research. Scientists belonging to other field of Biological, Pharmaceutical and Chemical research also collect and submit their voucher specimens of plants to Karachi University Herbarium for getting their plants identified/ verified and obtaining Voucher Specimen Number for further referencing/citation. KUH provided authentic reference material for scientific research like Playnology, Seed morphology, Genomics and Molecular Biology.

Herbarium research Library has more than 2000 books and 240 journals on Plant Sciences and a huge collection of national and international floras, proceedings and monographs for research scholars and students. Internet and audio visual facilities for the students are also available.

Scanning Electron Microscopy Lab This lab provides Scanning Electron Microscope facility to students, researchers and faculty members.

Karachi University Botanical Garden is a wide area of 35 Acres which is divided into 13 different zones as per vegetation variations. This garden is a hub of interesting learning and research opportunities for young researchers. The garden houses a wide range of plant species, including native and exotic plants, medicinal herbs, and rare species, providing a living repository for botanical diversity. The botanical garden is a vital resource for students and researchers from the university and beyond. It supports studies in botany, ecology, and environmental science, facilitating hands-on learning and research opportunities.

Seed Bank Another unique feature of Botanical Garden is the seed bank. It is established in the garden and contains about 500 sample of identified and documented seeds. These samples have been collected from various regions of Pakistan including wild, cultivated, ornament and medicinal plant seeds. Facilities of seed exchange program and many more are under process.

Tissue culture lab The Tissue Culture Lab at the Karachi University Botanical Garden is a specialized facility dedicated to the propagation and conservation of plants through tissue culture techniques. Tissue culture, also known as micropropagation, involves the cultivation of plant cells, tissues, or organs under sterile conditions on a nutrient culture medium. Tissue culture is an essential tool for conserving plant biodiversity. By propagating rare and endangered species, the lab helps in their preservation and reintroduction into their natural habitats.

Educational visits of Karachi University Botanical Garden The garden is open to the public on specialized events, offering a serene environment for visitors to enjoy nature. It also hosts educational programs, workshops, and guided tours to engage the community and promote environmental education. Special Garden tours are provided for school, colleges and university students/ research scholars for the preparation of Nature-based learning projects and presentations and organize field activities, workshops (Germination/ Organic gardening) and plantation drive.

Transport facility for field visits to various regions of Pakistan for research and training.

Other facilities The Center is well-equipped with advance laboratories i.e. Atomic absorption Lab, and Phytochemistry Lab.

INFRASTRUCTURE

Center for Plant Conservation consist of Two main buildings:

1. Karachi University Herbarium (KUH): It is double story building consisting of Herbarium, 4 labs, library, Scanning Electron Microscopy Room, and offices.
2. Prof. Dr. S.I. Ali Botanical Garden covers an area of 35 Acres. Karachi University Botanical Garden, located within the University of Karachi campus, is a significant green space in the bustling city of Karachi, Pakistan. It serves as a resource for education, research, conservation, and recreation. It is divided into 13 sections including Tropical Section, Endemic Plant Section, Aquatic Section, Bailey Hortorium, Wood land, Palm section, Local flora section, Halophytic Section, Cacti zone, Succulent Plant section, Medicinal Section, Propagation Unit & Organic Garden, Urban Forrest.

AVENUE FOR RESEARCH

Dean Grant, HEC grants, HEC Scholarships, Consultancy on national international level, Industrial linkage with pharmaceutical companies, Funding opportunities from international grant agencies.

CAREER OPPORTUNITIES

Students holding M. Phil. / Ph.D. degrees from Center for Plant Conservation can apply in different fields of Marine Sciences, Agriculture, Geography, Pharmaceutical Sciences, Forestry Departments, Plaeobotnical Research Center Botanical Sciences, Urban Forrest, Environmental Sciences Educational and Research Institutions, and Pharmaceutical Organizations (Government or Non-Government sectors).

PAKISTAN STUDY CENTER

INTRODUCTION OF THE CENTER

Pakistan Study Center was established under an Act of the parliament passed in 1976. The Center began functioning in 1983 and since then it has developed into a prestigious institute of higher learning and research. Apart from conducting courses at BS, M.Phil and Ph.D., levels, the Center has excelled in original in-house and contractual research that has received wide acclaim both at the national and international levels. The Center has also organized numerous international and national conferences apart from regular in-house seminars and symposia. The BS programme of the Center is based on multi-disciplinary courses which enables the students to compete in the civil service examination and also seek place in the teaching profession.

POSTGRADUATE PROGRAM AVAILABLE

1. M.Phil
2. Ph.D

PROGRAMS' OBJECTIVES

M.Phil / Ph.D Programme is designed to carry out intensive and in-depth research on any aspect of history, culture, literature, religion, education, Pakistan movement, sociology, international relations and politics of Pakistan.

AIMS & SCOPE OF THE PROGRAM

Aims

To promote post graduate research and produce scholars of Pakistan Studies.

Scope

The scholars will have scope in teaching and research institutions, Media, Civil Services, Other National & International Organization etc.

REGULAR FACULTY MEMBERS WITH THEIR AREAS OF SPECIALIZATION / EXPERTISE

1. Dr Erum Muzaffar (Foreign Relations, Environment & Geography)
2. Ms. Sadaf Masood (Economics)
3. Mr. Yasir Hanif (History)

RESEARCH FACILITIES AVAILABLE

Library and Internet Access

Infrastructure

Classrooms, multimedia, furniture etc.

AVENUES FOR RESEARCH

History, Politics, I.R, Economics, Sociology, Arts & Architecture, Culture, Literature, etc.

ANY OTHER RELEVANT INFORMATION

The Center regularly publishes a bi-annual research journal *Pakistan Perspectives* (since 1996) which is regarded as one of the most prestigious journals of social sciences published in Pakistan. The journal is recognized by the HEC in 'Y' category.

POSTGRADUATE ADMISSION POLICY – 2024

ADMISSION REQUIREMENTS AND PROCEDURE **FOR M.Phil / ME / MS**

Eligibility Criteria

- a) At least sixteen years of studies (Masters degree, M.A., M.Sc., M.B.A., M.P.A., B.B.A.(4 years), M.A.S., M.L.I.S., M. Com., B.S.(4 years), B.E. in Chemical Engineering, B. Pharm., Pharm. D., M.B.B.S., B.D.S., D.V.M.) or equivalent qualification with at least second division in the relevant field from an HEC recognized university/institution. However, candidates with M.B.B.S., B.D.S., D.V.M. or equivalent qualification will be required to undertake 15-credit hour courses (300-600 level), in addition to 24-credit hour courses prescribed for M.Phil program to be decided by the DRC and 30 credit hour courses for MS program. There is no requirement to submit Equivalence certificate for the graduates of Public Sector Universities.

Note: 6 credit hours of research work are mandatory for M.Phil. Those candidates who have done MS (with course work) are not eligible for admission in Ph.D

One M.Phil seat is especially allocated for the special person from each department / Center / Institute.

The admission on special persons' reserved seats is granted only after the medical examination and verification from the Social Welfare Department, Government of Sindh and / or civil surgeon.

b) Intradisciplinary Qualification

The intradisciplinary admission may only be allowed, if:

- a) **Departmental Research Committee allows, and**
- b) The applicant has a strong interest in purposing an MS/M.Phil/equivalent degree in a different discipline.
- c) **The applicant has passed GRE-Subject / equivalent test with minimum 50% marks in the discipline of admission and has taken 6-9 CH of deficiency courses of level 6.**
- d) The admission committee is satisfied that the applicant's knowledge of primary area (level 6) has sufficiently prepared him or her to undertake the course of studies of the MS/M.Phil/equivalent program (or, in the opinion of the admission committee, the preparation can be deemed satisfactory by taking a few additional courses after starting the program.

Admission Procedure

1. All candidates will be required to submit the following documents along with the **Entrance Test Application Form**
 - i. Consolidated transcript/marks certificate of the last degree examination.
 - ii. Matric / O levels Certificate
 - iii. Computerized National Identity Card.
 - iv. Recent photographs.
2. Appearance in Karachi University Entrance Test 2024 (KUET-2024) is mandatory for all applicants who wish to seek admission in M.Phil/MS program to be conducted by relevant departments/centers/institutes of the University of Karachi.
3. a) The test shall comprise of the following sections:

Section-I Subject knowledge	80%
Section-II English and Communication Skills	20%
- b) Candidates will be required to score at least 50% marks in KUET-2024 each section to be eligible for consideration for the *provisional admission*.

- c) If a candidate has already qualified the International (subject) GRE/HAT General/Equivalent test (GAT/NTS), he/she will be exempted from subject test component of KUET 2024. The validity of International (subject) GRE is 5 years and NTS/HAT General/Equivalent test (GAT/NTS) is for two years. The minimum passing score is 50%. However, the candidate has to appear in Karachi University Entrance Test 2024 (KUET-2024) English test moreover, interview will be mandatory

Note: *Qualifying the Karachi University Entrance Test (KUET-2024) is no guarantee for admission.*

4. All the students who qualify the Karachi University Entrance Test 2024 (KUET-2024) will be required to appear in person for an interview to be conducted by the respective Departmental Research Committee (DRC) to finalize the admission procedure. Distribution of interview marks are given in **Annexure-II**.

Provisional Admission for M.Phil /MS/ME

Applications for provisional admission will be considered on the basis of eligibility, merit, availability of seats and supervisors.

Final admission list will be prepared by the DRC according to the following formula:

- The marks obtained in the prerequisite degree examination will be given 20% weightage.
- The marks secured in the Entrance Test 2024 will be given 50% weightage.
- The marks obtained in the interview will be given 30% weightage.

Submission of Forms for Admission

Candidates selected for provisional admission will be required to complete admission form along with the submission of the following documents:

- Two recent photographs.
- Attested photocopies certificates/ degrees from S.S.C. to Masters or the prerequisite exam.
- Attested photocopy Computerized National Identity Card.
- A letter of leave for two years from the employer, if the candidate is in service or a certificate of non-employment stating that during research no employment shall be undertaken, executed on a stamp paper worth Rs.100/- (attested by Notary Public/ Oath Commissioner).
- Teachers and employees of organization where research work is conducted are exempted from leave. However, a NOC from the employer and a certificate stating that employment is relevant to his/her research from the Chairperson/Director of the concerned Department/ Institute/ Center will be required and should be submitted within three months after the date of provisional admission.
- In case of foreign students, NOC from the Ministry of Education, Economic Affairs Division and Ministry of Foreign Affairs, Government of Pakistan, Islamabad and a copy of valid visa and passport.
- Certificate from the Equivalence Committee, University of Karachi, in case the accreditation status of the prerequisite degrees is not clear.
- The Admission & Tuition fee is mentioned in **Annexure-I**.

Duration of Program

The duration of M.Phil and MS will be minimum of 1.5 years and maximum of four (4) years from the date of provisional admission. No further extension will be given.

Appointment of Research Supervisor

Faculty members of the University of Karachi / ASRB / HEC approved supervisors may be appointed as research supervisors.

Course Work

a) MS (with course work)

1. All selected candidates will be required to complete 10 courses of 30 credit hours (each of 3 credit hours in three semesters) in 1.5 year. See **Annexure-IV** for details.
2. Every candidate will be required to submit details of courses duly endorsed by the course Incharge and Chairperson of the parent department in the Semester Examination Section on prescribed course selection proforma (**Annexure-IV**).
3. Candidates are required to attend at least 75% classes of each course.
4. In the first year all candidate will be required to complete 8 courses of 24 credit hours (4 courses in each semester). In the second year candidate has to pass the 2 courses of 6 credit hours (2 course in 3rd semester) along with non-credit hours independent study (essay writing/project).
5. The candidate must clear the course work of 30 credit hours with minimum **CGPA 3.00**.
6. **A student who will be declared short of CGPR shall improve it in the next available single chance by appearing in any two courses having less than 70 marks i.e. B.**
7. If the student fails to attain a **CGPA of 3.00** or more even after the improvement chance, he/she will be given only a Post Graduate Diploma (PGD) instead of MS degree. Student can improve his/her CGPA in three years. Minimum **2.5 CGPA** is required for Post Graduate Diploma.
8. **Those candidates who have done MS (with course work) are not eligible for admission in Ph.D**

b) M.Phil / ME (With Course and Research Thesis)

1. All selected candidates will be required to complete 8 courses of 24 credit hours in two semesters (each course of 3 credit hours). See **Annexure-IV** for details.
2. Candidates are required to attend at least 75% classes of each course.
3. Every candidate will be required to submit details of courses duly endorsed by the course Incharge and Chairperson of the parent department in the Semester Examination Section on prescribed course selection proforma (**Annexure-IV**).
4. The student has to pass the course work with **CGPA 3.0** or more.
5. **A student who will be declared short of CGPR shall improve it in the next available single chance by appearing in two courses having less than 70 marks i.e. B.**
6. If the student fails to attain a CGPA of 3.00 or more even after improvement chances, after one year or overall within three years, he/she will be given only a Post Graduate Diploma (PGD) instead of M.Phil degree. Minimum **2.5 CGPA** is required for Post Graduate Diploma.
7. After passing the course work with **CGPA 3.0** or more, the student will be required to apply through proper channel for confirmation of admission and approval of research proposal within one year.
8. The student will submit a research proposal according to the approved format of the ASRB through proper channel along with a copy of the marks certificate issued by the Semester Examination Section.

Evaluation & Award of M.Phil and MS Degree

1. MS / ME (With Course Work)

The student will submit a mark Proforma of 30 credit hours course work with CGPA 3.0 or more along with clearance form through proper channel in ASRB for the award of MS degree (with course work). **They are not eligible for the admission in Ph.D.**

2. M.Phil (With Course and Research Thesis)

The student will submit a research thesis which will be evaluated by two external examiners (outside Karachi) appointed by the ASRB. The M.Phil degree shall be awarded by the ASRB subject to positive reports of the supervisor, the two external examiners and a successful oral defense of the thesis by the candidate.

ADMISSION REQUIREMENTS AND PROCEDURE FOR DOCTOR OF PHILOSOPHY (Ph.D)

Eligibility Criteria

- a) M.Phil., M.Pharm., M.S. (with research), MBA (with 4 years BBA) or equivalent qualification in the relevant field from HEC recognized University/Institution. There is no requirement to submit Equivalence certificate for the graduates of Public Sector Universities.

One Ph.D. seat is especially allocated for the special person from each department / Center / Institute.

The admission on special persons' reserved seats is granted only after the medical examination and verification from the Social Welfare Department, Government of Sindh and / or civil surgeon.

b) CGPA

- a) For admission in the Ph.D programs, a minimum CGPA of 3.0 (out of 4.0 in the semester system) in the M.Phil / equivalent degree being considered for admission, is required, whether such degree was obtained from Pakistani or foreign universities.
- b) If the CGPA/Percentage is not mentioned on the transcript, the candidate must produce equivalent weightage from the parent university.
- c) The students having strong demonstrated interest in obtaining Ph.D degree, but their CGPA is below 3.00 (minimum 2.6 CGPA) (out of 4.0 in the semester system) in the most recent degree obtained, may be admitted to a Ph.D program after fulfilling the following requirements:
 - i. Shall study additional courses of 9-12 CH of level 7 taking a zero semester at admission awarding university/HEI/DAI and score minimum 3.00 out of 4.00 GPA, and
 - ii. The admission committee is satisfied that the applicant's knowledge of primary area (level 7) has sufficiently prepared him or her to undertake the course of studies of the doctoral program.
 - iii. These requirements shall be in addition to any other requirements set in this policy for admission to a Ph.D program.

c) Intradisciplinary Qualification

The intradisciplinary admission may only be allowed, if:

- a) **Departmental Research Committee allows, and**
- b) The applicant has a strong interest in pursuing a Ph.D in a different discipline.
- c) **The applicant has passed GRE-Subject/Equivalent Test with minimum 60% marks in the discipline of admission and has taken 6-9 CH of deficiency courses of level 7.**
- d) The admission committee is satisfied that the applicant's knowledge of the primary area (level 7) has sufficiently prepared him or her to undertake the course of study of the doctoral program (or, in the opinion of the admissions committee, the preparation can be deemed satisfactory by taking a few additional courses after starting the program).

Admission Procedure (Ph.D)

1. All candidates will be required to submit the following documents along with the Entrance Test Application Form:
 - i. Consolidated transcript/marks certificate of the last degree examination.
 - ii. Matric / O levels Certificate
 - iii. Computerized National Identity Card.
 - iv. Recent photograph.
2. Appearance in Karachi University Entrance Test 2024 (KUET-2024) is mandatory for all applicants who wish to seek admission in Ph.D. program to be conducted by relevant departments/centers/institutes of the University of Karachi.
3. (a) The test shall comprise of the following sections;

Section-I Subject knowledge	80%
Section-II English and Communication Skills	20%

- (b) Candidates will be required to score at least 60% marks in each section of KUET-2024 to be eligible for consideration for the *provisional admission*.
- (c) If a candidate has already qualified the International (subject) GRE/GAT/NTS/HAT General, he/she will be exempted from subject test component of KUET 2024. The validity of International (subject) GRE is 5 years and NTS/GAT/HAT General is for two years. The passing score is 60%. However, the candidate has to appear in Karachi University Entrance Test 2024 (KUET-2024) English test Moreover, interview will be mandatory.

Note: *Qualifying the Karachi University Entrance Test (KUET-2024) is no guarantee for admission.
Admission will be granted on the basis of test and Interview.*

4. All the students who qualify the Karachi University Entrance Test 2024 (KUET-2024) will be required to appear in person for an interview to be conducted by the respective Departmental Research Committee (DRC) to finalize the admission procedure. Distribution of interview marks are given in **Annexure-III**.

Provisional Admission

Applications for provisional admission will be considered on the basis of eligibility, merit, availability of seats and supervisors. Final admission list will be prepared by the DRC, according to the marks secured in the Interview that will be given **50% weightage**. (See annexure-III)

Submission of Forms for Admission

Candidates selected for provisional admission will be required to complete admission form along with the submission of the following documents:

1. Two recent photographs, duly attested by the Chairperson / Director of the Departments/Institute/ Centers concerned.
2. Attested Photocopies of Certificates / degrees from S.S.C. to M.Phil. / MS or the prerequisite exam.
3. Attested photocopy of Computerized National Identity Card.
4. A letter of leave for three years from the employer, if the candidate is in service or an Affidavit of non-employment stating that during research no employment shall be undertaken, executed on a stamp paper worth Rs.100/-.
5. Teachers and employees of organization where research work is conducted are exempted from leave. However, a NOC from the employer and a certificate stating that employment is relevant to his/her research from the Chairperson/Director of the concerned Department/ Institute/ Center will be required and should submit within three months after the date of provisional admission.
6. In case of foreign students, NOC from the Ministry of Education, Economic Affairs Division and Ministry of Foreign Affairs, Government of Pakistan, Islamabad and a copy of valid visa and passport.
7. Certificate from the Equivalence Committee, University of Karachi, in case the accreditation status of the prerequisite degrees is not clear.
8. The Fee structure is mentioned in **Annexure-I**.

Duration of Ph.D

The duration of Ph.D is minimum of three (03) and maximum of eight (08) years from the date of provisional admission. No further extension will be given. Moreover minimum duration for Ph.D.

thesis submission is three (3) years from the date of synopsis submission in ASRB.

Course Work for Ph.D

1. All selected students will be required to complete 18 credit hours course work.
2. Candidates are required to attend at least 75% classes of each course.
3. The student will submit a research proposal according to the approved format of the ASRB through proper channel.

Comprehensive Examination

- i. Following the completion of coursework, every Ph.D student shall be required to pass the comprehensive examination to be granted candidacy as a Ph.D researcher, with the provision that the student failing to pass the comprehensive exam, shall be allowed to retake the exam once.

Appointment of Research Supervisor

Faculty members of University of Karachi/ASRB/HEC approved supervisors may be appointed as research supervisors.

Evaluation of Ph.D Research Thesis

During the research, the student must publish research paper/s from the thesis as per HEC-policy before the submission of Ph.D thesis in the ASRB Section duly verified by QEC, University of Karachi.

The thesis shall be evaluated by two foreign examiners. The Ph.D. degree will be awarded by the ASRB subject to the positive reports from the Research Supervisor, two foreign examiners and a successful public defense.

ADMISSION REQUIREMENTS AND PROCEDURE FOR LLM

Eligibility Criteria

At least sixteen years of studies (BA.LL.B/LL.B) or equivalent qualification with at least 'Second Division' in LLB having at least 55 percent aggregate or have a minimum CGPA of 3.00 from an HEC recognized university/institution. There is no requirement to submit Equivalence certificate for the graduates of Public Sector Universities.

Note: 6 credit hours of research work are mandatory for LL.M.

Admission Procedure

All candidates will be required to submit the following documents along with the **Entrance Test Application Form**

- i. Consolidated transcript/marks certificate of the last degree examination.
- ii. Matric / O levels Certificate
- iii. Computerized National Identity Card.
- iv. Recent photographs.

Appearance in Karachi University Entrance Test 2024 (KUET-2024) is mandatory for all applicants who wish to seek admission in LL.M. program to be conducted by School of Law University of Karachi.

The test shall comprise of the following sections:

Section-I Subject knowledge	80%
Section-II English and Communication Skills	20%

Candidates will be required to score at least 50% marks in each section to be eligible for consideration for the *provisional admission*.

Note: Qualifying the Karachi University Entrance Test (KUET-2024) is no guarantee for admission.

All the students who qualify the Karachi University Entrance Test 2024 (KUET- 2024) will be required to appear in person for an interview to be conducted by the respective Departmental Research Committee (DRC) to finalize the admission procedure. Distribution of interview marks are given in **Annexure-II**.

Provisional Admission for LLM

Applications for provisional admission will be considered on the basis of eligibility, merit, availability of seats and supervisors.

Final admission list will be prepared by the DRC according to the following formula:

- a) The marks obtained in the prerequisite degree examination will be given 50% weightage.
- b) The marks secured in the Karachi University Entrance Test 2024 (KUET- 2024) will be given 30% weightage.
- c) The marks obtained in the interview will be given 20% weightage.

Submission of Forms for Admission

Candidates selected for provisional admission will be required to complete admission form along with the submission of following documents:

1. Recent photograph.
2. Certificates/ degrees from S.S.C. to LL.B / B.A. LL.B.
3. Computerized National Identity Card.

4. A letter of leave for two years from the employer, if the candidate is in service or a certificate of non-employment stating that during research no employment shall be undertaken, executed on a stamp paper worth Rs. 100/- (attested by Notary Public/ Oath Commissioner).
5. Teachers and employees of organization where research work is conducted are exempted from leave. However, a NOC from the employer and a certificate stating that employment is relevant to his/her research from the Chairperson/Director of the concerned Department/ Institute/ Center will be required and should submit within six months after the date of provisional admission.
6. In case of foreign students, NOC from the Ministry of Education, Economic Affairs Division and Ministry of Foreign Affairs, Government of Pakistan, Islamabad and a copy of valid visa and passport.
7. Certificate from the Equivalence Committee, University of Karachi, in case the accreditation status of the prerequisite degrees is not clear.
8. The Admission & Tuition fee mentioned in **Annexure-I**.

Duration of program

The maximum duration of LL.M. program is four (4) years from the date of provisional admission. No further extension will be possible.

Appointment of Research Supervisor

Faculty members of University of Karachi / ASRB / HEC approved supervisors may be appointed as research supervisors.

LL.M. (With Course and Research Thesis)

1. All selected candidates will be required to complete the required 24 credit hours course work.
2. Candidates are required to attend at least 75% classes of each course.
3. The student has to pass the course work with CGPA 3.0 or more.
4. A student who will be declared short of CGPR shall improve it in the next available single chance by appearing in two courses having less than 70 marks i.e. B.
5. If the student fails to attain a CGPA of 3.00 or more even after improvement chances, after one year or overall within three years, he/she will be given only a Post Graduate Diploma (PGD) instead of LL.M. degree. Minimum 2.5 CGPA is required for Post Graduate Diploma.
6. After passing the course work with CGPA 3.0 or more, the student will be required to apply through proper channel for confirmation of admission and approval of research proposal.
7. The student will submit a research proposal according to approved format of the ASRB through proper channel along with a copy of the marks certificate issued by the Semester Examination Section

Evaluation & Award of LL.M. Degree

LL.M. (With Course and Research Thesis)

The student will submit a research thesis which will be evaluated by two external examiners (outside Karachi) appointed by the ASRB. The LL.M. degree shall be awarded by the ASRB subject to positive reports of the supervisor, the two external examiners and a successful oral defense of the thesis by the candidate.

ADMISSION REQUIREMENTS AND PROCEDURE FOR MS (Surgery) / MD (Medicine)

The MS (Surgery) / MD (Medicine) program at the Faculty of Medicine, University of Karachi runs under semester system of examinations.

Faculty of Medicine, University of Karachi offers admission in MS/MD program in the following fields:

Doctor of Medicine (MD)		Master of Surgery (M.S.)	
S.No	Specialty	S.No	Specialty
1	Anesthesiology	1	General Surgery
2	Community Medicine	2	Gynecology & Obstetrics
3	General Medicine	3	Ophthalmology
4	Nephrology	4	Orthopedic Surgery
5	Neurology	5	Otorhinolaryngology
6	Chest Medicine (Pulmonology)	6	Urology
7	Cardiology	7	Oral & Maxillofacial Surgery
8	Psychiatry	8	Operative Dentistry
9	Gastroenterology	9	Oral Pathology

Eligibility:

1. Bachelor of Medicine and Bachelor of Surgery (M.B.B.S.) or Bachelor of Dental Surgery B.D.S. or equivalent qualification with at least Second division from a Pakistan Medical Commission (PMC)/Pakistan Medical and Dental Council (PMDC) and Higher Education Commission (HEC) recognized University plus one year house job.
2. Copy of the valid registration certificate from Pakistan Medical Commission (PMC)/Pakistan Medical and Dental Council (PMDC).

Admission Procedure:

- i. All candidates are required to fill the application form for admission in MS/MD program at University of Karachi.
- ii. All required documents mentioned below must be attested by class-I Gazetted officer and submitted along with the Karachi University Entrance Test 2024 (KUET-2024) Application Form:
 - Photocopy of the last examination marks sheet.
 - Photocopy of the last degree.
 - Copy of the valid registration certificate from Pakistan Medical & Dental Council (PMDC) / Pakistan Medical Commission (PMC)
 - Photocopy of the Computerized National Identity Card.
 - Passport size photograph.
 - One year house job certificate.
 - Affidavit stating that candidate is not enrolled in any other regular postgraduate training program of any University or Degree awarding institution (DAI).
- iii. Candidates will be required to appear in the Karachi University Entrance Test (KUET-2024) conducted by the Faculty of Medicine, University of Karachi. The test shall have a total of 200 marks and comprise of the following sections:

Section I English and Communication Skills:	20%
Section II Subject Knowledge:	80%
(MCQ 50% + Video Projected Clinical assessment 30%)	

- iv. Candidates will be required to score at least 50% marks in section-I and 50% marks in section-II, separately to be eligible for consideration for Interview.

Note: *Qualifying the Karachi University Entrance Test (KUET-2024) is no guarantee for admission.*

- v. All the students who qualify the test will be required to appear in person for an interview to be conducted by the DRC/interview board of the Faculty of Medicine to finalize the admission procedure.

Exemption from Entrance Test:

Exemption from University of Karachi entry test 2024 may be granted by ASRB on case-to-case basis if the candidate has acquired qualifications from American Board of Medical Specialties or holds FCPS, FRCP, MRCOG, FRCOG, MRCP, FFDRCSI, FDSRCS or other major degree/equivalent qualification of relevant specialty recommended by the DRC. Any candidate willing to obtain exemption from University of Karachi Entry Test must apply for the same at the time of admission form submission. No request of exemption will be entertained after the conduction of University of Karachi Entry test 2024.

Format of Karachi University Entrance Test:

The entrance test shall have a total of 200 marks and shall comprise of the following sections:

- **Section I** (English and Communication Skills): 20% = 40 Marks
- **Section II** (Subject Knowledge): 80% = 160 Marks

Section-I

The section shall consist of 40 questions from English and Communication Skills.

Section-II

The section will comprise two sub-sections. Both sub-sections are separate passing heads:

- **MCQs 50% = 100 Marks**
- **Video projected clinical assessment 30% = 60 Marks**

A. MCQs Paper

The MCQs paper shall consist of 100 Multiple Choice Questions with single best answer. The MCQ's will be derived from different subjects as follows:

S.No.	Subject	No. of MCQs
i.	Basic Sciences	50
	a. Anatomy	10 from each subject
	b. Physiology & Biochemistry	
	c. Pathology	
	d. Pharmacology	
	e. Bio Statistics & Epidemiology	
ii. Any one	Medicine and Allied (MD)	50
	Surgery and Allied (MS)	50
	Dental Surgery (MS)	50

- Each correct answer will carry 1 mark
- An applicant scoring **50%** of marks will be considered pass in written paper.

B. Video-Projected Clinical Examination (VPCE)

- The Video - Projected Clinical Examination will consist of 30 videos/slides.
- The Videos/slides shall consist of data, videos and images of patient's pictures, Histopathology slides, X-rays, CT-Scan, Laboratory test, video clips of patients and pathophysiological videos.
- Each correct answer will carry 02 marks
- The applicant scoring 50 % of marks shall be considered pass in Video-Projected Clinical Examination.

Result Declaration of Karachi University Entrance Test:

The successful candidates will be required to score at least 50% marks in each section and/or subsection, to be eligible for consideration in the interview.

Interview:

Interviews of successful candidates will be held at the office of Dean, Faculty of Medicine by the DRC/Interview Board. Final credential of the candidate e.g., no. of attempts in each profession, any gold medals or distinctions, Diploma or minor qualification in relevant field. Work /research experience in a recognized institution. Number of research articles published in a National or International Journal will be assessed. Professional attitude of the candidates will also be assessed as per enclosed format in **Annexure-V**.

Final Admission List

1. Final admission list will be prepared by the Dean Faculty of Medicine according to the following formula:
 - i. The marks secured in the Entrance Test 2024 will be given 80% weightage.
 - ii. The marks obtained in the interview will be given 20% weightage.

Provisional Admission

Candidates selected for provisional admission will be required to submit admission form along with the following documents signed by the Dean of the Faculty of Medicine:

- i. Copy of the valid registration certificate from Pakistan Medical and Dental Council PMDC.
- ii. Degree holders of other Universities are required to submit Migration Certificate.
- iii. Computerized National Identity Card.
- iv. Candidates who are employed in Public/Private institutions require Four years leave/deputation for training and course work from the employer. This leave will be 5 years in case of sub-specialties while those who are self-employed will be required to submit the affidavit on Stamp paper of Rs. 100 stating that he/she will be attending the course work and training as per laid down criteria of University of Karachi accordingly.
- v. Teachers and employees of organization where training /research work conducted are exempted from leave. However, a NOC from the employer will be required.
- vi. Candidates who are not employed, a certificate of non-employment stating that during training no employment (other than training program) shall be undertaken/ executed on a stamp paper worth Rs. 100/- (Attested by Notary Public/ Oath Commissioner).
- vii. In case of foreign students, N.O.C. from the Ministry of Education, Economics Affairs Division and Ministry of Foreign Affairs, Government of Pakistan Islamabad and an attested copy of valid visa and passport.
- viii. Certificate from the Equivalence Committee, University of Karachi, in case, the accreditation status of the degrees is not clear.
- ix. Evidence of admission & Tuition fee, given in fee structure **Annexure-I**

Appointment of the Supervisor:

Every student will have to select a supervisor and training slot at the beginning of the program duly approved by the Dean Faculty of Medicine. The supervisor must fulfill the criteria of the University of Karachi.

ASRB shall approve the detailed plan of work on the subject or any of its branches and the supervisor on the recommendation of the Dean, Faculty of Medicine.

Criteria of Appointment of Supervisor:

1. The Supervisor should have major postgraduate degree/ diploma i.e. FCPS, Ph.D, FRCS, FRCP, MRCOG, MS, MD or equivalent with at least 5 years standing in relevant field. The Supervisor should be working in an permanent academic position as professor/Associate Professor/ Assistant Professor) whereas for the later two positions there must be at least five / seven years of experience respectively in a teaching hospital/ department in PMDC approved teaching institution.
2. The Supervisor of some other training/research program e.g. FCPS, should not have more than 8 trainees.
3. The Supervisor should follow the structured training program provided by the University of Karachi (ASRB) for his/her candidate, including one-year course work and is required to submit progress report to the ASRB of the candidates quarterly.
4. The Supervisor must have on his/her credit the below workshops being conducted by Faculty of Medicine University of Karachi / an institution recommended by Faculty of Medicine University of Karachi or by any recognized national or foreign university or Degree awarding institution (DAI) including Research Methods/ Biostatistics/Epidemiology, Medical Writing and Critical appraisal, Software based data Analysis and Evaluation
5. The Supervisor will guide his student in synopsis writing and check the synopsis before submission to the University of Karachi.

SCHEME OF STUDY**Duration of Program**

The duration of MD/MS program shall be minimum four (4) years and maximum of seven (7) years (in specialties) or minimum five (5) years and maximum eight (8) years (in subspecialties) with structured training in a University of Karachi approved institution under the guidance of an approved supervisor duly verified by the DRC to be submitted the office of Dean Faculty of Medicine.

Program Structure

The MS/MD program at the University of Karachi consists of three parts. To qualify for the award of MS or MD at the University of Karachi a candidate is required to pass all three parts independently. Moreover details regarding each part are stated below.

Part A: Course work in Basic Medical Sciences Part

B: Research & Thesis writing

Part C: Clinical Training (to commence after the selection of supervisor and training institution duly verified by the DRC and endorsed by the Dean Faculty of Medicine)

Part A: Course work in Basic Medical Sciences

All selected candidates will be required to complete 24 credit hours courses in two semesters. The candidates are required to attend at least 75% classes in each course. At the end of course work, the candidate will be required to secure CGPA 3.0 or more, And if a students is declared short of CGPR he/she shall improve it in the next available single chance by appearing in any two courses

having less than 70 marks i.e. B⁻.

If a student fails to attain a CGPA of 3.00 or more even after improvement chances, after one year or overall, within three years, he/she will be given a certificate of training only whereas the admission in the MS/MD will be terminated accordingly. A minimum of 2.5 CGPA is required for the award of course work certificate and training.

During the 1st year of the course work the student will be based in the departments of Basic Medical Sciences during semester 1 and in the department of research / epidemiology during the semester two as recommended by DRC and duly approved by Dean Faculty of Medicine University of Karachi.

A candidate may apply to ASRB through the Dean, Faculty of Medicine for exemption from Part-A and theory of Part-C assessment if he/she holds MRCP, FRCS MRCOG, FCPS or other major equivalent degrees. Their training period will be reduced to 2 years for specialty and 3 years for sub specialty. Candidates who have passed in the same sub specialty will be required 2 years of training. All other cases will be treated on the case to case basis.

Part B: Research & Thesis Writing

After passing the Part-A coursework with CGPA 3.0 or more, the student will be eligible for the submission of synopsis. The research proposal should be submitted according to approved format of ASRB along with a copy of the marks proforma issued by the Semester Examinations Section. The minimum duration for the submission of a thesis is one year from the date of synopsis approval.

A student must submit to ASRB before the award of the degree at least one published research paper from a thesis in a HEC recognized journal before the submission of thesis or a letter of acceptance of an ISI impact factor journal. He/she will also be required to submit the proof of the workshops being conducted by Faculty of Medicine University of Karachi / an institution recommended by Faculty of Medicine University of Karachi or by any recognized national or foreign university or Degree awarding institution (DAI) including Research Methods/ Biostatistics / Epidemiology, Medical Writing, Communication Skills, Software based data Analysis and Evaluation and Basic Life Support (BSL).

The thesis shall be examined by two foreign examiners appointed by ASRB on the recommendation of the Dean, Faculty of Medicine. The examiners shall report as to whether the thesis is accepted or rejected. In the event of a disagreement between the examiners, ASRB shall appoint a third examiner. After approval of the thesis, the thesis defense examination shall be held at the office of the Dean, Faculty of Medicine. Upon successful acceptance of thesis and satisfactory thesis defense, the candidate shall be permitted to appear in Part-C examination.

Part C: Theory and Clinical/OSCE/Viva/Examination

All candidates admitted in MS/MD courses shall appear in Part-C examination at the end of structured training program (Fourth year/Fifth year) and having declared Pass in Part-A and Part-B examinations. The examination shall be held on a biannual basis. The candidate who fails to pass the examination within 7 years/8 years of enrollment shall be dropped from the course. A total of maximum five attempts shall be allowed within three years of completion of training to Pass Part-C of MS/MD program.

The Part-C examination shall comprise of the following three components. All three components are separate passing heads to be declared as Pass in Part-C and secure a minimum of 60% in each component:

Written Examination: 300 marks

- Three papers of 100 marks each.
- The examiner shall be approved by ASRB on the recommendation of Dean, Faculty of Medicine.
- A candidate declared Pass in the first component (written examination) will be eligible to appear in second component (Oral & practical/clinical examination).

Clinical/OSCE/Viva Voce Examination: 300 marks

- 1 long case 100 marks
- 4 short cases 100 marks
- OSCE/Viva Voce 100 marks

The candidate will be required to appear against a panel of three examiners approved by the ASRB on the recommendation of the Dean, Faculty of Medicine for the second component of Part-C (Oral & Practical/Clinical Examination). A maximum of three attempts to Pass in Oral & Practical/Clinical Examinations will be given to a candidate failing which the candidate will have to reappear in both First and Second component of Part-C Examinations.

Log Book Evaluation 200 Marks (50 marks per year)

A candidate has to maintain the Log book throughout the length of the course, which has to be duly certified and evaluated by the Supervisor (in consultation with the Co-Supervisor, if appointed). The developed Log Book will be approved by ASRB on the recommendation of the Dean, Faculty of Medicine. The performance of the candidate shall be evaluated on annual basis i.e. 50 marks/year. The log book shall reflect the performance of the candidate based on his/her competence of skills, assignments, attitude, behavior, journal club, lectures, presentations and clinico-pathologic conferences attended.

Part C: Theory and Clinical /OSCE/Viva Voce Examination (For sub-specialty)

Part C examination for sub-specialty shall be divided into two parts:

- Part C-I
- Part C-II

Part C-I shall be held after completion of two years of training in the following format:

- Written 100 Marks
- Clinical/OSCE Exam 100 Marks
- Log Book 100 Marks (50 each year)

Part C-II shall be held at the end of 5 years training and in the following format:

- Written 300 Marks
- Oral, & Clinical 300 Marks
- Log Book 150 Marks (50 each year)

Award of MS/MD Degree

After successful completion of the structured courses of MS/MD and qualifying Part-A, Part- B and Part-C examinations, the degree with the title MS/MD shall be awarded.

POSTGRADUATE ADMISSIONS – 2024

SCHEDULE

M.Phil, ME, MS, LLM, Ph.D, MS (Surgery), & MD (Medicine)

1.	Date of Advertisement	Sunday, June 23, 2024
2.	Issuance & submission of Entrance Test Forms (Form can be downloaded for KU website and submitted to the respective department)	June 24, 2024 to July 05, 2024
3.	List of eligible candidates for KU Entrance Test will be displayed on KU website and Departmental notice boards	July 14, 2024
4.	Collection of admit cards by the candidates eligible for KU Entrance Test from the office of the Department concerned	July 18, 2024 to July 22, 2024
5.	Karachi University Entrance Test	Sunday, July 28, 2024
6.	List of successful candidates of Karachi University KUET will be displayed on KU website.	August 04, 2024
7.	Interview of Successful Candidates at their respective departments/ centers/ institutes	August 05, 2024 to August 09, 2024
8.	Final Merit List of successful candidates for provisional admission will be displayed on KU website	August 18, 2024
9.	Issuance and Submission of Admission Form and Fee Voucher (Voucher duly endorsed by ASRB) with all documents to the Postgraduate Admission Office, Psychology Department 2 nd Floor	August 20, 2024 to August 30, 2024
10.	Commencement of Semester	September 01, 2024

POSTGRADUATE ADMISSIONS – 2024

ACADEMIC CALENDAR

First Semester

Teaching	September 01, 2024	December 30, 2024
Semester Examinations	January 01, 2025	January 10, 2025
Semester Break	January 11, 2025	January 14, 2025

Second Semester

Teaching	January 15, 2025	May 16, 2025
Semester Examinations	May 20, 2025	May 28, 2025
Semester Break	May 29, 2025	May 31, 2025

Annexure-I

FEE STRUCTURE – 2024

Department/Program	Processing Fee	Admission Fee (M.Phil/ Ph.D)	M.Phil Semester Fee (Per Six Months)	M.Phil Thesis Evaluation/ Viva Examination Fee	Ph.D Semester Fee (Per Six Months)	Ph.D Thesis Evaluation/ Viva Examination Fee
Islamic History	7,500	30,000	65,000	50,000	80,000	85,000
Philosophy	7,500	30,000	65,000	50,000	80,000	85,000
Arabic	7,500	30,000	65,000	50,000	80,000	85,000
Sindhi	7,500	30,000	65,000	50,000	80,000	85,000
Persian	7,500	30,000	65,000	50,000	80,000	85,000
Urdu	7,500	30,000	65,000	50,000	80,000	85,000
Usool-ud-din	7,500	30,000	65,000	50,000	80,000	85,000
Islamic learning	7,500	30,000	65,000	50,000	80,000	85,000
Quran and Sunnah	7,500	30,000	65,000	50,000	80,000	85,000
Pakistan Study Center	7,500	30,000	65,000	50,000	--	--
Center of Excellence for Women's Studies	7,500	30,000	65,000	50,000	80,000	85,000
Social work	7,500	30,000	65,000	50,000	80,000	85,000
Education	7,500	30,000	65,000	50,000	80,000	85,000
History	7,500	30,000	65,000	50,000	80,000	85,000
Special Education	7,500	30,000	65,000	50,000	80,000	85,000
Library and Information Science	7,500	30,000	65,000	50,000	---	---
Political Science	7,500	30,000	65,000	50,000	80,000	85,000
International Relations	7,500	30,000	75,000	50,000	90,000	85,000
Mass Communication	7,500	30,000	75,000	50,000	90,000	85,000
Economics	7,500	30,000	75,000	50,000	90,000	85,000
Sociology	7,500	30,000	75,000	50,000	90,000	85,000
English Literature	7,500	30,000	75,000	50,000	90,000	85,000
ELT & Applied Linguistics (English)	7,500	30,000	75,000	50,000	--	--
Teacher Education	7,500	30,000	75,000	50,000	90,000	85,000
Botany	7,500	30,000	75,000	50,000	90,000	85,000
Physics	7,500	30,000	75,000	50,000	90,000	85,000
Geography	7,500	30,000	75,000	50,000	90,000	85,000
Genetics	7,500	30,000	75,000	50,000	90,000	85,000
Institute of Marine Science	7,500	30,000	75,000	50,000	90,000	85,000
Marine Reference Collection and Resource Center	7,500	30,000	75,000	50,000	---	---
Zoology	7,500	30,000	75,000	50,000	90,000	85,000
Geology	7,500	30,000	75,000	50,000	90,000	85,000
Applied Physics	7,500	30,000	75,000	50,000	90,000	85,000
Biochemistry	7,500	30,000	75,000	50,000	90,000	85,000
Microbiology	7,500	30,000	75,000	50,000	90,000	85,000
Biotechnology	7,500	30,000	75,000	50,000	90,000	85,000
Institute of Space Science and Technology	7,500	30,000	75,000	50,000	90,000	85,000

Postgraduate Admissions – 2024
Prospectus

Physiology	7,500	30,000	75,000	50,000	90,000	85,000
Applied Chemistry and Chemical Technology	7,500	30,000	75,000	50,000	90,000	85,000
Food Science and Technology	7,500	30,000	75,000	50,000	90,000	85,000
Chemistry	7,500	30,000	75,000	50,000	90,000	85,000
Commerce	7,500	30,000	75,000	50,000	90,000	85,000
Mathematics	7,500	30,000	75,000	50,000	90,000	85,000
Dr. M. Ajmal Khan Institute of Sustainable Halophyte Utilization	7,500	30,000	75,000	50,000	90,000	85,000
Dr. A. Q. Khan Institute of Biotechnology & Genetic Engineering	7,500	30,000	75,000	50,000	90,000	85,000
Center for Plant Conservation	7,500	30,000	75,000	50,000	90,000	85,000
Chemistry (HEJ)	7,500	30,000	75,000	50,000	90,000	85,000
Dr. Zafar H. Zaidi Center for Proteomics	7,500	30,000	75,000	50,000	90,000	85,000
Health, Physical Education and Sports Sciences	7,500	30,000	90,000	50,000	110,000	85,000
Institute of Environmental Studies	7,500	30,000	90,000	50,000	110,000	85,000
Statistics	7,500	30,000	90,000	50,000	110,000	85,000
Agriculture and Agribusiness Management	7,500	30,000	90,000	50,000	110,000	85,000
Chemical Engineering	7,500	30,000	90,000	50,000	110,000	85,000
Computer Science	7,500	30,000	90,000	50,000	110,000	85,000
Karachi University Business School	7,500	30,000	90,000	50,000	110,000	85,000
National Nematological Research Center	7,500	30,000	90,000	50,000	110,000	85,000
Psychology	7,500	30,000	100,000	50,000	120,000	85,000
Criminology	7,500	30,000	100,000	50,000	120,000	85,000
Public Administration	7,500	30,000	100,000	50,000	120,000	85,000
Dr. Panjwani Center for Molecular Medicine and Drug Research	7,500	30,000	100,000	50,000	120,000	85,000
Pharmaceutics	7,500	30,000	100,000	50,000	120,000	85,000
Pharmacognocny	7,500	30,000	100,000	50,000	120,000	85,000
Pharmaceutical Chemistry	7,500	30,000	100,000	50,000	120,000	85,000
Pharmacy Practice	7,500	30,000	100,000	50,000	--	--
Pharmacology	7,500	30,000	100,000	50,000	120,000	85,000
Physical Therapy	7,500	30,000	120,000	50,000	150,000	85,000

Faculty of Law (LLM)

Department/Program	Processing Fee	Admission Fee LLM	LLM Semester Fee (Per Six Months)	LLM Thesis Evaluation/ Viva Examination Fee
School of Law	15,000	75,000	100,000	50,000

Faculty of Medicine: MS (Surgery) & MD (Medicine)						
Department / Program	Processing Fee	Admission Fee MS / MD	MS/MD Semester Fee (Per Six Months)	MS/MD Thesis Evaluation / Viva Examination Fee	MS / MD Entrance Test Exemption	MS / MD Exemption for Training
MS/MD	15,000	75,000	150,000	85,000	20000	75000

Foreign Students			
Department/Program	Processing Fee	Admission Fee	Semester Fee (Per Six Months)
All Programs	7500	30,000	200,000

Annexure-II



Advanced Studies & Research Board University of Karachi

INTERVIEW CRITERIA FOR M.Phil / LLM / ME						
	Research, Knowledge, Experience and Aptitude			Oral Communication	Proposed Research Plan/Project	Total
	Research Experience (thesis/project)	Research Dissemination (poster/presentation)	Publication			
Max Marks	04	04	02	10	10	30
Marks Obtained						
Comments						
Marks obtained in the Interview (out of 30)						
Marks obtained from Last Degree (20%)						
Marks obtained from test (50%)						
Total Marks Obtained (Out of 100)						
Name of Student:						
Form No:						
Department:						

Name and Signature of DRC Members

1. _____

4. _____

2. _____

5. _____

3. _____

6. _____

Name and Signature of the Chairperson



Advanced Studies & Research Board University of Karachi

INTERVIEW CRITERIA FOR Ph.D						
	Research, Knowledge, Experience and Aptitude			Oral Communication	Proposed Research Plan/Project	Total
	Research Experience (thesis/project)	Research Dissemination (poster/presentation)	Publication			
Max Marks	10	05	05	15	15	50
Marks Obtained						
Comments						
Total Marks Obtained in the interview out of 50						
Marks obtained from Test (50%)						
Total Marks Obtained (Out of 100)						
Name of Student:						
Form No:						
Department/Institute/Center						

Name and Signature of DRC Members

- | | |
|----------|----------|
| 1. _____ | 4. _____ |
| 2. _____ | 5. _____ |
| 3. _____ | 6. _____ |

Name and Signature of the Chairperson



Advanced Studies & Research Board University of Karachi

No. ASRB/Exams/

1. As per University of Karachi rules all M.Phil Students are required to complete eight courses of 3 credit hours each (total 24 credit hours) in two semesters. With minimum CGPA of 3.
2. As per University of Karachi rules all MS students are required to complete ten courses of 3 credit hours each in four semesters (total of 30 Credit hours). In 1st year 8 courses (4 courses per semester) and in the second year 2 courses (1 course per semester).
3. Each student will be required to take 8 courses. There is no restriction to take 50% interdisciplinary courses. However, DRC of the respective department may reconsider this policy according to its specific need.
4. All newly admitted M.Phil, MS and MS (Surgery)/ MD (Medicine) students are required to submit the following information duly endorsed by the course Incharge and Chairperson/Director of parent department/ Institute/ center in the Semester Examination Section.

SEMESTER - I / II

S.No	Course No.	Course Title	Name of the Department offering the courses	Signature of the course Incharge
1				
2				
3				
4				
5				
6				

Name of Student: _____ Signature: _____

Name of Department (Major): _____

Date: _____

**Chairperson/ Director
(Parent Department)**



Advanced Studies & Research Board University of Karachi

INTERVIEW CRITERIA FOR MS / M.D

	Clinical / Subject / Research Knowledge / skills and Work Experience				Professional Attitude	
	No. of attempts in Prof. Exams / Distinctions / Gold Medal / Diploma / Workshops Field / ACLS / BLS / Work Experience in Relevant Field	Clinical / Subject Knowledge / Letter of Recommendations / Good standing Certificates	Research Publication / Research Orientation	Surgical Cases / Procedures / performed Assist / Case Presentation	Motivation / Oral Communication / Behavior / Personality	Total
Max Marks	30	20	10	20	20	100
Marks obtained						
Comments:						

Total Marks Obtained out of 100 = _____

Name of Student	
Form Number	
Specialty (MD / MS / Dentistry)	
Total marks obtained in interview out of 100	

Name and signature of members DRC/Interview Board

1. _____ 2. _____

3. _____ 4. _____

Name and Signature of Dean: _____